# RecoTrackerEntranceCompare: entrance seeding diagnostic

A: original saved trajectory. B: truth position only. C: full truth boundary.
D: truth momentum only, when enabled. No truth constraint upstream of entrance.
Control: fresh backward propagation from all-reconstructed boundary quantities.
All distances: mm. Momenta: MeV/c. Times: ns. Phase: degrees. Pitch: mrad.
Signed width s68 = (q84.1345-q15.8655)/2. Standard deviation includes tails.
Nonnegative errors use 68% and 95% containment, not signed Gaussian widths.
All statistics use unbinned finite values, including histogram under/overflow.
NaN values and unavailable states are counted, not converted to zero.
dt is corrected for the common proton-bunch clock; dt_beam includes that clock.
Fixed-z comparisons do not measure an independent longitudinal resolution.
Target branch is unique, has initial production direction, and precedes entrance.
Principal target/foil results require completed transport; partial roots are counted.
Residual plots use available valid states; always compare their sample sizes.
Event efficiencies keep the identical generated-primary event denominator.
Full-truth residuals can include stochastic material effects and transport error.
Linear angular moments depend on wrapping; consult the circular statistics.
Input and propagated covariance-invalid counts are separate; legacy status bits overlap.
ROOT histograms remain in the input file. JSON/CSV retain full-precision tables.

# A: tracker entrance / baseline_direct

```
ent_dx         N=     14 NaN=    0
  mean=4.8223 med=3.2414 std=5.705 s68=4.9245
  |err| q68=7.8131 q95=13.871 q99=17.214 tail |err|>40: 0 (0.000%)
ent_dy         N=     14 NaN=    0
  mean=1.8938 med=1.3315 std=5.9954 s68=4.7027
  |err| q68=5.5939 q95=12.109 q99=13.306 tail |err|>40: 0 (0.000%)
ent_dr         N=     14 NaN=    0
  mean=7.9902 med=7.2668 std=5.6115 s68=6.5158
  |err| q68=9.8191 q95=16.344 q99=17.749 tail |err|>40: 0 (0.000%)
ent_dp         N=     14 NaN=    0
  mean=-0.012168 med=-0.058663 std=0.2003 s68=0.12819
  |err| q68=0.14224 q95=0.42859 q99=0.44266 tail |err|>2: 0 (0.000%)
ent_dpz        N=     14 NaN=    0
  mean=-0.0047099 med=0.036083 std=0.38263 s68=0.28402
  |err| q68=0.31559 q95=0.7342 q99=0.79296 tail |err|>2: 0 (0.000%)
ent_pitch      N=     14 NaN=    0
  mean=-0.13011 med=-0.14815 std=7.6142 s68=7.4009
  |err| q68=7.8808 q95=14.656 q99=15.804 tail |err|>20: 0 (0.000%)
ent_dt         N=     14 NaN=    0
  mean=0.18222 med=0.41588 std=0.56102 s68=0.30346
  |err| q68=0.56417 q95=1.0275 q99=1.2627 tail |err|>3: 0 (0.000%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.17488 med=-0.11659 std=0.76236 s68=0.79992
  |err| q68=0.98923 q95=1.2329 q99=1.261 tail |err|>3: 0 (0.000%)
```

# A: tracker entrance / baseline_reflected

```
ent_dx         N=     14 NaN=    0
  mean=-17.81 med=2.6304 std=80.203 s68=4.1333
  |err| q68=6.4336 q95=135.75 q99=269.08 tail |err|>40: 2 (14.286%)
ent_dy         N=     14 NaN=    0
  mean=-25.458 med=-1.8042 std=90.498 s68=6.5499
  |err| q68=6.684 q95=130.83 q99=307.06 tail |err|>40: 1 (7.143%)
ent_dr         N=     14 NaN=    0
  mean=43.509 med=7.5704 std=117.02 s68=11.603
  |err| q68=9.278 q95=192.1 q99=409.14 tail |err|>40: 2 (14.286%)
ent_dp         N=     14 NaN=    0
  mean=0.12702 med=-0.025854 std=0.5587 s68=0.13923
  |err| q68=0.15751 q95=0.88625 q99=1.8549 tail |err|>2: 1 (7.143%)
ent_dpz        N=     14 NaN=    0
  mean=-0.89445 med=0.029303 std=2.8803 s68=0.3643
  |err| q68=0.35132 q95=4.5017 q99=9.8678 tail |err|>2: 1 (7.143%)
ent_pitch      N=     14 NaN=    0
  mean=22.762 med=-1.1969 std=69.101 s68=9.14
  |err| q68=8.211 q95=124.94 q99=238.51 tail |err|>20: 2 (14.286%)
ent_dt         N=     14 NaN=    0
  mean=0.31794 med=0.38511 std=1.3602 s68=0.40233
  |err| q68=0.68861 q95=3.4289 q99=3.4852 tail |err|>3: 2 (14.286%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.16352 med=-0.2705 std=1.1934 s68=0.38663
  |err| q68=0.48489 q95=2.6213 q99=2.9785 tail |err|>3: 1 (7.143%)
```

# A: tracker entrance / control_direct

```
ent_dx         N=     14 NaN=    0
  mean=4.8223 med=3.2414 std=5.705 s68=4.9245
  |err| q68=7.8131 q95=13.871 q99=17.214 tail |err|>40: 0 (0.000%)
ent_dy         N=     14 NaN=    0
  mean=1.8938 med=1.3315 std=5.9954 s68=4.7027
  |err| q68=5.5939 q95=12.109 q99=13.306 tail |err|>40: 0 (0.000%)
ent_dr         N=     14 NaN=    0
  mean=7.9902 med=7.2668 std=5.6115 s68=6.5158
  |err| q68=9.8191 q95=16.344 q99=17.749 tail |err|>40: 0 (0.000%)
ent_dp         N=     14 NaN=    0
  mean=-0.012168 med=-0.058663 std=0.2003 s68=0.12819
  |err| q68=0.14224 q95=0.42859 q99=0.44266 tail |err|>2: 0 (0.000%)
ent_dpz        N=     14 NaN=    0
  mean=-0.0047099 med=0.036083 std=0.38263 s68=0.28402
  |err| q68=0.31559 q95=0.7342 q99=0.79296 tail |err|>2: 0 (0.000%)
ent_pitch      N=     14 NaN=    0
  mean=-0.13011 med=-0.14815 std=7.6142 s68=7.4009
  |err| q68=7.8808 q95=14.656 q99=15.804 tail |err|>20: 0 (0.000%)
ent_dt         N=     14 NaN=    0
  mean=0.18222 med=0.41588 std=0.56102 s68=0.30346
  |err| q68=0.56417 q95=1.0275 q99=1.2627 tail |err|>3: 0 (0.000%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.17488 med=-0.11659 std=0.76236 s68=0.79992
  |err| q68=0.98923 q95=1.2329 q99=1.261 tail |err|>3: 0 (0.000%)
```

# A: tracker entrance / control_reflected

```
ent_dx         N=     14 NaN=    0
  mean=-17.81 med=2.6304 std=80.203 s68=4.1333
  |err| q68=6.4336 q95=135.75 q99=269.08 tail |err|>40: 2 (14.286%)
ent_dy         N=     14 NaN=    0
  mean=-25.458 med=-1.8042 std=90.498 s68=6.5499
  |err| q68=6.684 q95=130.83 q99=307.06 tail |err|>40: 1 (7.143%)
ent_dr         N=     14 NaN=    0
  mean=43.509 med=7.5704 std=117.02 s68=11.603
  |err| q68=9.278 q95=192.1 q99=409.14 tail |err|>40: 2 (14.286%)
ent_dp         N=     14 NaN=    0
  mean=0.12702 med=-0.025854 std=0.5587 s68=0.13923
  |err| q68=0.15751 q95=0.88625 q99=1.8549 tail |err|>2: 1 (7.143%)
ent_dpz        N=     14 NaN=    0
  mean=-0.89445 med=0.029303 std=2.8803 s68=0.3643
  |err| q68=0.35132 q95=4.5017 q99=9.8678 tail |err|>2: 1 (7.143%)
ent_pitch      N=     14 NaN=    0
  mean=22.762 med=-1.1969 std=69.101 s68=9.14
  |err| q68=8.211 q95=124.94 q99=238.51 tail |err|>20: 2 (14.286%)
ent_dt         N=     14 NaN=    0
  mean=0.31794 med=0.38511 std=1.3602 s68=0.40233
  |err| q68=0.68861 q95=3.4289 q99=3.4852 tail |err|>3: 2 (14.286%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.16352 med=-0.2705 std=1.1934 s68=0.38663
  |err| q68=0.48489 q95=2.6213 q99=2.9785 tail |err|>3: 1 (7.143%)
```

# A: tracker entrance / momentum_direct

```
ent_dx         N=     14 NaN=    0
  mean=4.8223 med=3.2414 std=5.705 s68=4.9245
  |err| q68=7.8131 q95=13.871 q99=17.214 tail |err|>40: 0 (0.000%)
ent_dy         N=     14 NaN=    0
  mean=1.8938 med=1.3315 std=5.9954 s68=4.7027
  |err| q68=5.5939 q95=12.109 q99=13.306 tail |err|>40: 0 (0.000%)
ent_dr         N=     14 NaN=    0
  mean=7.9902 med=7.2668 std=5.6115 s68=6.5158
  |err| q68=9.8191 q95=16.344 q99=17.749 tail |err|>40: 0 (0.000%)
ent_dp         N=     14 NaN=    0
  mean=-0.012168 med=-0.058663 std=0.2003 s68=0.12819
  |err| q68=0.14224 q95=0.42859 q99=0.44266 tail |err|>2: 0 (0.000%)
ent_dpz        N=     14 NaN=    0
  mean=-0.0047099 med=0.036083 std=0.38263 s68=0.28402
  |err| q68=0.31559 q95=0.7342 q99=0.79296 tail |err|>2: 0 (0.000%)
ent_pitch      N=     14 NaN=    0
  mean=-0.13011 med=-0.14815 std=7.6142 s68=7.4009
  |err| q68=7.8808 q95=14.656 q99=15.804 tail |err|>20: 0 (0.000%)
ent_dt         N=     14 NaN=    0
  mean=0.18222 med=0.41588 std=0.56102 s68=0.30346
  |err| q68=0.56417 q95=1.0275 q99=1.2627 tail |err|>3: 0 (0.000%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.17488 med=-0.11659 std=0.76236 s68=0.79992
  |err| q68=0.98923 q95=1.2329 q99=1.261 tail |err|>3: 0 (0.000%)
```

# A: tracker entrance / momentum_reflected

```
ent_dx         N=     14 NaN=    0
  mean=-17.81 med=2.6304 std=80.203 s68=4.1333
  |err| q68=6.4336 q95=135.75 q99=269.08 tail |err|>40: 2 (14.286%)
ent_dy         N=     14 NaN=    0
  mean=-25.458 med=-1.8042 std=90.498 s68=6.5499
  |err| q68=6.684 q95=130.83 q99=307.06 tail |err|>40: 1 (7.143%)
ent_dr         N=     14 NaN=    0
  mean=43.509 med=7.5704 std=117.02 s68=11.603
  |err| q68=9.278 q95=192.1 q99=409.14 tail |err|>40: 2 (14.286%)
ent_dp         N=     14 NaN=    0
  mean=0.12702 med=-0.025854 std=0.5587 s68=0.13923
  |err| q68=0.15751 q95=0.88625 q99=1.8549 tail |err|>2: 1 (7.143%)
ent_dpz        N=     14 NaN=    0
  mean=-0.89445 med=0.029303 std=2.8803 s68=0.3643
  |err| q68=0.35132 q95=4.5017 q99=9.8678 tail |err|>2: 1 (7.143%)
ent_pitch      N=     14 NaN=    0
  mean=22.762 med=-1.1969 std=69.101 s68=9.14
  |err| q68=8.211 q95=124.94 q99=238.51 tail |err|>20: 2 (14.286%)
ent_dt         N=     14 NaN=    0
  mean=0.31794 med=0.38511 std=1.3602 s68=0.40233
  |err| q68=0.68861 q95=3.4289 q99=3.4852 tail |err|>3: 2 (14.286%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.16352 med=-0.2705 std=1.1934 s68=0.38663
  |err| q68=0.48489 q95=2.6213 q99=2.9785 tail |err|>3: 1 (7.143%)
```

# A: tracker entrance / position_direct

```
ent_dx         N=     14 NaN=    0
  mean=4.8223 med=3.2414 std=5.705 s68=4.9245
  |err| q68=7.8131 q95=13.871 q99=17.214 tail |err|>40: 0 (0.000%)
ent_dy         N=     14 NaN=    0
  mean=1.8938 med=1.3315 std=5.9954 s68=4.7027
  |err| q68=5.5939 q95=12.109 q99=13.306 tail |err|>40: 0 (0.000%)
ent_dr         N=     14 NaN=    0
  mean=7.9902 med=7.2668 std=5.6115 s68=6.5158
  |err| q68=9.8191 q95=16.344 q99=17.749 tail |err|>40: 0 (0.000%)
ent_dp         N=     14 NaN=    0
  mean=-0.012168 med=-0.058663 std=0.2003 s68=0.12819
  |err| q68=0.14224 q95=0.42859 q99=0.44266 tail |err|>2: 0 (0.000%)
ent_dpz        N=     14 NaN=    0
  mean=-0.0047099 med=0.036083 std=0.38263 s68=0.28402
  |err| q68=0.31559 q95=0.7342 q99=0.79296 tail |err|>2: 0 (0.000%)
ent_pitch      N=     14 NaN=    0
  mean=-0.13011 med=-0.14815 std=7.6142 s68=7.4009
  |err| q68=7.8808 q95=14.656 q99=15.804 tail |err|>20: 0 (0.000%)
ent_dt         N=     14 NaN=    0
  mean=0.18222 med=0.41588 std=0.56102 s68=0.30346
  |err| q68=0.56417 q95=1.0275 q99=1.2627 tail |err|>3: 0 (0.000%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.17488 med=-0.11659 std=0.76236 s68=0.79992
  |err| q68=0.98923 q95=1.2329 q99=1.261 tail |err|>3: 0 (0.000%)
```

# A: tracker entrance / position_reflected

```
ent_dx         N=     14 NaN=    0
  mean=-17.81 med=2.6304 std=80.203 s68=4.1333
  |err| q68=6.4336 q95=135.75 q99=269.08 tail |err|>40: 2 (14.286%)
ent_dy         N=     14 NaN=    0
  mean=-25.458 med=-1.8042 std=90.498 s68=6.5499
  |err| q68=6.684 q95=130.83 q99=307.06 tail |err|>40: 1 (7.143%)
ent_dr         N=     14 NaN=    0
  mean=43.509 med=7.5704 std=117.02 s68=11.603
  |err| q68=9.278 q95=192.1 q99=409.14 tail |err|>40: 2 (14.286%)
ent_dp         N=     14 NaN=    0
  mean=0.12702 med=-0.025854 std=0.5587 s68=0.13923
  |err| q68=0.15751 q95=0.88625 q99=1.8549 tail |err|>2: 1 (7.143%)
ent_dpz        N=     14 NaN=    0
  mean=-0.89445 med=0.029303 std=2.8803 s68=0.3643
  |err| q68=0.35132 q95=4.5017 q99=9.8678 tail |err|>2: 1 (7.143%)
ent_pitch      N=     14 NaN=    0
  mean=22.762 med=-1.1969 std=69.101 s68=9.14
  |err| q68=8.211 q95=124.94 q99=238.51 tail |err|>20: 2 (14.286%)
ent_dt         N=     14 NaN=    0
  mean=0.31794 med=0.38511 std=1.3602 s68=0.40233
  |err| q68=0.68861 q95=3.4289 q99=3.4852 tail |err|>3: 2 (14.286%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.16352 med=-0.2705 std=1.1934 s68=0.38663
  |err| q68=0.48489 q95=2.6213 q99=2.9785 tail |err|>3: 1 (7.143%)
```

# A: tracker entrance / truth_direct

```
ent_dx         N=     14 NaN=    0
  mean=4.8223 med=3.2414 std=5.705 s68=4.9245
  |err| q68=7.8131 q95=13.871 q99=17.214 tail |err|>40: 0 (0.000%)
ent_dy         N=     14 NaN=    0
  mean=1.8938 med=1.3315 std=5.9954 s68=4.7027
  |err| q68=5.5939 q95=12.109 q99=13.306 tail |err|>40: 0 (0.000%)
ent_dr         N=     14 NaN=    0
  mean=7.9902 med=7.2668 std=5.6115 s68=6.5158
  |err| q68=9.8191 q95=16.344 q99=17.749 tail |err|>40: 0 (0.000%)
ent_dp         N=     14 NaN=    0
  mean=-0.012168 med=-0.058663 std=0.2003 s68=0.12819
  |err| q68=0.14224 q95=0.42859 q99=0.44266 tail |err|>2: 0 (0.000%)
ent_dpz        N=     14 NaN=    0
  mean=-0.0047099 med=0.036083 std=0.38263 s68=0.28402
  |err| q68=0.31559 q95=0.7342 q99=0.79296 tail |err|>2: 0 (0.000%)
ent_pitch      N=     14 NaN=    0
  mean=-0.13011 med=-0.14815 std=7.6142 s68=7.4009
  |err| q68=7.8808 q95=14.656 q99=15.804 tail |err|>20: 0 (0.000%)
ent_dt         N=     14 NaN=    0
  mean=0.18222 med=0.41588 std=0.56102 s68=0.30346
  |err| q68=0.56417 q95=1.0275 q99=1.2627 tail |err|>3: 0 (0.000%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.17488 med=-0.11659 std=0.76236 s68=0.79992
  |err| q68=0.98923 q95=1.2329 q99=1.261 tail |err|>3: 0 (0.000%)
```

# A: tracker entrance / truth_reflected

```
ent_dx         N=     14 NaN=    0
  mean=-17.81 med=2.6304 std=80.203 s68=4.1333
  |err| q68=6.4336 q95=135.75 q99=269.08 tail |err|>40: 2 (14.286%)
ent_dy         N=     14 NaN=    0
  mean=-25.458 med=-1.8042 std=90.498 s68=6.5499
  |err| q68=6.684 q95=130.83 q99=307.06 tail |err|>40: 1 (7.143%)
ent_dr         N=     14 NaN=    0
  mean=43.509 med=7.5704 std=117.02 s68=11.603
  |err| q68=9.278 q95=192.1 q99=409.14 tail |err|>40: 2 (14.286%)
ent_dp         N=     14 NaN=    0
  mean=0.12702 med=-0.025854 std=0.5587 s68=0.13923
  |err| q68=0.15751 q95=0.88625 q99=1.8549 tail |err|>2: 1 (7.143%)
ent_dpz        N=     14 NaN=    0
  mean=-0.89445 med=0.029303 std=2.8803 s68=0.3643
  |err| q68=0.35132 q95=4.5017 q99=9.8678 tail |err|>2: 1 (7.143%)
ent_pitch      N=     14 NaN=    0
  mean=22.762 med=-1.1969 std=69.101 s68=9.14
  |err| q68=8.211 q95=124.94 q99=238.51 tail |err|>20: 2 (14.286%)
ent_dt         N=     14 NaN=    0
  mean=0.31794 med=0.38511 std=1.3602 s68=0.40233
  |err| q68=0.68861 q95=3.4289 q99=3.4852 tail |err|>3: 2 (14.286%)
ent_dt_beam    N=     14 NaN=    0
  mean=-0.16352 med=-0.2705 std=1.1934 s68=0.38663
  |err| q68=0.48489 q95=2.6213 q99=2.9785 tail |err|>3: 1 (7.143%)
```

# C: target production plane / baseline_direct

```
tar_dx         N=     11 NaN=    0
  mean=33.364 med=-6.2891 std=207.93 s68=161.13
  |err| q68=127.87 q95=397.63 q99=412.64 tail |err|>400: 1 (9.091%)
tar_dy         N=     11 NaN=    0
  mean=-14.419 med=-14.864 std=116.89 s68=64.721
  |err| q68=71.204 q95=245.72 q99=295.14 tail |err|>400: 0 (0.000%)
tar_dr         N=     11 NaN=    0
  mean=186.28 med=118.41 std=153.36 s68=171.7
  |err| q68=289.22 q95=406.84 q99=414.69 tail |err|>400: 1 (9.091%)
tar_phase      N=     11 NaN=    0
  mean=-18.548 med=-1.7866 std=74.226 s68=72.55
  |err| q68=90.025 q95=136.55 q99=145.61 tail |err|>90: 4 (36.364%)
tar_center     N=     11 NaN=    0
  mean=6.9398 med=5.3228 std=6.083 s68=2.4418
  |err| q68=6.4073 q95=17.401 q99=23.307 tail |err|>40: 0 (0.000%)
tar_radius     N=     11 NaN=    0
  mean=-3.3214 med=-0.66135 std=6.0245 s68=2.9971
  |err| q68=4.3958 q95=13.422 q99=19.095 tail |err|>20: 1 (9.091%)
tar_dp         N=     11 NaN=    0
  mean=-0.59397 med=-0.22121 std=1.2574 s68=0.48047
  |err| q68=0.42566 q95=2.7342 q99=4.036 tail |err|>2: 1 (9.091%)
tar_dpz        N=     11 NaN=    0
  mean=1.4636 med=0.379 std=3.1059 s68=2.2494
  |err| q68=1.2245 q95=7.4967 q99=9.0005 tail |err|>2: 3 (27.273%)
tar_pitch      N=     11 NaN=    0
  mean=-30.53 med=-7.1281 std=59.413 s68=43.238
  |err| q68=23.504 q95=145.92 q99=173.99 tail |err|>20: 7 (63.636%)
tar_dt         N=     11 NaN=    0
  mean=1.0744 med=0.65275 std=1.4461 s68=1.2896
```

# C: target production plane / baseline_direct

```
  |err| q68=0.97114 q95=3.8047 q99=3.9455 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     11 NaN=    0
  mean=0.69283 med=0.24681 std=1.4361 s68=1.1621
  |err| q68=1.0968 q95=3.4883 q99=3.5077 tail |err|>10: 0 (0.000%)
```

# C: target production plane / baseline_reflected

```
tar_dx         N=      9 NaN=    0
  mean=-10.765 med=-20.757 std=196.16 s68=166.31
  |err| q68=204.36 q95=354.14 q99=391.04 tail |err|>400: 1 (11.111%)
tar_dy         N=      9 NaN=    0
  mean=78.312 med=-9.7066 std=169.97 s68=170.19
  |err| q68=174.35 q95=345.39 q99=381.17 tail |err|>400: 0 (0.000%)
tar_dr         N=      9 NaN=    0
  mean=215.23 med=279.75 std=165.21 s68=183.43
  |err| q68=347.42 q95=404.96 q99=409.15 tail |err|>400: 1 (11.111%)
tar_phase      N=      9 NaN=    0
  mean=-16.643 med=-6.162 std=81.134 s68=83.813
  |err| q68=96.499 q95=135.79 q99=144.28 tail |err|>90: 3 (33.333%)
tar_center     N=      9 NaN=    0
  mean=6.8342 med=6.7013 std=3.7 s68=3.1346
  |err| q68=8.497 q95=12.148 q99=13.757 tail |err|>40: 0 (0.000%)
tar_radius     N=      9 NaN=    0
  mean=-1.5722 med=-0.77822 std=4.4867 s68=3.7047
  |err| q68=4.1905 q95=8.8338 q99=10.369 tail |err|>20: 0 (0.000%)
tar_dp         N=      9 NaN=    0
  mean=-0.30444 med=0.021385 std=1.0966 s68=0.61127
  |err| q68=0.71949 q95=2.2523 q99=2.7976 tail |err|>2: 1 (11.111%)
tar_dpz        N=      9 NaN=    0
  mean=-0.85573 med=0.26652 std=3.206 s68=3.412
  |err| q68=2.4667 q95=6.3688 q99=6.6277 tail |err|>2: 4 (44.444%)
tar_pitch      N=      9 NaN=    0
  mean=18.357 med=-5.3977 std=59.776 s68=60.658
  |err| q68=48.246 q95=120.72 q99=123.51 tail |err|>20: 6 (66.667%)
tar_dt         N=      9 NaN=    0
  mean=0.78196 med=0.22704 std=2.5676 s68=2.7845
```

# C: target production plane / baseline_reflected

```
  |err| q68=3.1271 q95=4.4754 q99=4.755 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=      9 NaN=    0
  mean=0.056979 med=-0.19268 std=2.8122 s68=3.1493
  |err| q68=3.5437 q95=4.2583 q99=4.4328 tail |err|>10: 0 (0.000%)
```

# C: target production plane / control_direct

```
tar_dx         N=     12 NaN=    0
  mean=36.654 med=4.5396 std=199.83 s68=142.36
  |err| q68=120.39 q95=395.79 q99=412.28 tail |err|>400: 1 (8.333%)
tar_dy         N=     12 NaN=    0
  mean=-13.226 med=-13.373 std=112.09 s68=59.989
  |err| q68=65.662 q95=239.54 q99=293.89 tail |err|>400: 0 (0.000%)
tar_dr         N=     12 NaN=    0
  mean=179.06 med=103.25 std=148.16 s68=170.8
  |err| q68=229.82 q95=405.85 q99=414.51 tail |err|>400: 1 (8.333%)
tar_phase      N=     12 NaN=    0
  mean=-18.897 med=-9.448 std=71.021 s68=70.115
  |err| q68=69.165 q95=135.42 q99=145.4 tail |err|>90: 4 (33.333%)
tar_center     N=     12 NaN=    0
  mean=7.6407 med=6.2441 std=5.7597 s68=3.0006
  |err| q68=7.5219 q95=16.825 q99=23.197 tail |err|>40: 0 (0.000%)
tar_radius     N=     12 NaN=    0
  mean=-3.2143 med=-1.0664 std=5.7616 s68=2.7397
  |err| q68=3.4762 q95=12.724 q99=18.96 tail |err|>20: 1 (8.333%)
tar_dp         N=     12 NaN=    0
  mean=-0.53773 med=-0.17019 std=1.2146 s68=0.44305
  |err| q68=0.40049 q95=2.5682 q99=4.0028 tail |err|>2: 1 (8.333%)
tar_dpz        N=     12 NaN=    0
  mean=1.4641 med=0.54793 std=2.948 s68=1.8895
  |err| q68=1.1542 q95=7.3084 q99=8.9632 tail |err|>2: 3 (25.000%)
tar_pitch      N=     12 NaN=    0
  mean=-30.399 med=-14.122 std=56.399 s68=36.081
  |err| q68=23.148 q95=142.41 q99=173.29 tail |err|>20: 7 (58.333%)
tar_dt         N=     12 NaN=    0
  mean=0.87448 med=0.64988 std=1.5312 s68=1.2869
```

# C: target production plane / control_direct

```
  |err| q68=1.126 q95=3.7871 q99=3.9421 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     12 NaN=    0
  mean=0.32025 med=-0.048074 std=1.5369 s68=1.2076
  |err| q68=1.2056 q95=3.4858 q99=3.5071 tail |err|>10: 0 (0.000%)
```

# C: target production plane / control_reflected

```
tar_dx         N=      9 NaN=    0
  mean=-11.627 med=-23.696 std=196.03 s68=165.15
  |err| q68=202.71 q95=354.48 q99=390.9 tail |err|>400: 1 (11.111%)
tar_dy         N=      9 NaN=    0
  mean=82.291 med=-10.13 std=174.55 s68=175.66
  |err| q68=190.88 q95=345.78 q99=381.39 tail |err|>400: 0 (0.000%)
tar_dr         N=      9 NaN=    0
  mean=219.04 med=311.01 std=166.81 s68=183.58
  |err| q68=348.48 q95=404.99 q99=409 tail |err|>400: 1 (11.111%)
tar_phase      N=      9 NaN=    0
  mean=-15.783 med=-6.1591 std=82.243 s68=86.831
  |err| q68=97.333 q95=135.92 q99=144.39 tail |err|>90: 3 (33.333%)
tar_center     N=      9 NaN=    0
  mean=7.0345 med=6.7048 std=3.4114 s68=2.1577
  |err| q68=8.452 q95=12.146 q99=13.756 tail |err|>40: 0 (0.000%)
tar_radius     N=      9 NaN=    0
  mean=-1.1691 med=-0.77311 std=4.2542 s68=2.8506
  |err| q68=3.5722 q95=8.3559 q99=10.28 tail |err|>20: 0 (0.000%)
tar_dp         N=      9 NaN=    0
  mean=-0.21001 med=0.04716 std=1.196 s68=0.96426
  |err| q68=1.1402 q95=2.2772 q99=2.8011 tail |err|>2: 1 (11.111%)
tar_dpz        N=      9 NaN=    0
  mean=-0.89682 med=0.26668 std=3.2863 s68=3.572
  |err| q68=2.4651 q95=6.5391 q99=6.6621 tail |err|>2: 4 (44.444%)
tar_pitch      N=      9 NaN=    0
  mean=18.632 med=-5.7521 std=60.149 s68=61.374
  |err| q68=48.274 q95=121.61 q99=123.69 tail |err|>20: 6 (66.667%)
tar_dt         N=      9 NaN=    0
  mean=0.77663 med=0.22698 std=2.5807 s68=2.8055
```

# C: target production plane / control_reflected

```
  |err| q68=3.126 q95=4.4918 q99=4.7813 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=      9 NaN=    0
  mean=0.051648 med=-0.2018 std=2.8245 s68=3.1695
  |err| q68=3.5748 q95=4.2706 q99=4.4363 tail |err|>10: 0 (0.000%)
```

# C: target production plane / momentum_direct

```
tar_dx         N=     12 NaN=    0
  mean=-30.498 med=-4.9971 std=198.65 s68=135.45
  |err| q68=120.13 q95=410.39 q99=438.98 tail |err|>400: 1 (8.333%)
tar_dy         N=     12 NaN=    0
  mean=-45.211 med=-36.86 std=98.503 s68=60.1
  |err| q68=80.679 q95=212.46 q99=244.15 tail |err|>400: 0 (0.000%)
tar_dr         N=     12 NaN=    0
  mean=160.3 med=70.348 std=162.62 s68=188.53
  |err| q68=174.2 q95=432.99 q99=453.3 tail |err|>400: 3 (25.000%)
tar_phase      N=     12 NaN=    0
  mean=37.397 med=11.27 std=71.819 s68=77.624
  |err| q68=45.911 q95=163.28 q99=164.36 tail |err|>90: 3 (25.000%)
tar_center     N=     12 NaN=    0
  mean=9.8576 med=8.8249 std=6.3717 s68=7.2808
  |err| q68=11.789 q95=19.618 q99=20.502 tail |err|>40: 0 (0.000%)
tar_radius     N=     12 NaN=    0
  mean=-3.2381 med=-2.2153 std=5.6835 s68=2.933
  |err| q68=3.5203 q95=12.491 q99=18.772 tail |err|>20: 1 (8.333%)
tar_dp         N=     12 NaN=    0
  mean=-0.56488 med=-0.14359 std=1.1179 s68=0.33569
  |err| q68=0.3798 q95=2.41 q99=3.7903 tail |err|>2: 1 (8.333%)
tar_dpz        N=     12 NaN=    0
  mean=1.2511 med=0.6617 std=2.1905 s68=1.9248
  |err| q68=1.8837 q95=5.2083 q99=5.9882 tail |err|>2: 4 (33.333%)
tar_pitch      N=     12 NaN=    0
  mean=-26.508 med=-20.591 std=42.724 s68=38.227
  |err| q68=41.048 q95=100.47 q99=116.19 tail |err|>20: 7 (58.333%)
tar_dt         N=     12 NaN=    0
  mean=0.82493 med=0.56406 std=1.1551 s68=1.2709
```

# C: target production plane / momentum_direct

```
  |err| q68=0.95855 q95=2.6666 q99=2.7418 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     12 NaN=    0
  mean=0.27069 med=-0.020289 std=1.0833 s68=0.98065
  |err| q68=0.93079 q95=2.3472 q99=2.4483 tail |err|>10: 0 (0.000%)
```

# C: target production plane / momentum_reflected

```
tar_dx         N=     10 NaN=    0
  mean=-32.489 med=-106.28 std=185.95 s68=162.57
  |err| q68=198.47 q95=315.5 q99=378.52 tail |err|>400: 0 (0.000%)
tar_dy         N=     10 NaN=    0
  mean=80.511 med=38.978 std=179.55 s68=197.18
  |err| q68=173.46 q95=363.14 q99=386.44 tail |err|>400: 0 (0.000%)
tar_dr         N=     10 NaN=    0
  mean=233.67 med=228.58 std=140.54 s68=147.67
  |err| q68=309.29 q95=428.81 q99=445.26 tail |err|>400: 2 (20.000%)
tar_phase      N=     10 NaN=    0
  mean=25.721 med=45.653 std=79.476 s68=62.06
  |err| q68=87.184 q95=142.43 q99=159.45 tail |err|>90: 3 (30.000%)
tar_center     N=     10 NaN=    0
  mean=12.169 med=11.326 std=6.2921 s68=6.1169
  |err| q68=13.011 q95=23.208 q99=24.677 tail |err|>40: 0 (0.000%)
tar_radius     N=     10 NaN=    0
  mean=-2.5431 med=-1.1565 std=5.199 s68=2.7617
  |err| q68=4.3615 q95=11.078 q99=14.182 tail |err|>20: 0 (0.000%)
tar_dp         N=     10 NaN=    0
  mean=-0.34394 med=-0.15016 std=1.2323 s68=0.65074
  |err| q68=0.79057 q95=2.6154 q99=3.0756 tail |err|>2: 1 (10.000%)
tar_dpz        N=     10 NaN=    0
  mean=-1.6107 med=-0.67954 std=2.3555 s68=2.2915
  |err| q68=2.301 q95=5.6807 q99=6.289 tail |err|>2: 4 (40.000%)
tar_pitch      N=     10 NaN=    0
  mean=33.26 med=18.151 std=43.433 s68=44.24
  |err| q68=36.43 q95=108.51 q99=119.42 tail |err|>20: 5 (50.000%)
tar_dt         N=     10 NaN=    0
  mean=0.38031 med=0.76051 std=2.5603 s68=2.0094
```

# C: target production plane / momentum_reflected

```
  |err| q68=2.6085 q95=4.8927 q99=5.3177 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     10 NaN=    0
  mean=-0.15823 med=-0.02456 std=2.4375 s68=2.1157
  |err| q68=2.2904 q95=4.4645 q99=4.517 tail |err|>10: 0 (0.000%)
```

# C: target production plane / position_direct

```
tar_dx         N=     12 NaN=    0
  mean=42.568 med=6.9928 std=197.65 s68=142.23
  |err| q68=118.83 q95=390.83 q99=410.74 tail |err|>400: 1 (8.333%)
tar_dy         N=     12 NaN=    0
  mean=-12.182 med=-12.514 std=110.62 s68=68.256
  |err| q68=68.579 q95=223.12 q99=288.66 tail |err|>400: 0 (0.000%)
tar_dr         N=     12 NaN=    0
  mean=180.02 med=107.72 std=144.42 s68=164.45
  |err| q68=231.68 q95=408.3 q99=414.45 tail |err|>400: 2 (16.667%)
tar_phase      N=     12 NaN=    0
  mean=-20.562 med=-9.6788 std=70.067 s68=68.613
  |err| q68=67.66 q95=139.94 q99=146.17 tail |err|>90: 4 (33.333%)
tar_center     N=     12 NaN=    0
  mean=10.347 med=6.6276 std=8.8412 s68=6.4966
  |err| q68=8.7172 q95=26.222 q99=33.616 tail |err|>40: 0 (0.000%)
tar_radius     N=     12 NaN=    0
  mean=-3.2537 med=-1.0457 std=5.6765 s68=2.7499
  |err| q68=3.3644 q95=12.624 q99=18.764 tail |err|>20: 1 (8.333%)
tar_dp         N=     12 NaN=    0
  mean=-0.55377 med=-0.21741 std=1.2034 s68=0.44551
  |err| q68=0.39372 q95=2.5608 q99=3.9859 tail |err|>2: 1 (8.333%)
tar_dpz        N=     12 NaN=    0
  mean=1.4113 med=0.39987 std=2.9941 s68=1.7353
  |err| q68=1.1594 q95=7.4013 q99=9.1364 tail |err|>2: 2 (16.667%)
tar_pitch      N=     12 NaN=    0
  mean=-29.459 med=-11.991 std=57.233 s68=32.97
  |err| q68=22.262 q95=144.25 q99=176.73 tail |err|>20: 5 (41.667%)
tar_dt         N=     12 NaN=    0
  mean=0.84979 med=0.64792 std=1.5568 s68=1.2575
```

# C: target production plane / position_direct

```
  |err| q68=1.183 q95=3.8252 q99=4.0157 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     12 NaN=    0
  mean=0.29556 med=-0.033856 std=1.5719 s68=1.2279
  |err| q68=1.2575 q95=3.5288 q99=3.543 tail |err|>10: 0 (0.000%)
```

# C: target production plane / position_reflected

```
tar_dx         N=     10 NaN=    0
  mean=-44.496 med=-26.46 std=186.55 s68=132.79
  |err| q68=166.15 q95=352.58 q99=386.94 tail |err|>400: 0 (0.000%)
tar_dy         N=     10 NaN=    0
  mean=56.895 med=-12.221 std=142.96 s68=151.72
  |err| q68=118.57 q95=287.52 q99=313.48 tail |err|>400: 0 (0.000%)
tar_dr         N=     10 NaN=    0
  mean=185.31 med=113 std=161.61 s68=173.38
  |err| q68=355.54 q95=394.36 q99=403.56 tail |err|>400: 1 (10.000%)
tar_phase      N=     10 NaN=    0
  mean=-4.291 med=-5.1182 std=71.501 s68=74.373
  |err| q68=101.64 q95=114.44 q99=115.32 tail |err|>90: 4 (40.000%)
tar_center     N=     10 NaN=    0
  mean=7.8571 med=6.5657 std=4.9336 s68=4.1145
  |err| q68=7.9309 q95=16.831 q99=19.001 tail |err|>40: 0 (0.000%)
tar_radius     N=     10 NaN=    0
  mean=-2.1703 med=-2.0907 std=3.7379 s68=3.7297
  |err| q68=4.2587 q95=7.6594 q99=8.6208 tail |err|>20: 0 (0.000%)
tar_dp         N=     10 NaN=    0
  mean=-0.47896 med=-0.43884 std=0.87629 s68=0.7415
  |err| q68=1.2282 q95=1.7064 q99=1.9766 tail |err|>2: 1 (10.000%)
tar_dpz        N=     10 NaN=    0
  mean=-0.35941 med=0.37044 std=3.0902 s68=3.0134
  |err| q68=2.5022 q95=5.9311 q99=6.55 tail |err|>2: 5 (50.000%)
tar_pitch      N=     10 NaN=    0
  mean=9.0533 med=-7.6532 std=58.282 s68=57.956
  |err| q68=45.761 q95=114.31 q99=122.43 tail |err|>20: 7 (70.000%)
tar_dt         N=     10 NaN=    0
  mean=0.91461 med=0.35013 std=2.5911 s68=2.8561
```

# C: target production plane / position_reflected

```
  |err| q68=3.1752 q95=4.5608 q99=4.8063 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     10 NaN=    0
  mean=0.23619 med=-0.26391 std=2.8361 s68=3.1591
  |err| q68=3.802 q95=4.2569 q99=4.4422 tail |err|>10: 0 (0.000%)
```

# C: target production plane / truth_direct

```
tar_dx         N=     12 NaN=    0
  mean=-28.094 med=-2.8818 std=199.21 s68=132.19
  |err| q68=125.71 q95=408.25 q99=445.82 tail |err|>400: 1 (8.333%)
tar_dy         N=     12 NaN=    0
  mean=-43.441 med=-33.153 std=103.89 s68=53.705
  |err| q68=65.324 q95=224.48 q99=246.48 tail |err|>400: 0 (0.000%)
tar_dr         N=     12 NaN=    0
  mean=160 med=62.361 std=166 s68=191.2
  |err| q68=179.24 q95=438.99 q99=456.2 tail |err|>400: 3 (25.000%)
tar_phase      N=     12 NaN=    0
  mean=36.465 med=10.593 std=73.354 s68=77.372
  |err| q68=49.986 q95=163.88 q99=171.42 tail |err|>90: 3 (25.000%)
tar_center     N=     12 NaN=    0
  mean=7.3325 med=5.4825 std=6.4273 s68=2.704
  |err| q68=7.0077 q95=18.657 q99=25.185 tail |err|>40: 0 (0.000%)
tar_radius     N=     12 NaN=    0
  mean=-3.3253 med=-1.9294 std=5.9146 s68=2.5946
  |err| q68=3.4552 q95=12.924 q99=19.76 tail |err|>20: 1 (8.333%)
tar_dp         N=     12 NaN=    0
  mean=-0.58737 med=-0.14553 std=1.1889 s68=0.35715
  |err| q68=0.40569 q95=2.5308 q99=4.0285 tail |err|>2: 1 (8.333%)
tar_dpz        N=     12 NaN=    0
  mean=1.2207 med=0.57552 std=2.2143 s68=1.8368
  |err| q68=1.8239 q95=5.1559 q99=6.3196 tail |err|>2: 4 (33.333%)
tar_pitch      N=     12 NaN=    0
  mean=-25.973 med=-17.878 std=42.978 s68=36.336
  |err| q68=40.418 q95=99.346 q99=122.33 tail |err|>20: 7 (58.333%)
tar_dt         N=     12 NaN=    0
  mean=0.63227 med=0.1927 std=0.91978 s68=0.9961
```

# C: target production plane / truth_direct

```
  |err| q68=0.53134 q95=2.2431 q99=2.2884 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     12 NaN=    0
  mean=0.078032 med=0.054759 std=0.9631 s68=0.97406
  |err| q68=0.89604 q95=1.8378 q99=1.8915 tail |err|>10: 0 (0.000%)
```

# C: target production plane / truth_reflected

```
tar_dx         N=     11 NaN=    0
  mean=-37.937 med=-42.178 std=202.82 s68=168.21
  |err| q68=214.84 q95=372.13 q99=385.64 tail |err|>400: 0 (0.000%)
tar_dy         N=     11 NaN=    0
  mean=76.317 med=20.198 std=169.29 s68=157.39
  |err| q68=174.9 q95=354.8 q99=401.09 tail |err|>400: 1 (9.091%)
tar_dr         N=     11 NaN=    0
  mean=230.48 med=224.35 std=154.72 s68=186.47
  |err| q68=279.27 q95=464.15 q99=465.07 tail |err|>400: 2 (18.182%)
tar_phase      N=     11 NaN=    0
  mean=27.727 med=40.91 std=83.425 s68=60.668
  |err| q68=82.68 q95=160.43 q99=167.24 tail |err|>90: 3 (27.273%)
tar_center     N=     11 NaN=    0
  mean=6.4279 med=5.7615 std=3.1926 s68=2.7711
  |err| q68=6.6231 q95=12.066 q99=13.551 tail |err|>40: 0 (0.000%)
tar_radius     N=     11 NaN=    0
  mean=-2.0677 med=-0.94024 std=4.4774 s68=2.7414
  |err| q68=3.8065 q95=9.5021 q99=12.765 tail |err|>20: 0 (0.000%)
tar_dp         N=     11 NaN=    0
  mean=-0.3784 med=0.00061142 std=0.91028 s68=0.62125
  |err| q68=0.42886 q95=1.9901 q99=2.6917 tail |err|>2: 1 (9.091%)
tar_dpz        N=     11 NaN=    0
  mean=-1.1858 med=-0.22851 std=2.5736 s68=2.3818
  |err| q68=2.3786 q95=5.5709 q99=6.2788 tail |err|>2: 4 (36.364%)
tar_pitch      N=     11 NaN=    0
  mean=24.739 med=14.882 std=49.358 s68=45.137
  |err| q68=48.77 q95=107.11 q99=119.35 tail |err|>20: 6 (54.545%)
tar_dt         N=     11 NaN=    0
  mean=0.26084 med=0.4377 std=2.5373 s68=2.3657
```

# C: target production plane / truth_reflected

```
  |err| q68=2.6408 q95=4.5918 q99=4.6956 tail |err|>10: 0 (0.000%)
tar_dt_beam    N=     11 NaN=    0
  mean=-0.18953 med=-0.065143 std=2.582 s68=2.6842
  |err| q68=3.1363 q95=4.1878 q99=4.4759 tail |err|>10: 0 (0.000%)
```

# C: phase statistics, wrapping-aware

```
baseline_direct        N=11
  circular mean=-5.575 deg, R=0.40608, circular std=76.922 deg
baseline_reflected     N=9
  circular mean=-9.455 deg, R=0.29328, circular std=89.742 deg
control_direct         N=12
  circular mean=-8.7021 deg, R=0.45375, circular std=72.029 deg
control_reflected      N=9
  circular mean=-9.1996 deg, R=0.27655, circular std=91.865 deg
momentum_direct        N=12
  circular mean=11.347 deg, R=0.46616, circular std=70.788 deg
momentum_reflected     N=10
  circular mean=54.924 deg, R=0.50115, circular std=67.349 deg
position_direct        N=12
  circular mean=-8.5469 deg, R=0.46138, circular std=71.265 deg
position_reflected     N=10
  circular mean=-6.2407 deg, R=0.4285, circular std=74.593 deg
truth_direct           N=12
  circular mean=9.2258 deg, R=0.45641, circular std=71.763 deg
truth_reflected        N=11
  circular mean=51.079 deg, R=0.43773, circular std=73.649 deg
```

# C: common-cohort configuration comparison

```
Same tracks valid in every enabled configuration; N=19
This sensitivity cohort supplements, and never replaces, failure-inclusive efficiencies.
baseline_direct
  tar_dr N=10 med=129.6 s68=166.96 |err|q95=407.82
  tar_phase N=10 med=-1.2065 s68=74.985 |err|q95=137.68
  tar_dt N=10 med=0.73741 s68=1.3165 |err|q95=3.8223
baseline_reflected
  tar_dr N=9 med=279.75 s68=183.43 |err|q95=404.96
  tar_phase N=9 med=-6.162 s68=83.813 |err|q95=135.79
  tar_dt N=9 med=0.22704 s68=2.7845 |err|q95=4.4754
control_direct
  tar_dr N=10 med=129.62 s68=167.29 |err|q95=407.82
  tar_phase N=10 med=-1.0451 s68=74.985 |err|q95=137.69
  tar_dt N=10 med=0.7376 s68=1.3157 |err|q95=3.8223
control_reflected
  tar_dr N=9 med=311.01 s68=183.58 |err|q95=404.99
  tar_phase N=9 med=-6.1591 s68=86.831 |err|q95=135.92
  tar_dt N=9 med=0.22698 s68=2.8055 |err|q95=4.4918
momentum_direct
  tar_dr N=10 med=97.58 s68=188.89 |err|q95=437.61
  tar_phase N=10 med=11.27 s68=86.401 |err|q95=163.52
  tar_dt N=10 med=0.59225 s68=1.1922 |err|q95=2.6837
momentum_reflected
  tar_dr N=9 med=262.81 s68=144.77 |err|q95=431.09
  tar_phase N=9 med=57.495 s68=58.523 |err|q95=144.79
  tar_dt N=9 med=0.81037 s68=1.3441 |err|q95=4.4545
position_direct
  tar_dr N=10 med=132.2 s68=165.16 |err|q95=409.69
  tar_phase N=10 med=-1.2117 s68=76.471 |err|q95=141.36
```

# C: common-cohort configuration comparison

```
  tar_dt N=10 med=0.72808 s68=1.2881 |err|q95=3.8685
position_reflected
  tar_dr N=9 med=78.575 s68=171.42 |err|q95=395.64
  tar_phase N=9 med=-6.1403 s68=55.636 |err|q95=114.56
  tar_dt N=9 med=0.2349 s68=3.0839 |err|q95=4.5949
truth_direct
  tar_dr N=10 med=94.962 s68=190.3 |err|q95=442.9
  tar_phase N=10 med=10.593 s68=85.157 |err|q95=165.6
  tar_dt N=10 med=0.1927 s68=1.0252 |err|q95=2.2534
truth_reflected
  tar_dr N=9 med=277.79 s68=148.49 |err|q95=464.38
  tar_phase N=9 med=60.573 s68=65.777 |err|q95=162.13
  tar_dt N=9 med=0.4377 s68=1.7044 |err|q95=4.0985
```

# B: event crossing efficiencies

```
baseline: read=20, common denominator=20
  any_plane                  14 /       20  fraction=0.7
  any_annulus                 5 /       20  fraction=0.25
  both_plane                  8 /       20  fraction=0.4
  both_annulus                1 /       20  fraction=0.05
  both_compatible             0 /       20  fraction=0
control: read=20, common denominator=20
  any_plane                  13 /       20  fraction=0.65
  any_annulus                 4 /       20  fraction=0.2
  both_plane                  8 /       20  fraction=0.4
  both_annulus                1 /       20  fraction=0.05
  both_compatible             0 /       20  fraction=0
momentum: read=20, common denominator=20
  any_plane                  14 /       20  fraction=0.7
  any_annulus                 6 /       20  fraction=0.3
  both_plane                  8 /       20  fraction=0.4
  both_annulus                2 /       20  fraction=0.1
  both_compatible             0 /       20  fraction=0
position: read=20, common denominator=20
  any_plane                  13 /       20  fraction=0.65
  any_annulus                 6 /       20  fraction=0.3
  both_plane                  9 /       20  fraction=0.45
  both_annulus                2 /       20  fraction=0.1
  both_compatible             0 /       20  fraction=0
truth: read=20, common denominator=20
  any_plane                  15 /       20  fraction=0.75
  any_annulus                 7 /       20  fraction=0.35
  both_plane                  8 /       20  fraction=0.4
  both_annulus                2 /       20  fraction=0.1
```

# B: event crossing efficiencies

```
both_compatible             0 /       20  fraction=0
```

# E: event availability and failure flags, including zero-track events

```
baseline
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
control
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
momentum
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
position
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
truth
  missing_truth_events            0
  missing_input_events            0
  not_two_primary_events          0
```

# E: status, missing states and covariance failures

```
baseline_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        5
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  3
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      14
  branch_missing           3
  covariance_invalid       5
  ent_valid                14
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
```

# E: status, missing states and covariance failures

```
  propagated_covariance_invalid 5
  propagation_complete     14
  selected                 14
  status_0                 6
  status_16384             3
  status_4096              5
  tar_valid                11
baseline_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        1
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  6
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      1
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
```

# E: status, missing states and covariance failures

```
  all                       15
  branch_missing            5
  covariance_invalid        1
  ent_valid                 14
  input_covariance_invalid  0
  partial_target_available  0
  product_missing           0
  propagated_covariance_invalid 1
  propagation_complete      14
  selected                  14
  status_0                  8
  status_16384              5
  status_16640              1
  status_4096               1
  tar_valid                 9
control_direct
  AmbiguousAssociation      0
  AmbiguousProductionBranch 0
  BoundaryMismatch          0
  ConfigurationDisabled     0
  CrossingSearchFailure     0
  Exception                 0
  InvalidCovariance         5
  MissingAssociation        0
  MissingClock              0
  MissingInput              0
  MissingMCEntrance         0
  MissingProductionBranch   1
  MissingRecoEntrance       0
```

# E: status, missing states and covariance failures

```
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       1
  UnsupportedTrajectory    0
  all                      14
  branch_missing           1
  covariance_invalid       5
  ent_valid                14
  input_covariance_invalid 0
  partial_target_available 1
  product_missing          0
  propagated_covariance_invalid 5
  propagation_complete     13
  selected                 14
  status_0                 8
  status_20480             1
  status_4096              4
  status_8192              1
  tar_valid                12
control_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
```

# E: status, missing states and covariance failures

```
Exception                3
InvalidCovariance        3
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  5
MissingRecoEntrance      0
MissingTruth             0
MultipleMCEntrances      1
MultipleRecoEntrances    0
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       0
UnsupportedTrajectory    0
all                      15
branch_missing           5
covariance_invalid       3
ent_valid                14
input_covariance_invalid 0
partial_target_available 0
product_missing          0
propagated_covariance_invalid 3
propagation_complete     14
selected                 14
status_0                 7
status_16384             4
status_20480             1
```

# E: status, missing states and covariance failures

```
  status_256               1
  status_4096              2
  tar_valid                9
momentum_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        6
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  1
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       1
  UnsupportedTrajectory    0
  all                      14
  branch_missing           1
  covariance_invalid       6
  ent_valid                14
```

# E: status, missing states and covariance failures

```
  input_covariance_invalid 0
  partial_target_available 1
  product_missing          0
  propagated_covariance_invalid 6
  propagation_complete     13
  selected                 14
  status_0                 6
  status_16384             1
  status_4096              6
  status_8192              1
  tar_valid                12
momentum_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        5
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  4
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      1
  MultipleRecoEntrances    0
  NotElectron              0
```

# E: status, missing states and covariance failures

```
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       1
  UnsupportedTrajectory    0
  all                      15
  branch_missing           4
  covariance_invalid       5
  ent_valid                14
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
  propagated_covariance_invalid 5
  propagation_complete     13
  selected                 14
  status_0                 7
  status_16384             2
  status_20480             1
  status_256               1
  status_28672             1
  status_4096              3
  tar_valid                10
position_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        8
```

# E: status, missing states and covariance failures

```
MissingAssociation       0
MissingClock             0
MissingInput             0
MissingMCEntrance        0
MissingProductionBranch  1
MissingRecoEntrance      0
MissingTruth             0
MultipleMCEntrances      0
MultipleRecoEntrances    0
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       1
UnsupportedTrajectory    0
all                      14
branch_missing           1
covariance_invalid       8
ent_valid                14
input_covariance_invalid 0
partial_target_available 1
product_missing          0
propagated_covariance_invalid 8
propagation_complete     13
selected                 14
status_0                 5
status_20480             1
status_4096              7
status_8192              1
tar_valid                12
```

# E: status, missing states and covariance failures

```
position_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        7
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  4
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      1
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
  PropagationFailure       0
  UnsupportedTrajectory    0
  all                      15
  branch_missing           4
  covariance_invalid       7
  ent_valid                14
  input_covariance_invalid 0
  partial_target_available 0
  product_missing          0
```

# E: status, missing states and covariance failures

```
  propagated_covariance_invalid 7
  propagation_complete     14
  selected                 14
  status_0                 4
  status_16384             3
  status_20480             1
  status_256               1
  status_4096              6
  tar_valid                10
truth_direct
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        4
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
  MissingProductionBranch  1
  MissingRecoEntrance      0
  MissingTruth             0
  MultipleMCEntrances      0
  MultipleRecoEntrances    0
  NotElectron              0
  NotPrimary               0
  OutsideTarget            0
```

# E: status, missing states and covariance failures

```
  PropagationFailure       1
  UnsupportedTrajectory    0
  all                      14
  branch_missing           1
  covariance_invalid       4
  ent_valid                14
  input_covariance_invalid 0
  partial_target_available 1
  product_missing          0
  propagated_covariance_invalid 4
  propagation_complete     13
  selected                 14
  status_0                 8
  status_16384             1
  status_4096              4
  status_8192              1
  tar_valid                12
truth_reflected
  AmbiguousAssociation     0
  AmbiguousProductionBranch 0
  BoundaryMismatch         0
  ConfigurationDisabled    0
  CrossingSearchFailure    0
  Exception                0
  InvalidCovariance        5
  MissingAssociation       0
  MissingClock             0
  MissingInput             0
  MissingMCEntrance        0
```

# E: status, missing states and covariance failures

```
MissingProductionBranch  3
MissingRecoEntrance      0
MissingTruth             0
MultipleMCEntrances      1
MultipleRecoEntrances    0
NotElectron              0
NotPrimary               0
OutsideTarget            0
PropagationFailure       0
UnsupportedTrajectory    0
all                      15
branch_missing           3
covariance_invalid       5
ent_valid                14
input_covariance_invalid 0
partial_target_available 0
product_missing          0
propagated_covariance_invalid 5
propagation_complete     14
selected                 14
status_0                 6
status_16384             3
status_256               1
status_4096              5
tar_valid                11
```

# D: two-electron timing and vertex / baseline

```
pair rows=20 denominator=10 valid=7
time_difference: N=7, NaN=0
  mean=0.3499 med=0.41987 std=3.3849 s68=3.5588
  |err| q68=3.7248 q95=4.562; tail >10: 0.000%
time_difference_error: N=7, NaN=0
  mean=0.3499 med=0.41987 std=3.3849 s68=3.5588
  |err| q68=3.7248 q95=4.562; tail >10: 0.000%
separation: N=7, NaN=0
  mean=361.43 med=392.96 std=223.63 s68=153.1
  |err| q68=408.39 q95=693.78; tail >400: 42.857%
opening_error_deg: N=7, NaN=0
  mean=66.116 med=44.264 std=51.112 s68=47.785
  |err| q68=90.088 q95=144.05; tail >30: 71.429%
max_vertex_dr: N=7, NaN=0
  mean=288.79 med=310.03 std=127.1 s68=136.58
  |err| q68=396.21 q95=414.72; tail >400: 28.571%
max_vertex_dt: N=7, NaN=0
  mean=3.1126 med=3.1712 std=1.2185 s68=0.80394
  |err| q68=3.6545 q95=4.5627; tail >10: 0.000%
```

# D: two-electron timing and vertex / control

```
pair rows=20 denominator=10 valid=8
time_difference: N=8, NaN=0
  mean=0.28631 med=0.10769 std=3.184 s68=3.4556
  |err| q68=3.6275 q95=4.5293; tail >10: 0.000%
time_difference_error: N=8, NaN=0
  mean=0.28631 med=0.10769 std=3.184 s68=3.4556
  |err| q68=3.6275 q95=4.5293; tail >10: 0.000%
separation: N=8, NaN=0
  mean=377.06 med=413.68 std=211.39 s68=136.06
  |err| q68=441.25 q95=677.98; tail >400: 62.500%
opening_error_deg: N=8, NaN=0
  mean=70.395 med=70.197 std=48.232 s68=44.836
  |err| q68=92.208 q95=141.7; tail >30: 62.500%
max_vertex_dr: N=8, NaN=0
  mean=306.49 med=353.47 std=123.93 s68=124.51
  |err| q68=396.89 q95=414.34; tail >400: 25.000%
max_vertex_dt: N=8, NaN=0
  mean=2.7511 med=3.1304 std=1.5089 s68=1.5292
  |err| q68=3.5177 q95=4.5371; tail >10: 0.000%
```

# D: two-electron timing and vertex / momentum

```
pair rows=20 denominator=10 valid=8
time_difference: N=8, NaN=0
  mean=0.16396 med=0.12191 std=2.8324 s68=1.8674
  |err| q68=2.0849 q95=5.2891; tail >10: 0.000%
time_difference_error: N=8, NaN=0
  mean=0.16396 med=0.12191 std=2.8324 s68=1.8674
  |err| q68=2.0849 q95=5.2891; tail >10: 0.000%
separation: N=8, NaN=0
  mean=364.95 med=341.48 std=269.69 s68=286.58
  |err| q68=442.93 q95=785.32; tail >400: 50.000%
opening_error_deg: N=8, NaN=0
  mean=68.699 med=57.533 std=39.816 s68=21.259
  |err| q68=80.701 q95=131.29; tail >30: 87.500%
max_vertex_dr: N=8, NaN=0
  mean=289.16 med=322.51 std=147.87 s68=166.29
  |err| q68=400.57 q95=455.22; tail >400: 37.500%
max_vertex_dt: N=8, NaN=0
  mean=2.1164 med=1.8606 std=1.5329 s68=1.1172
  |err| q68=2.5814 q95=4.5757; tail >10: 0.000%
```

# D: two-electron timing and vertex / position

```
pair rows=20 denominator=10 valid=9
time_difference: N=9, NaN=0
  mean=-0.12627 med=-0.38833 std=3.2723 s68=3.655
  |err| q68=3.739 q95=4.4121; tail >10: 0.000%
time_difference_error: N=9, NaN=0
  mean=-0.12627 med=-0.38833 std=3.2723 s68=3.655
  |err| q68=3.739 q95=4.4121; tail >10: 0.000%
separation: N=9, NaN=0
  mean=364.67 med=421.89 std=240.1 s68=244.45
  |err| q68=442.4 q95=736.32; tail >400: 55.556%
opening_error_deg: N=9, NaN=0
  mean=61.402 med=51.949 std=45.018 s68=49.066
  |err| q68=82.624 q95=131.52; tail >30: 66.667%
max_vertex_dr: N=9, NaN=0
  mean=270.66 med=353.17 std=137.98 s68=141.33
  |err| q68=389.86 q95=411.94; tail >400: 33.333%
max_vertex_dt: N=9, NaN=0
  mean=2.7563 med=3.1044 std=1.5018 s68=1.5628
  |err| q68=3.6586 q95=4.5949; tail >10: 0.000%
```

# D: two-electron timing and vertex / truth

```
pair rows=20 denominator=10 valid=8
time_difference: N=8, NaN=0
  mean=0.059654 med=0.35734 std=2.8529 s68=2.1659
  |err| q68=2.4292 q95=5.1044; tail >10: 0.000%
time_difference_error: N=8, NaN=0
  mean=0.059654 med=0.35734 std=2.8529 s68=2.1659
  |err| q68=2.4292 q95=5.1044; tail >10: 0.000%
separation: N=8, NaN=0
  mean=388.3 med=389.09 std=244.67 s68=237.79
  |err| q68=476.28 q95=757.1; tail >400: 50.000%
opening_error_deg: N=8, NaN=0
  mean=79.35 med=70.851 std=45.718 s68=42.804
  |err| q68=103.37 q95=144.98; tail >30: 87.500%
max_vertex_dr: N=8, NaN=0
  mean=319.3 med=343.27 std=146.88 s68=151.24
  |err| q68=447.64 q95=464.5; tail >400: 50.000%
max_vertex_dt: N=8, NaN=0
  mean=2.0133 med=1.9067 std=1.3641 s68=1.2695
  |err| q68=2.07 q95=4.1763; tail >10: 0.000%
```

# E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
baseline_direct:ent_dpz:tar_phase
  n=11, circular_linear_rho=0.13043
baseline_direct:ent_dr:tar_dr
  n=11, pearson_r=0.40112
baseline_direct:ent_dt:tar_dt
  n=11, pearson_r=0.59199
baseline_direct:ent_pitch:tar_phase
  n=11, circular_linear_rho=0.30643
baseline_direct:tar_phase:tar_dr
  n=11, circular_linear_rho=0.98745
baseline_reflected:ent_dpz:tar_phase
  n=9, circular_linear_rho=0.7224
baseline_reflected:ent_dr:tar_dr
  n=9, pearson_r=0.2264
baseline_reflected:ent_dt:tar_dt
  n=9, pearson_r=0.10073
baseline_reflected:ent_pitch:tar_phase
  n=9, circular_linear_rho=0.70139
baseline_reflected:tar_phase:tar_dr
  n=9, circular_linear_rho=0.96771
control_direct:ent_dpz:tar_phase
  n=12, circular_linear_rho=0.20846
control_direct:ent_dr:tar_dr
  n=12, pearson_r=-0.023111
control_direct:ent_dt:tar_dt
  n=12, pearson_r=0.66001
control_direct:ent_pitch:tar_phase
```

# E: entrance-to-target correlations

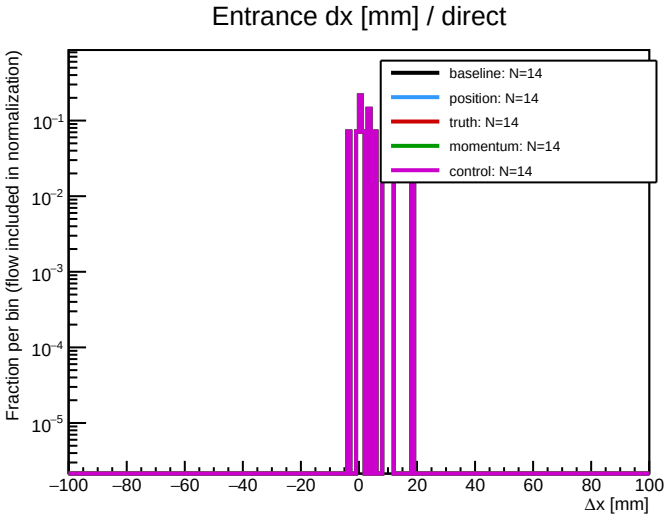
```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
  n=12, circular_linear_rho=0.20173
control_direct:tar_phase:tar_dr
  n=12, circular_linear_rho=0.98917
control_reflected:ent_dpz:tar_phase
  n=9, circular_linear_rho=0.72099
control_reflected:ent_dr:tar_dr
  n=9, pearson_r=0.23392
control_reflected:ent_dt:tar_dt
  n=9, pearson_r=0.1001
control_reflected:ent_pitch:tar_phase
  n=9, circular_linear_rho=0.70027
control_reflected:tar_phase:tar_dr
  n=9, circular_linear_rho=0.96894
momentum_direct:ent_dpz:tar_phase
  n=12, circular_linear_rho=0.24525
momentum_direct:ent_dr:tar_dr
  n=12, pearson_r=-0.098007
momentum_direct:ent_dt:tar_dt
  n=12, pearson_r=0.61709
momentum_direct:ent_pitch:tar_phase
  n=12, circular_linear_rho=0.26192
momentum_direct:tar_phase:tar_dr
  n=12, circular_linear_rho=0.99044
momentum_reflected:ent_dpz:tar_phase
  n=10, circular_linear_rho=0.17424
momentum_reflected:ent_dr:tar_dr
  n=10, pearson_r=0.44228
```

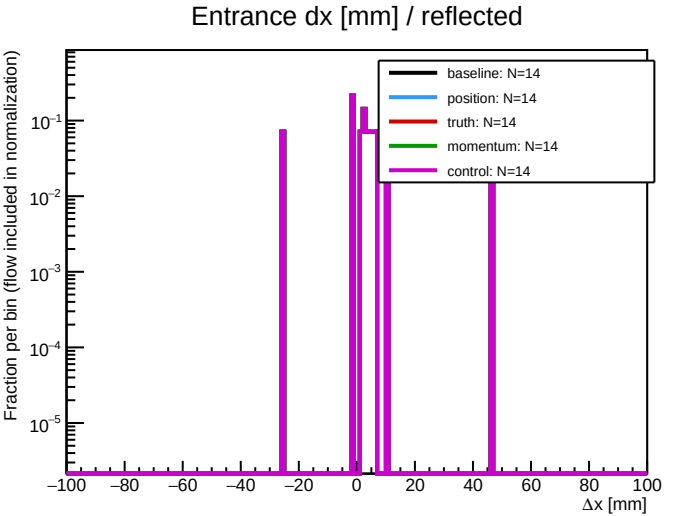
# E: entrance-to-target correlations

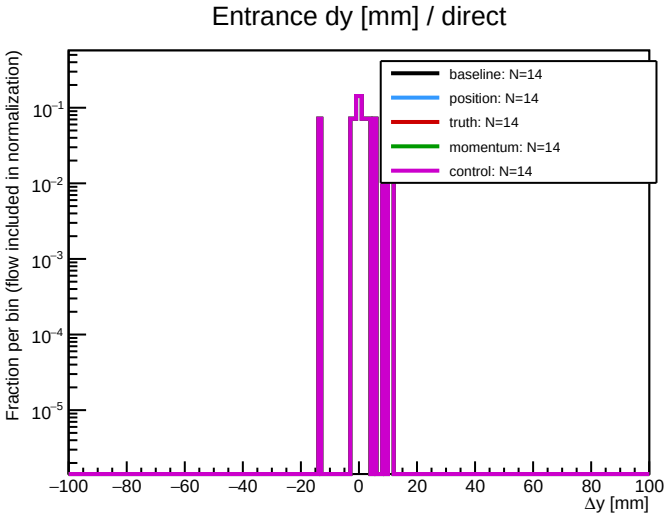
```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
momentum_reflected:ent_dt:tar_dt
  n=10, pearson_r=0.15816
momentum_reflected:ent_pitch:tar_phase
  n=10, circular_linear_rho=0.23012
momentum_reflected:tar_phase:tar_dr
  n=10, circular_linear_rho=0.98775
position_direct:ent_dpz:tar_phase
  n=12, circular_linear_rho=0.20384
position_direct:ent_dr:tar_dr
  n=12, pearson_r=0.0092014
position_direct:ent_dt:tar_dt
  n=12, pearson_r=0.66095
position_direct:ent_pitch:tar_phase
  n=12, circular_linear_rho=0.18295
position_direct:tar_phase:tar_dr
  n=12, circular_linear_rho=0.98697
position_reflected:ent_dpz:tar_phase
  n=10, circular_linear_rho=0.51975
position_reflected:ent_dr:tar_dr
  n=10, pearson_r=-0.096022
position_reflected:ent_dt:tar_dt
  n=10, pearson_r=0.058724
position_reflected:ent_pitch:tar_phase
  n=10, circular_linear_rho=0.5304
position_reflected:tar_phase:tar_dr
  n=10, circular_linear_rho=0.99041
truth_direct:ent_dpz:tar_phase
```

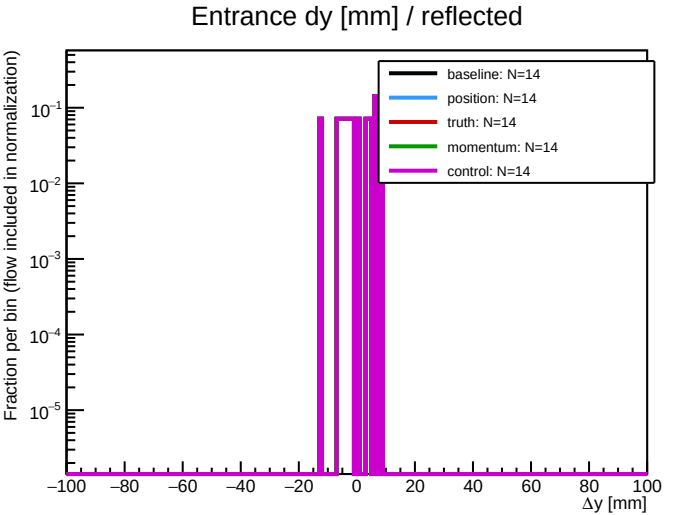
# E: entrance-to-target correlations

```
Circular rho is unsigned association with sin/cos phase; not a slope.
Correlations use pairwise finite values; no truncation of tails.
  n=12, circular_linear_rho=0.26154
truth_direct:ent_dr:tar_dr
  n=12, pearson_r=-0.10247
truth_direct:ent_dt:tar_dt
  n=12, pearson_r=0.14437
truth_direct:ent_pitch:tar_phase
  n=12, circular_linear_rho=0.27985
truth_direct:tar_phase:tar_dr
  n=12, circular_linear_rho=0.99046
truth_reflected:ent_dpz:tar_phase
  n=11, circular_linear_rho=0.4042
truth_reflected:ent_dr:tar_dr
  n=11, pearson_r=-0.31451
truth_reflected:ent_dt:tar_dt
  n=11, pearson_r=-0.29588
truth_reflected:ent_pitch:tar_phase
  n=11, circular_linear_rho=0.3984
truth_reflected:tar_phase:tar_dr
  n=11, circular_linear_rho=0.98463
```

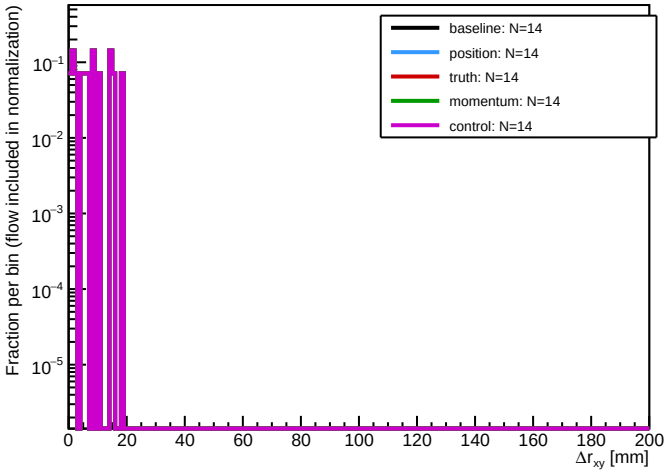

Entrance dx [mm] / direct
baseline: N=14
position: N=14
truth: N=14
momentum: N=14
control: N=14
Fraction per bin (flow included in normalization)
Δx [mm]


Entrance dx [mm] / reflected
baseline: N=14
position: N=14
truth: N=14
momentum: N=14
control: N=14
Fraction per bin (flow included in normalization)
Δx [mm]

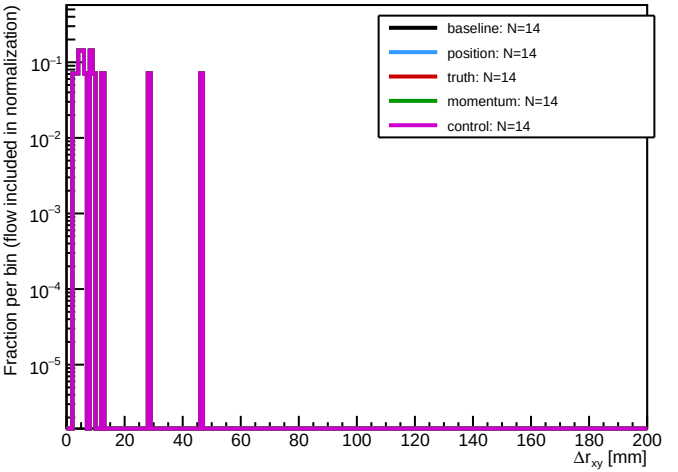

Entrance dy [mm] / direct
baseline: N=14
position: N=14
truth: N=14
momentum: N=14
control: N=14
Fraction per bin (flow included in normalization)
Δy [mm]


Entrance dy [mm] / reflected
baseline: N=14
position: N=14
truth: N=14
momentum: N=14
control: N=14
Fraction per bin (flow included in normalization)
Δy [mm]

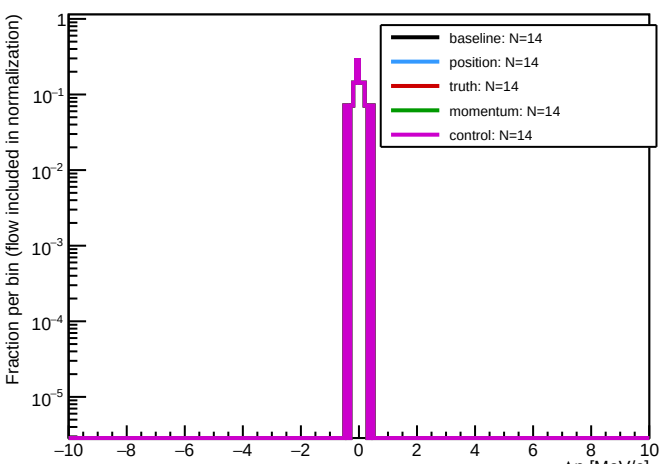
Entrance transverse error [mm] / direct



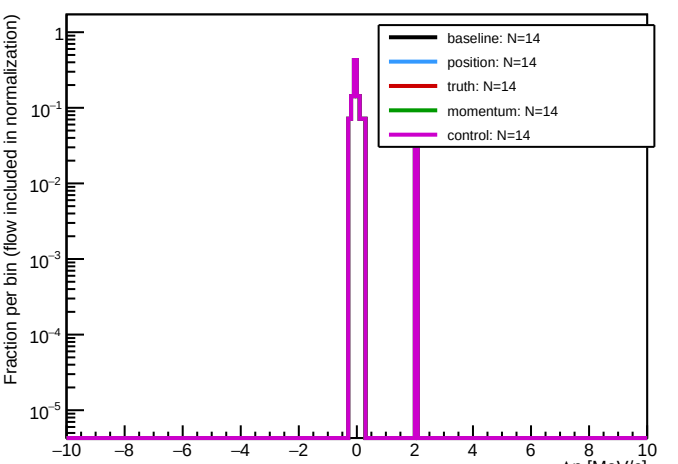

Entrance transverse error [mm] / reflected



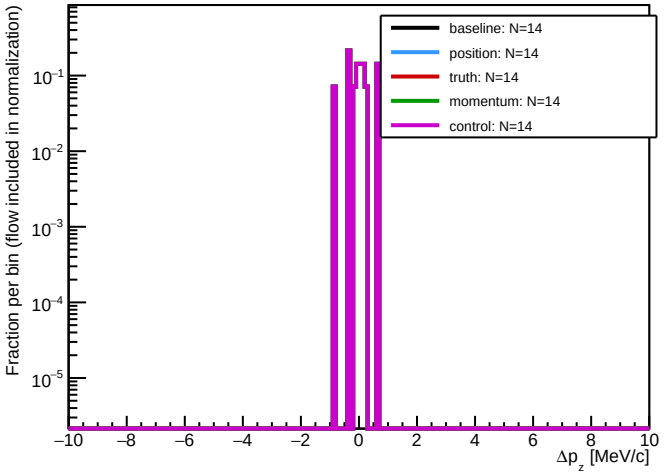

Entrance momentum error [MeV/c] / direct



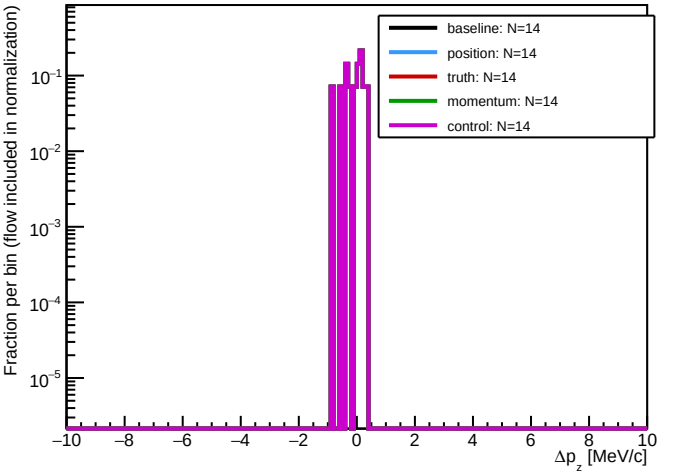

Entrance momentum error [MeV/c] / reflected



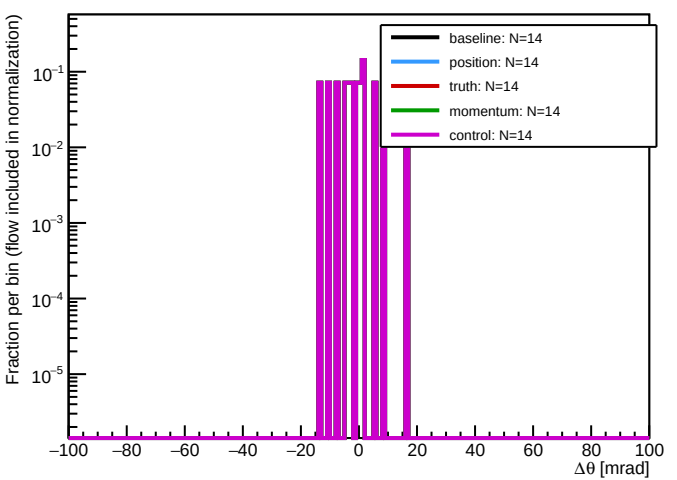
Entrance pz error [MeV/c] / direct



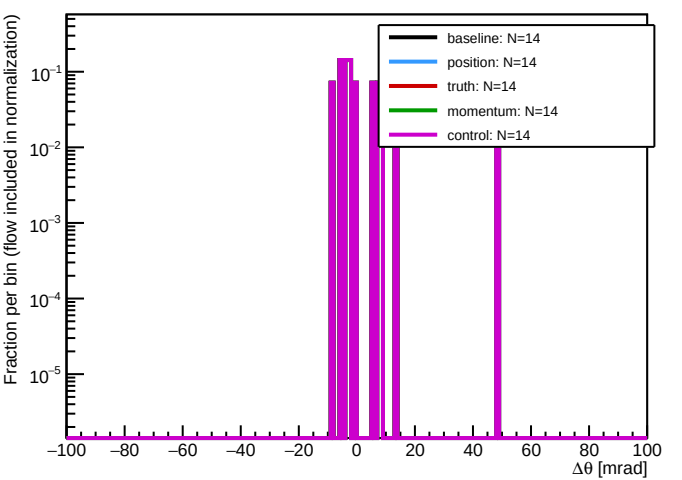

Entrance pz error [MeV/c] / reflected



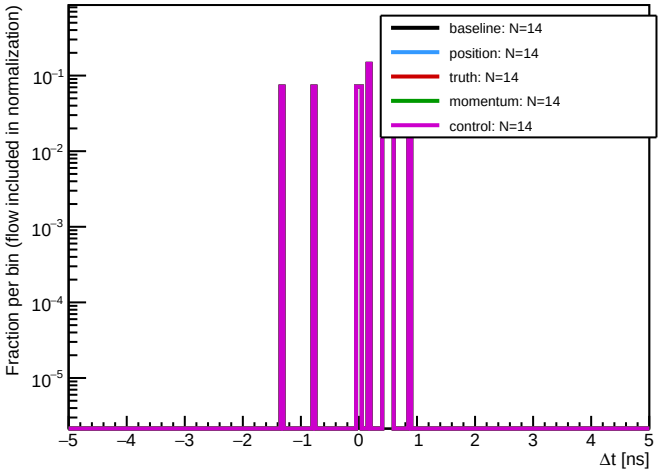

Entrance pitch error [mrad] / direct



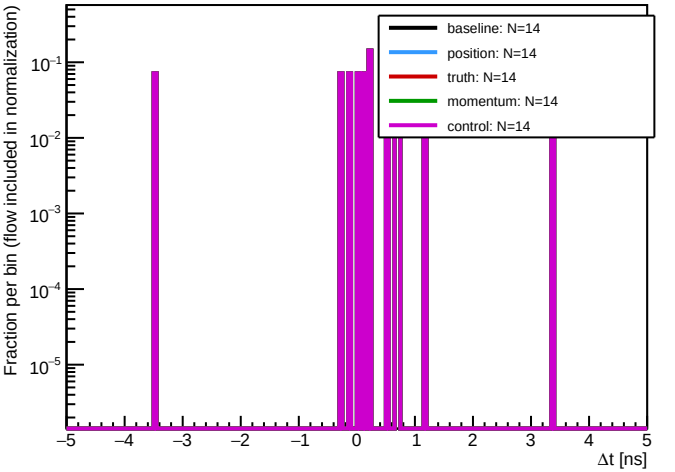

Entrance pitch error [mrad] / reflected



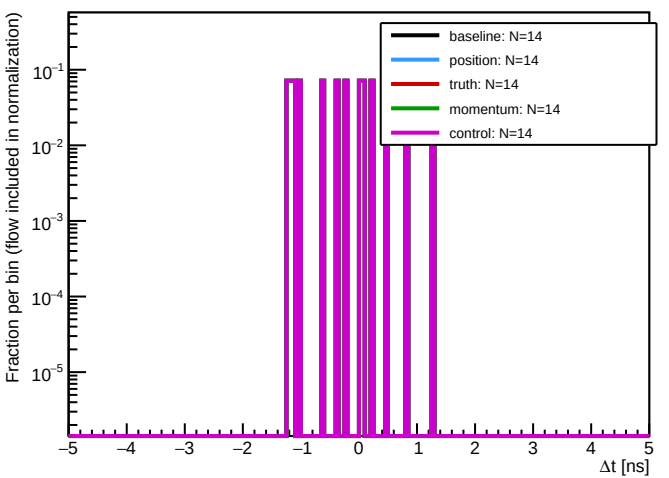
Entrance time error, clock corrected [ns] / direct



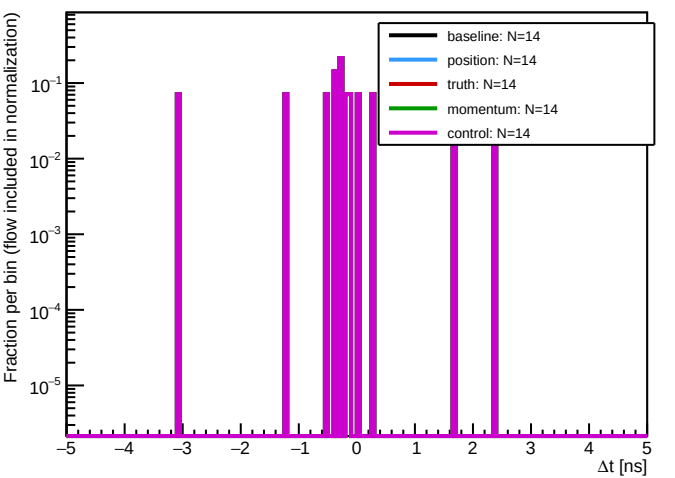

Entrance time error, clock corrected [ns] / reflected

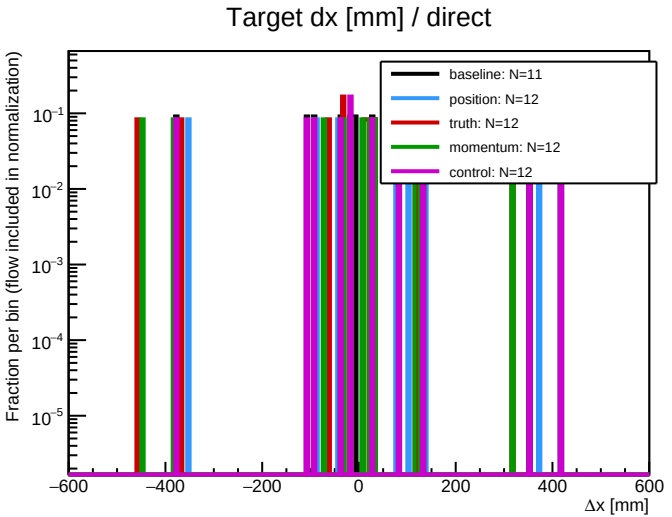


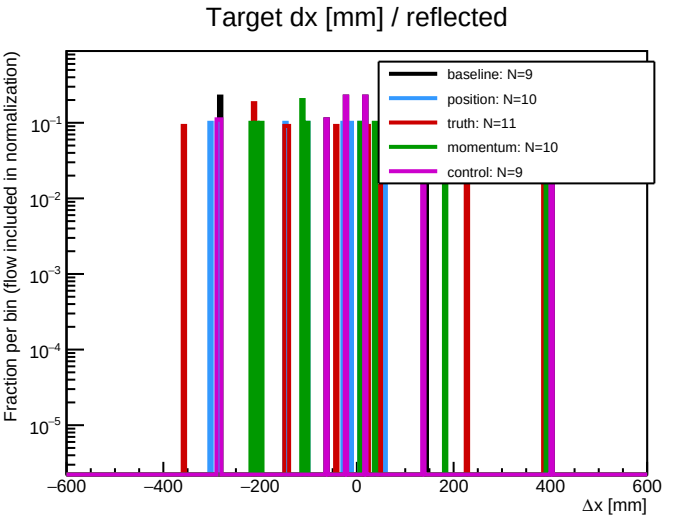

Entrance time error, bunch clock included [ns] / direct

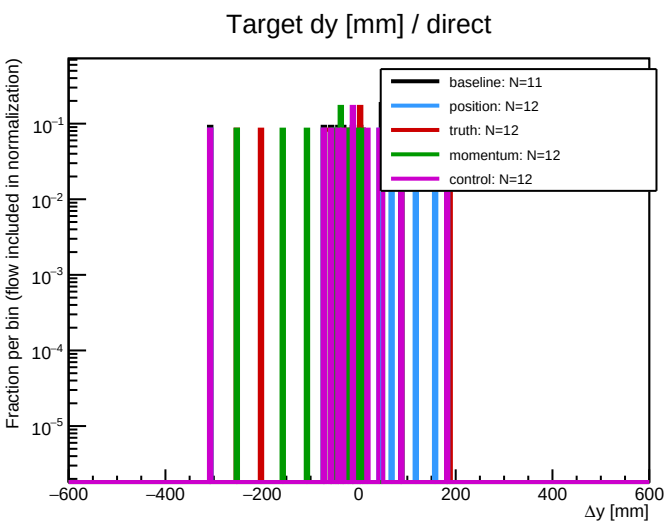


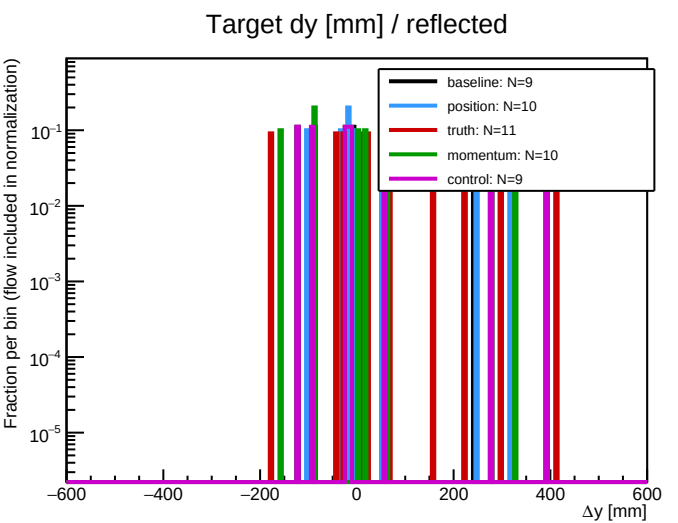

Entrance time error, bunch clock included [ns] / reflected



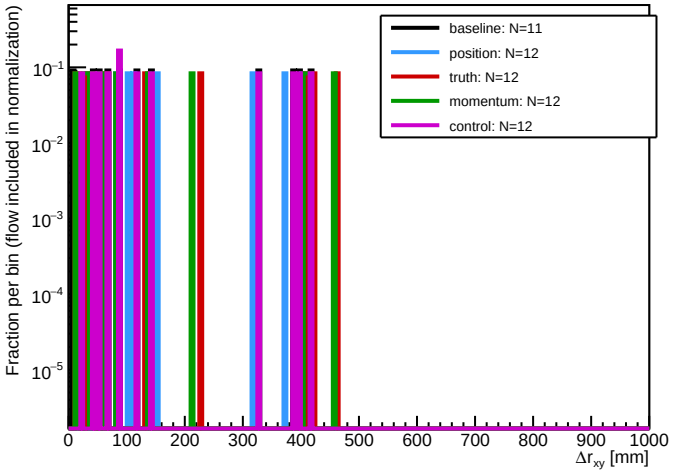

Target dx [mm] / direct
baseline: N=11
position: N=12
truth: N=12
momentum: N=12
control: N=12
Fraction per bin (flow included in normalization)
Δx [mm]


Target dx [mm] / reflected
baseline: N=9
position: N=10
truth: N=11
momentum: N=10
control: N=9
Fraction per bin (flow included in normalization)
Δx [mm]

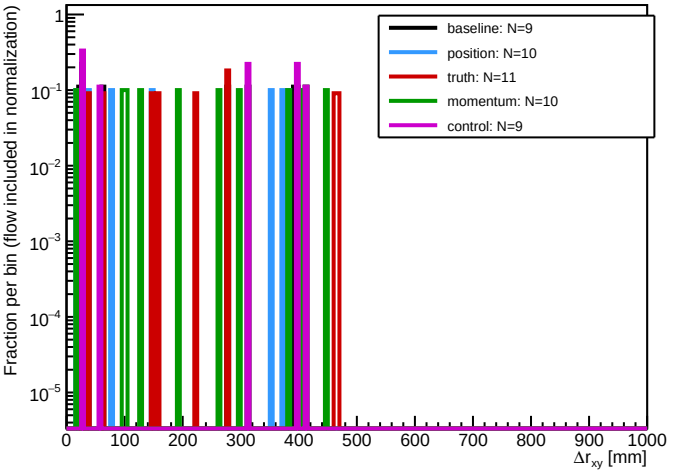

Target dy [mm] / direct
baseline: N=11
position: N=12
truth: N=12
momentum: N=12
control: N=12
Fraction per bin (flow included in normalization)
Δy [mm]


Target dy [mm] / reflected
baseline: N=9
position: N=10
truth: N=11
momentum: N=10
control: N=9
Fraction per bin (flow included in normalization)
Δy [mm]

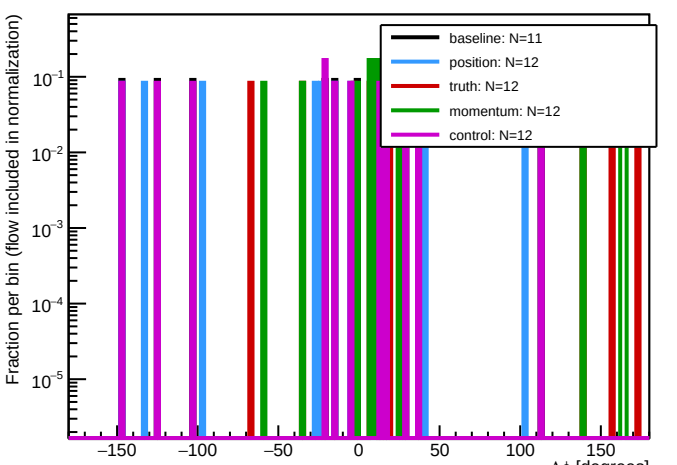
Target transverse error [mm] / direct



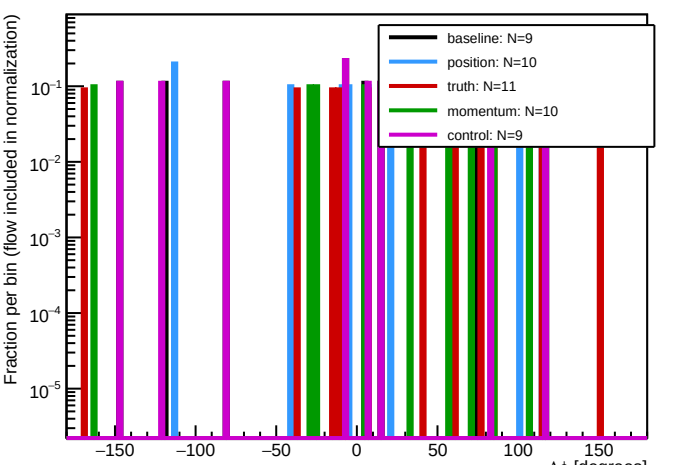

Target transverse error [mm] / reflected

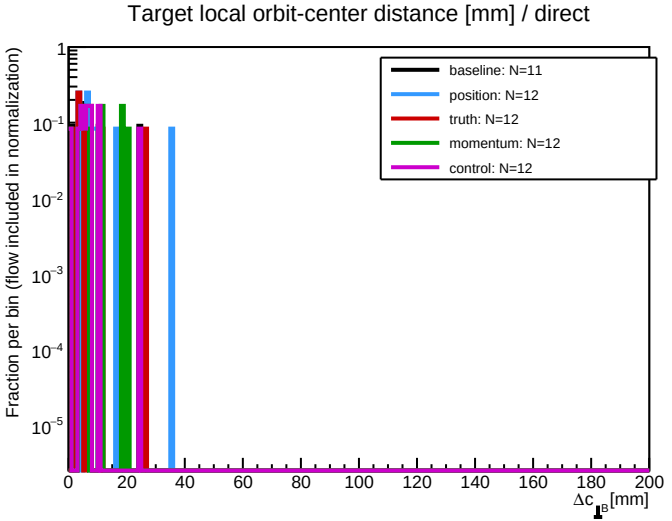


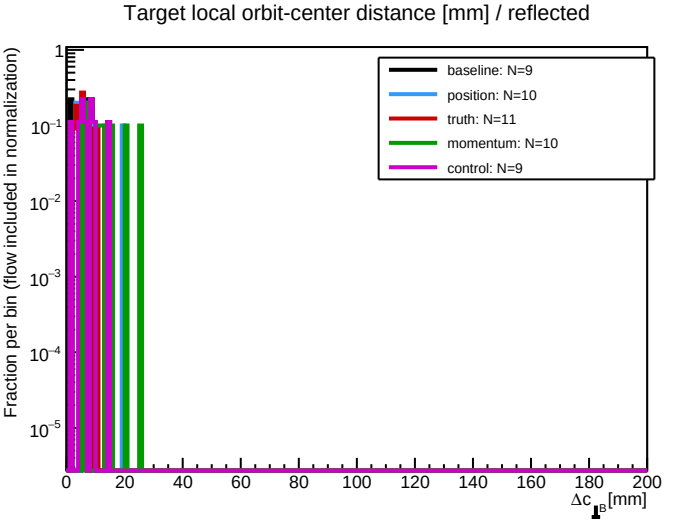

Target wrapped phase error [degrees] / direct

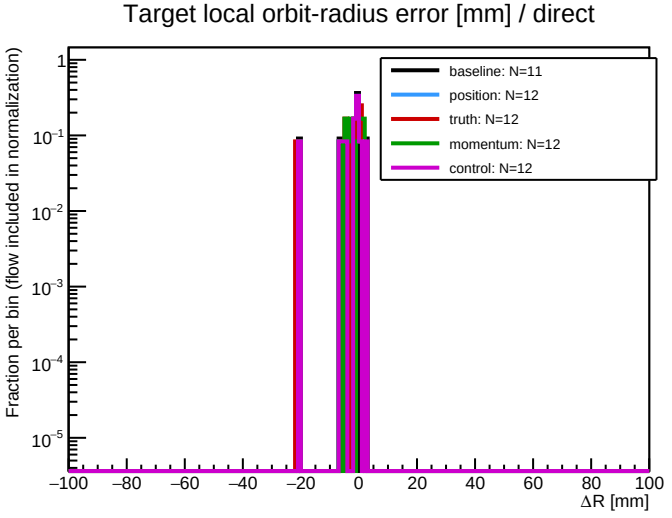


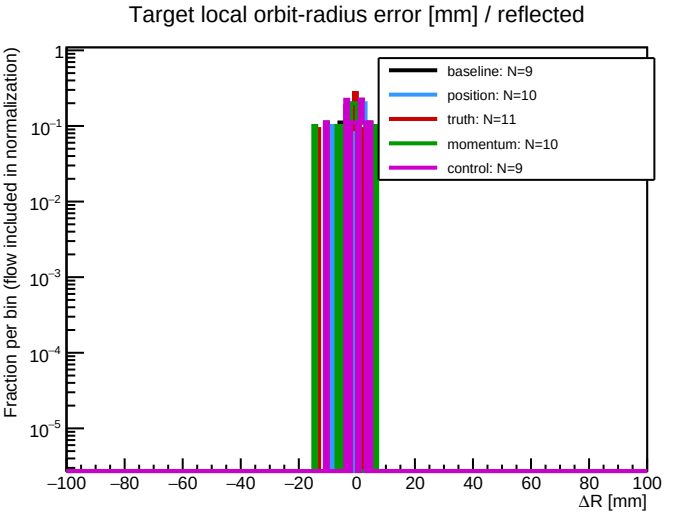

Target wrapped phase error [degrees] / reflected



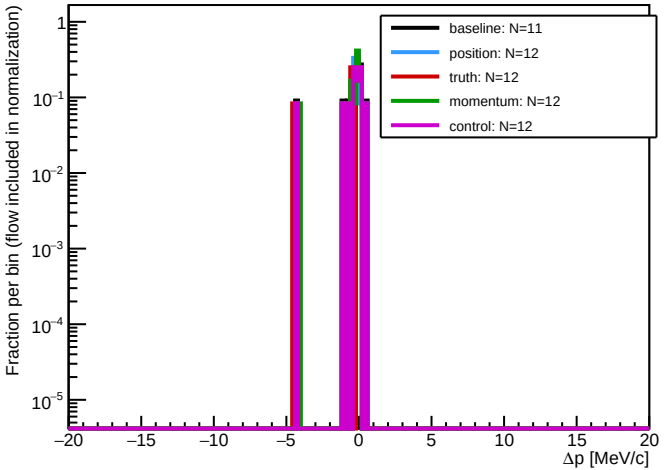

Target local orbit-center distance [mm] / direct
Fraction per bin (flow included in normalization)
baseline: N=11
position: N=12
truth: N=12
momentum: N=12
control: N=12
Δc_B [mm]


Target local orbit-center distance [mm] / reflected
Fraction per bin (flow included in normalization)
baseline: N=9
position: N=10
truth: N=11
momentum: N=10
control: N=9
Δc_B [mm]

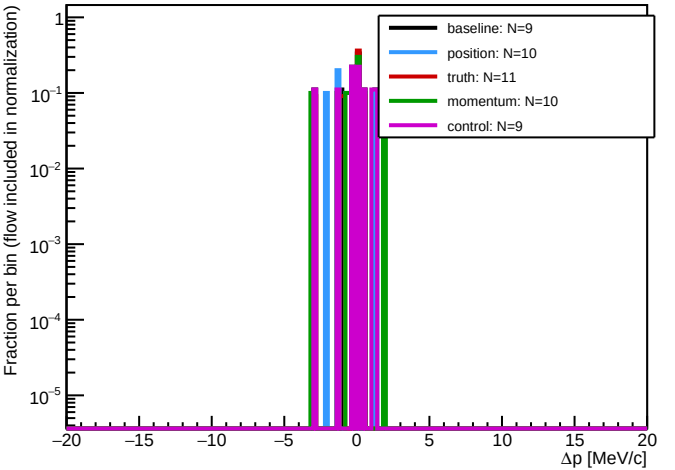

Target local orbit-radius error [mm] / direct
Fraction per bin (flow included in normalization)
baseline: N=11
position: N=12
truth: N=12
momentum: N=12
control: N=12
ΔR [mm]


Target local orbit-radius error [mm] / reflected
Fraction per bin (flow included in normalization)
baseline: N=9
position: N=10
truth: N=11
momentum: N=10
control: N=9
ΔR [mm]

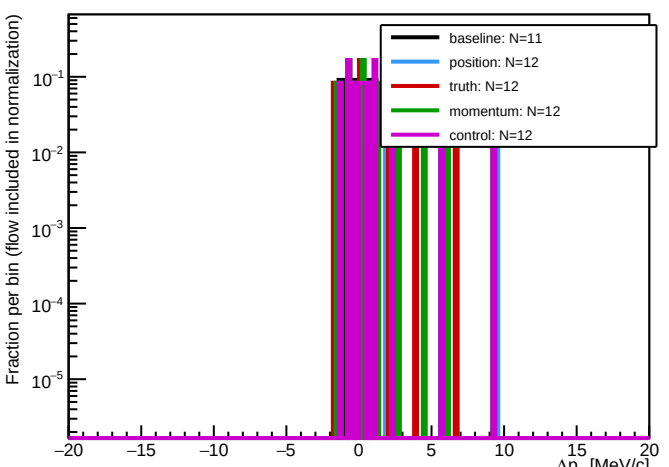
Target momentum error [MeV/c] / direct



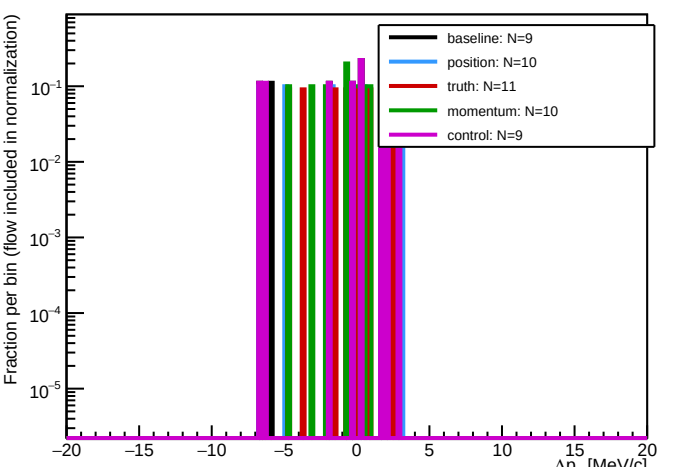

Target momentum error [MeV/c] / reflected



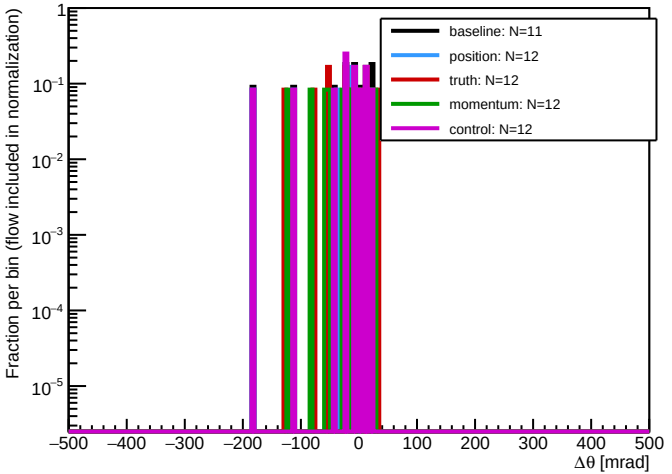

Target pz error [MeV/c] / direct



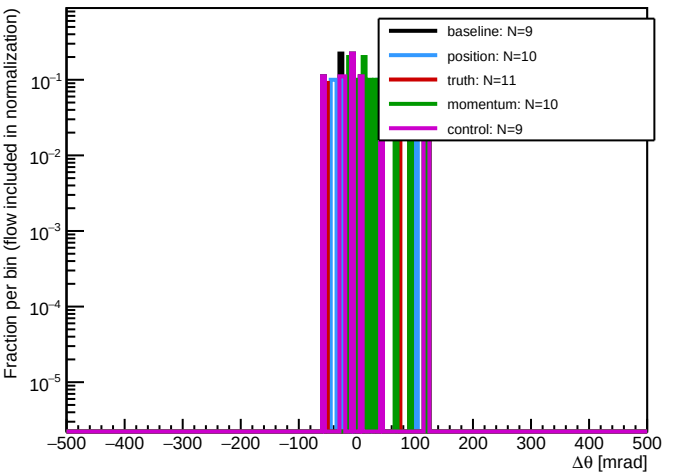

Target pz error [MeV/c] / reflected



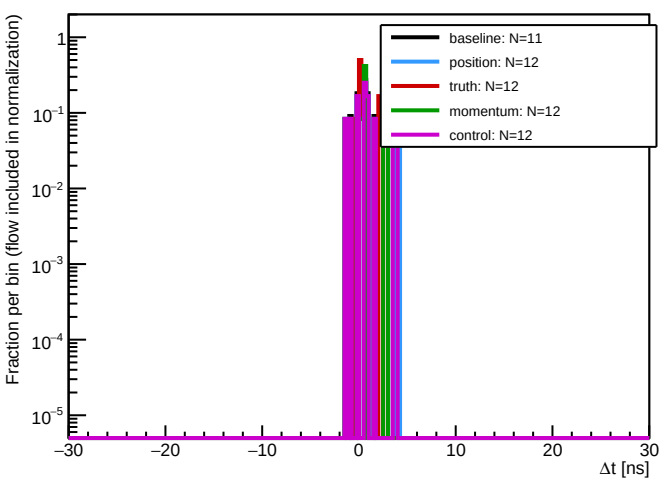
Target pitch error [mrad] / direct



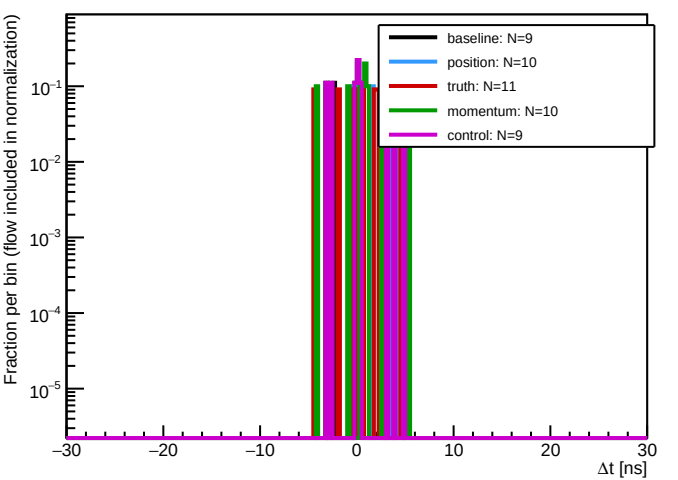

Target pitch error [mrad] / reflected

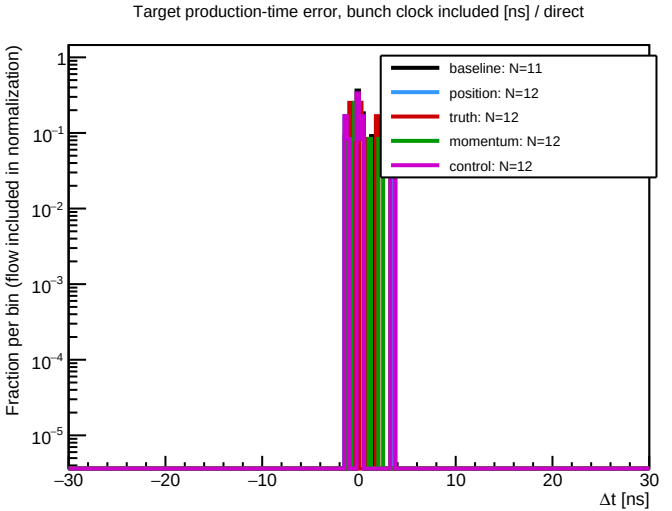


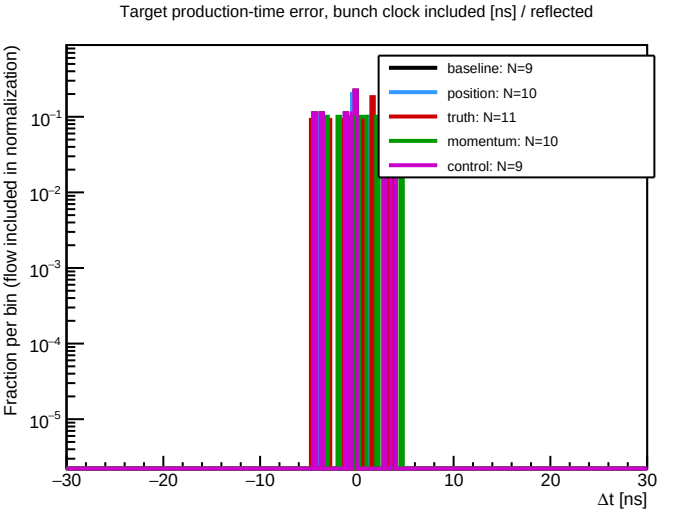

Target production-time error, clock corrected [ns] / direct

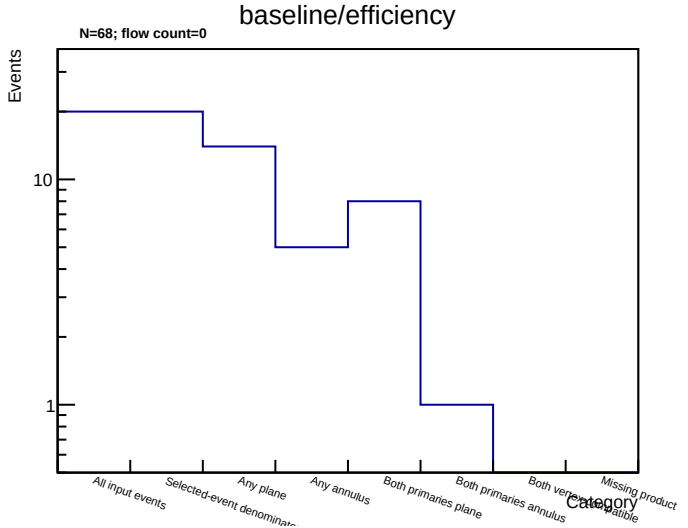


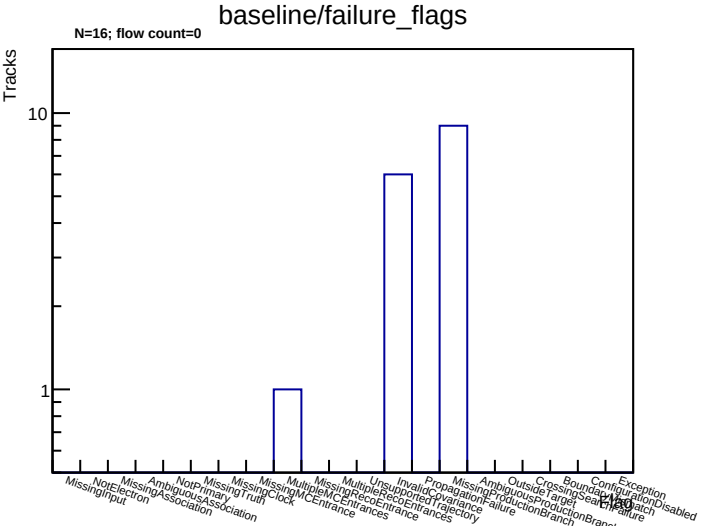

Target production-time error, clock corrected [ns] / reflected

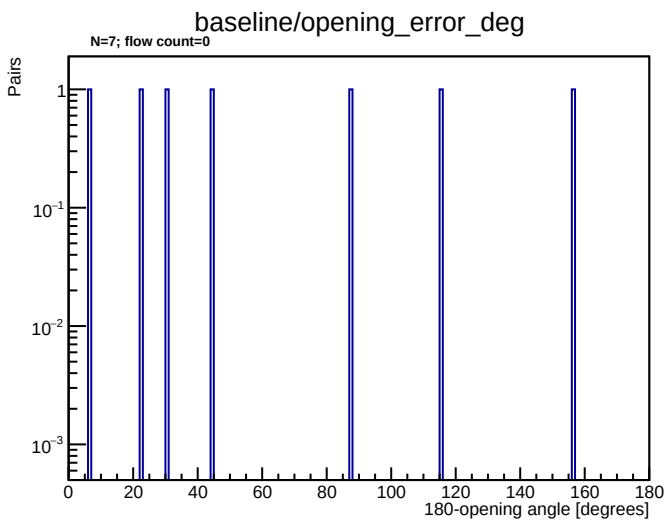


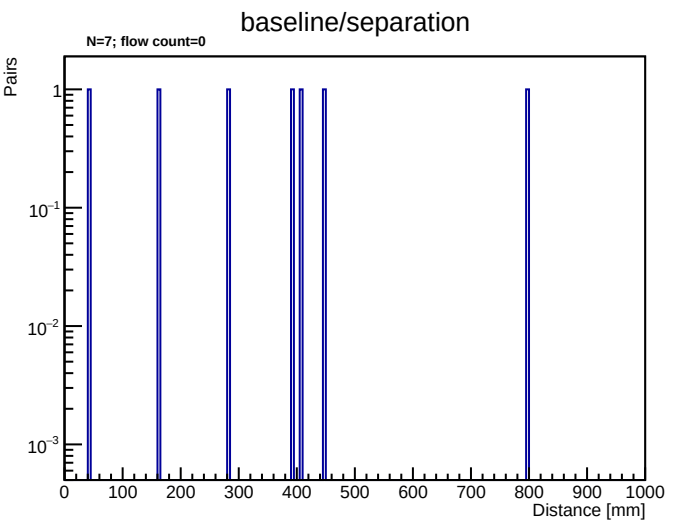

Target production-time error, bunch clock included [ns] / direct
Fraction per bin (flow included in normalization)
Δt [ns]
baseline: N=11
position: N=12
truth: N=12
momentum: N=12
control: N=12

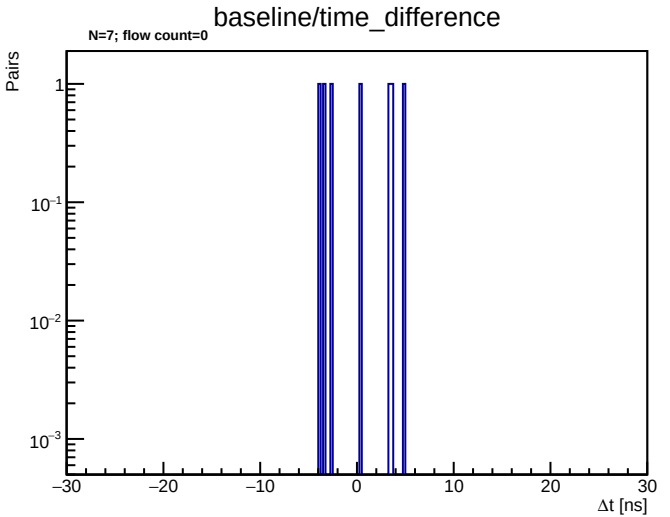

Target production-time error, bunch clock included [ns] / reflected
Fraction per bin (flow included in normalization)
Δt [ns]
baseline: N=9
position: N=10
truth: N=11
momentum: N=10
control: N=9

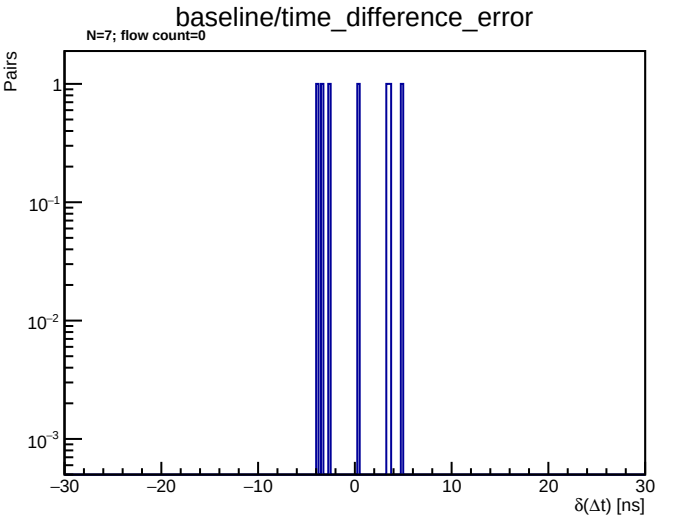

baseline/efficiency
N=68; flow count=0
Events
10
1
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both ver
Missing product
Category

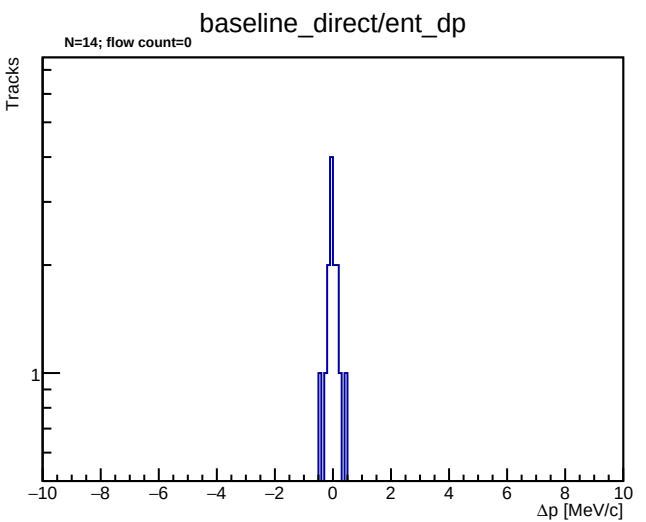

baseline/failure_flags
N=16; flow count=0
Tracks
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimary
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleRecoEntrance
MissingRecoEntrances
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeptum
BoundaryMatch
Configuration
Exception
Disabled
Failure
Flag

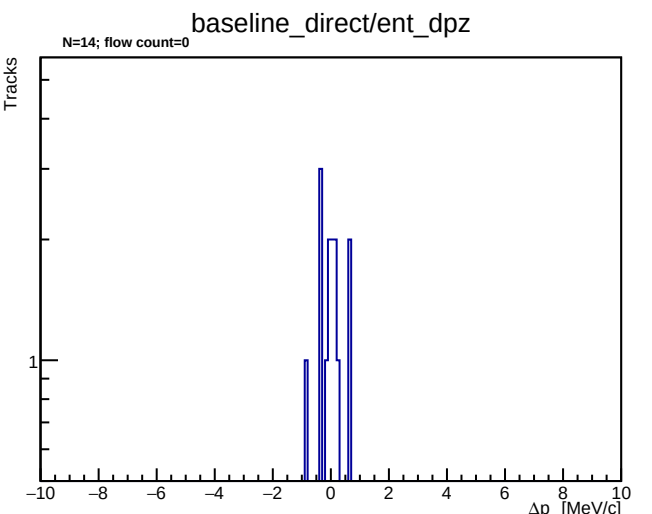

baseline/opening_error_deg
N=7; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

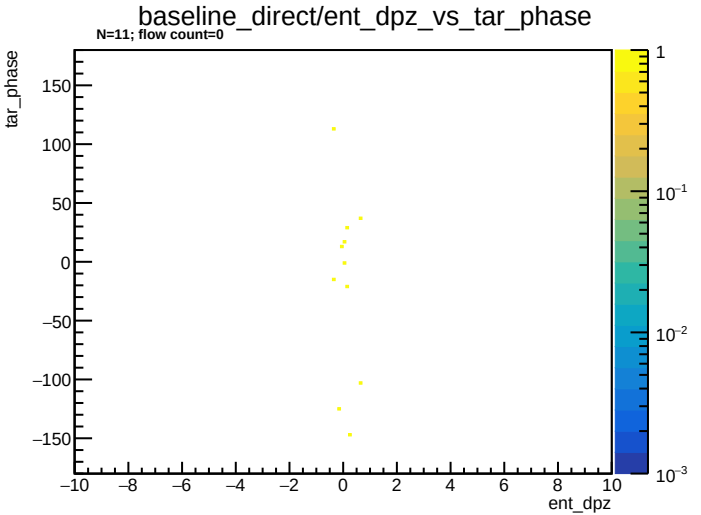

baseline/separation
N=7; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

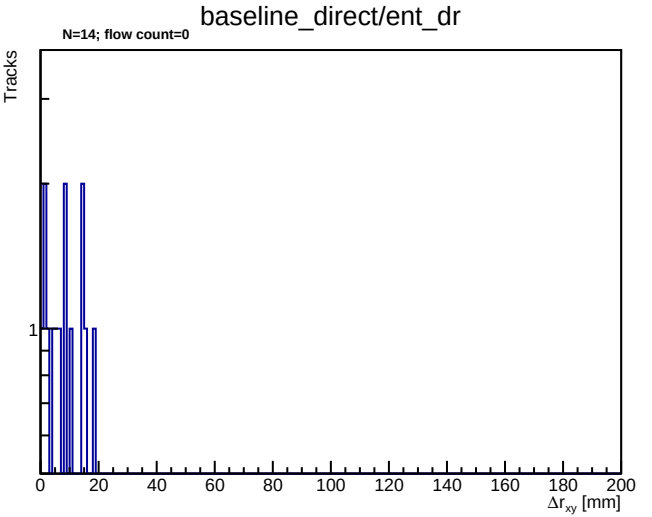

baseline/time_difference
N=7; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−30 −20 −10 0 10 20 30
Δt [ns]

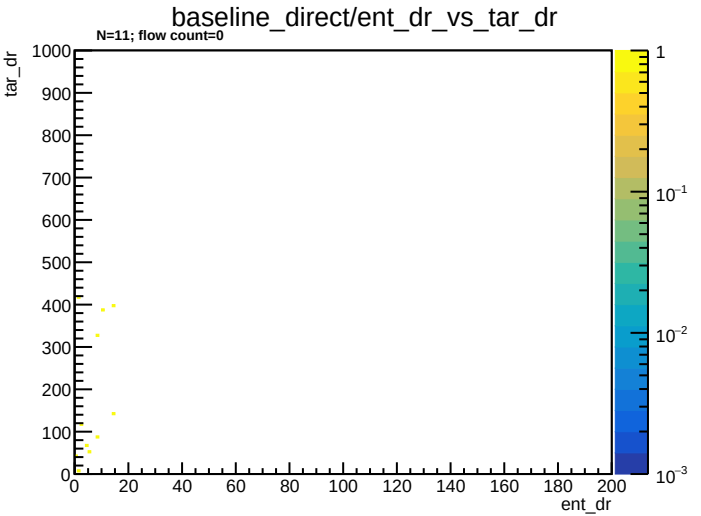

baseline/time_difference_error
N=7; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−30 −20 −10 0 10 20 30
δ(Δt) [ns]

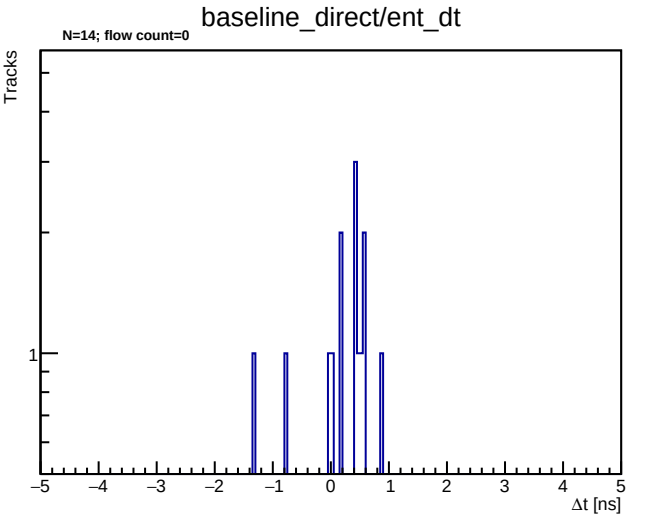

baseline_direct/ent_dp
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

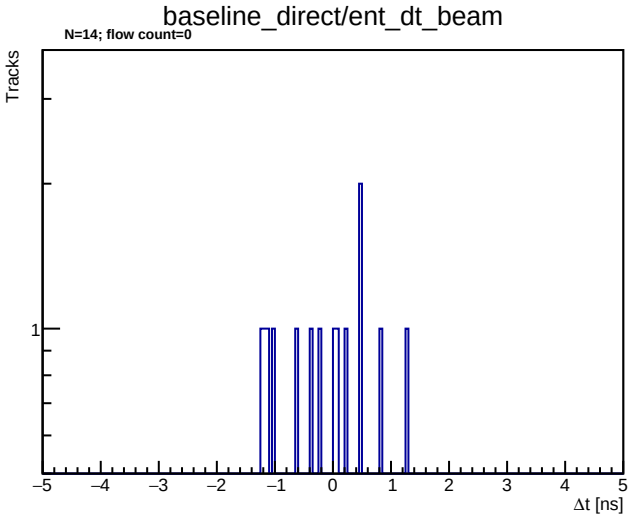

baseline_direct/ent_dpz
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp$_z$ [MeV/c]

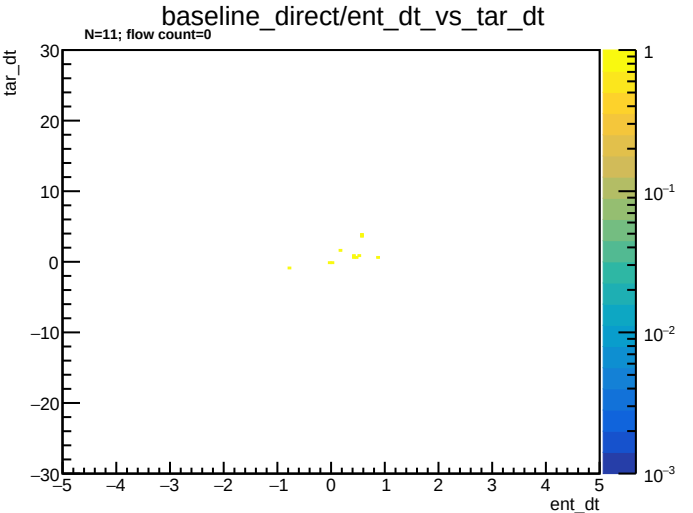

baseline_direct/ent_dpz_vs_tar_phase
N=11; flow count=0
tar_phase
ent_dpz

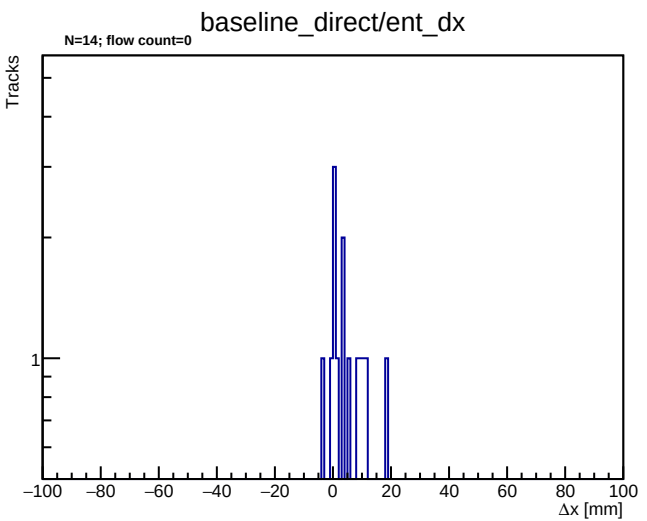

baseline_direct/ent_dr
N=14; flow count=0
Tracks
Δr_xy [mm]

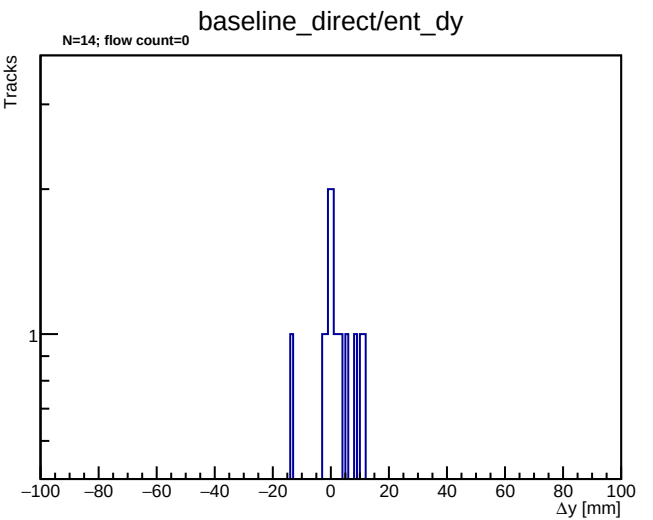

baseline_direct/ent_dr_vs_tar_dr
N=11; flow count=0
tar_dr
ent_dr

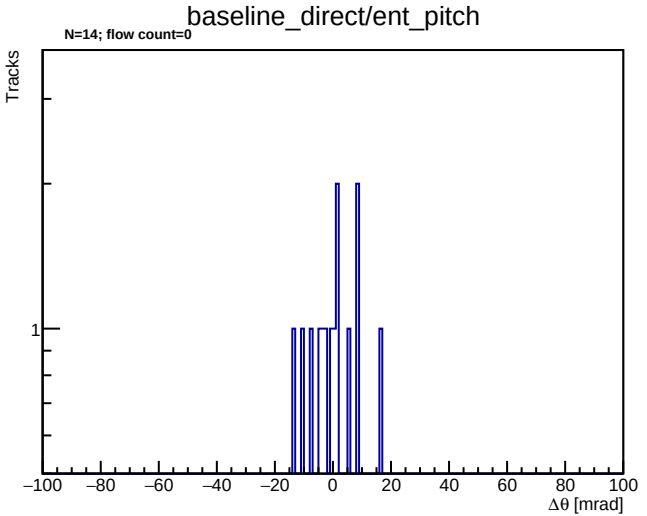

baseline_direct/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

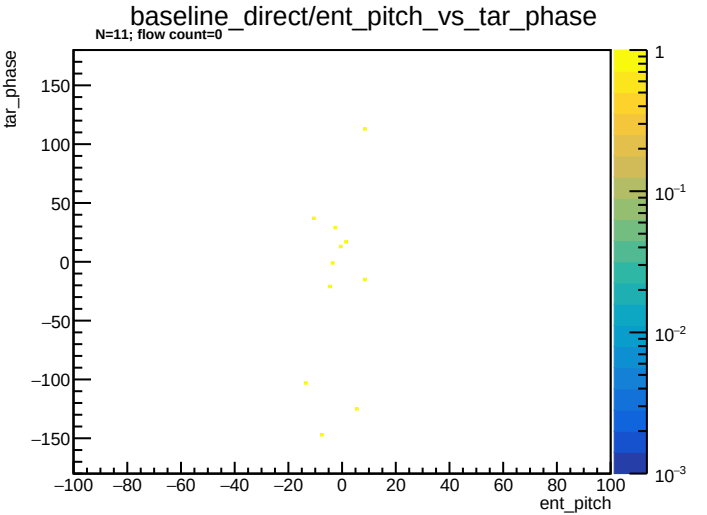

baseline_direct/ent_dt_beam
N=14; flow count=0
Tracks
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

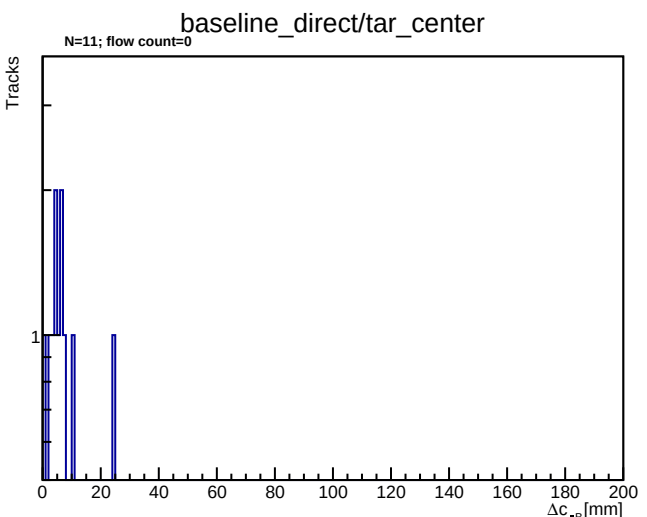

baseline_direct/ent_dt_vs_tar_dt
N=11; flow count=0
tar_dt
30 20 10 0 −10 −20 −30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^-1
10^-2
10^-3

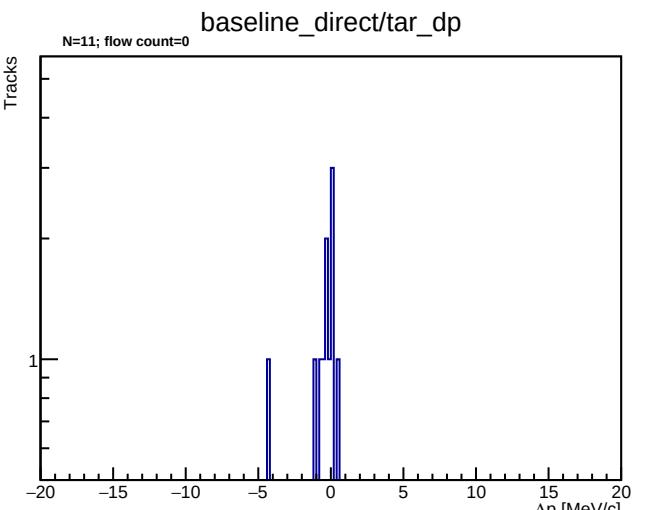

baseline_direct/ent_dx
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

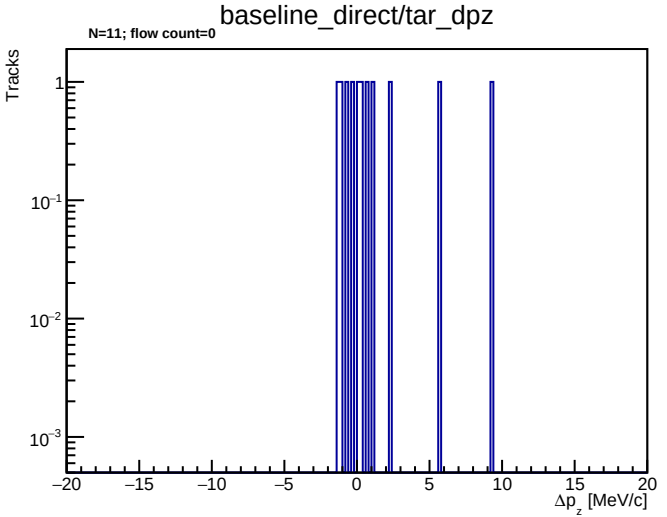

baseline_direct/ent_dy
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

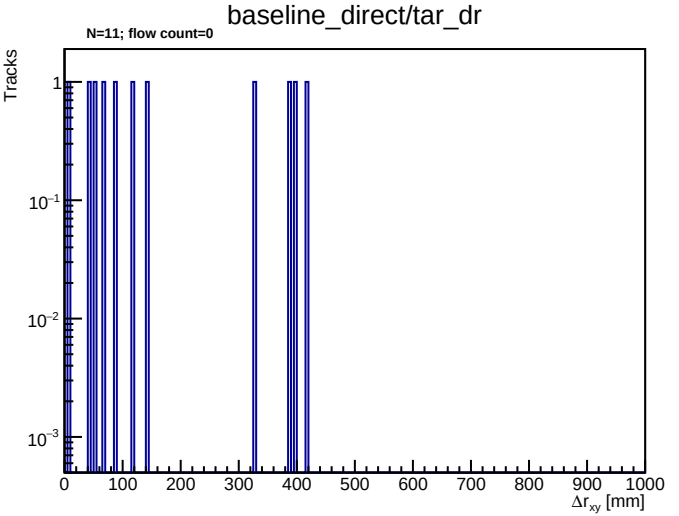

baseline_direct/ent_pitch
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

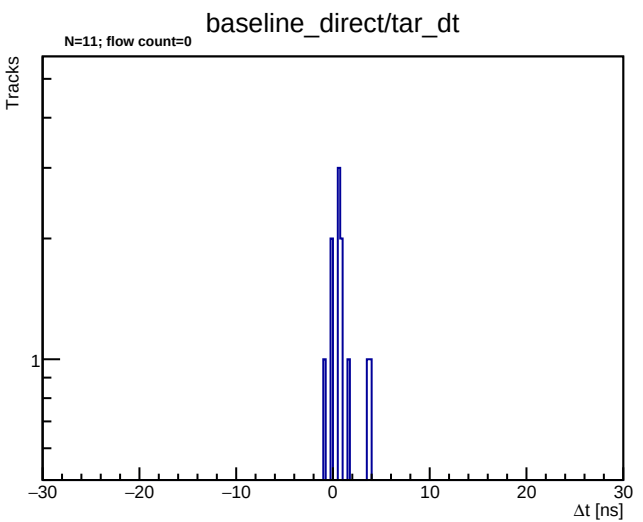

baseline_direct/ent_pitch_vs_tar_phase
N=11; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²
10⁻³

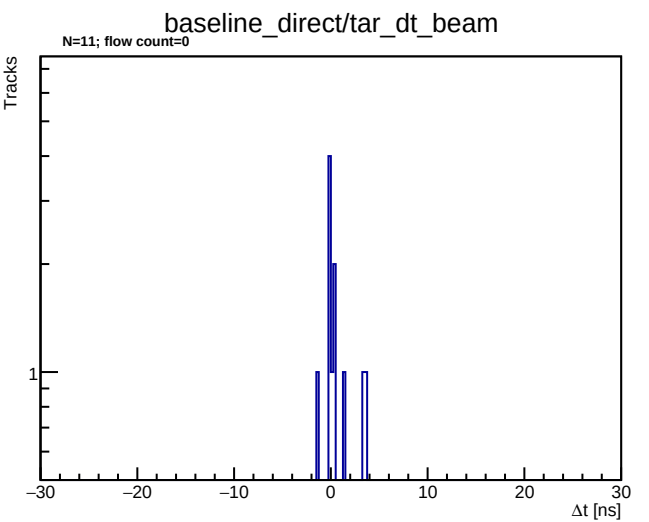

baseline_direct/tar_center
N=11; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

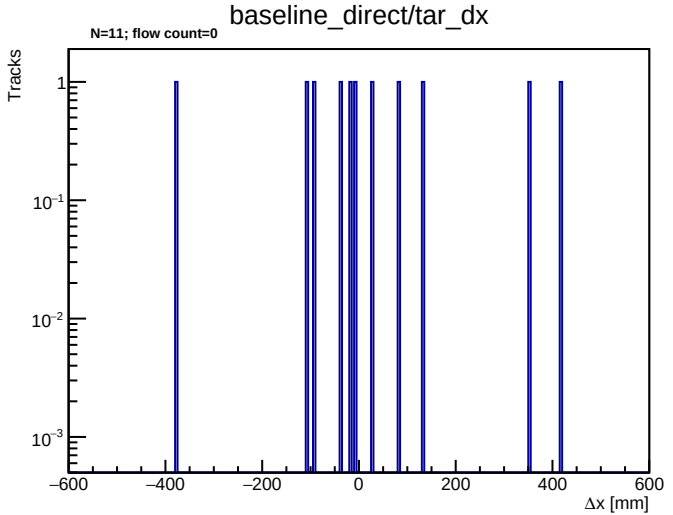

baseline_direct/tar_dp
N=11; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

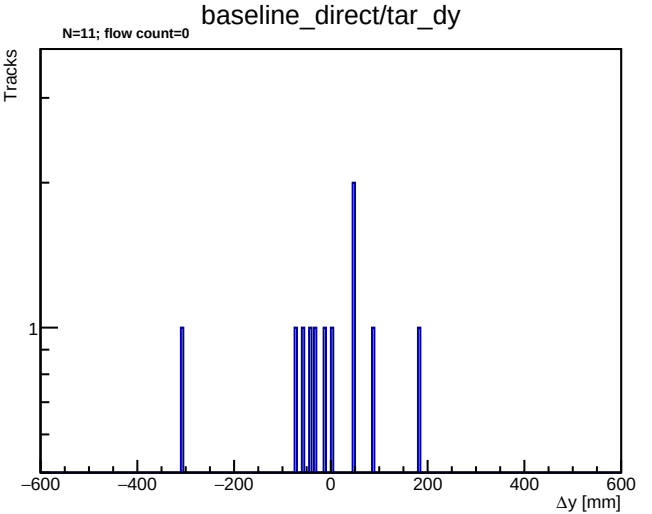

baseline_direct/tar_dpz
N=11; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−20 −15 −10 −5 0 5 10 15 20
$\Delta p_z$ [MeV/c]

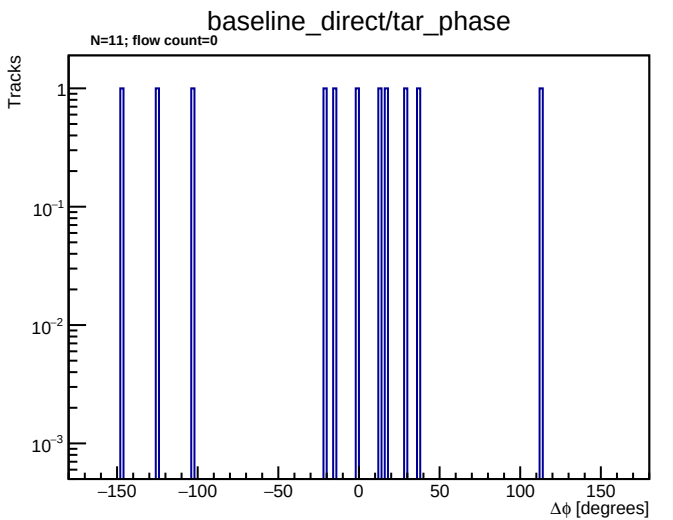

baseline_direct/tar_dr
N=11; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0 100 200 300 400 500 600 700 800 900 1000
$\Delta r_{xy}$ [mm]

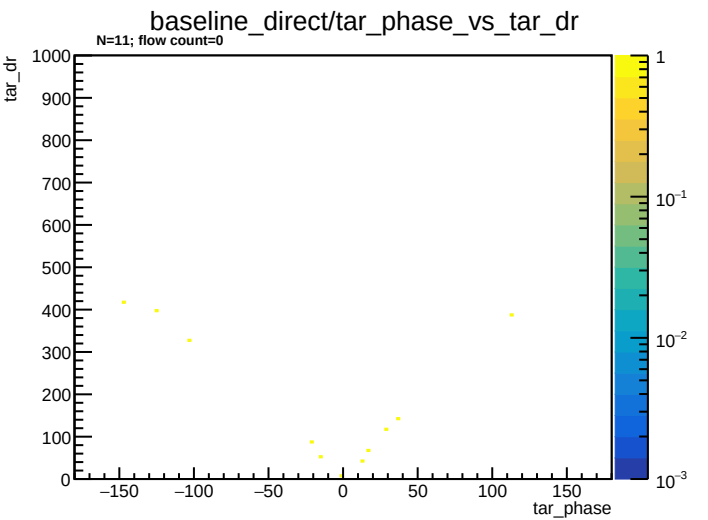

baseline_direct/tar_dt
N=11; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

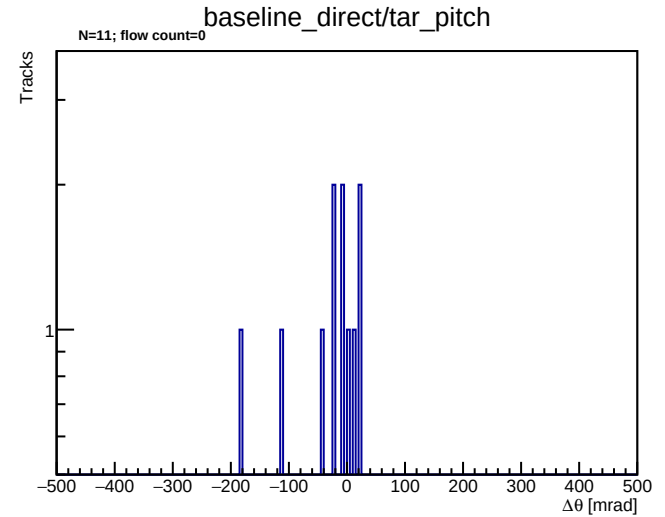

baseline_direct/tar_dt_beam
N=11; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

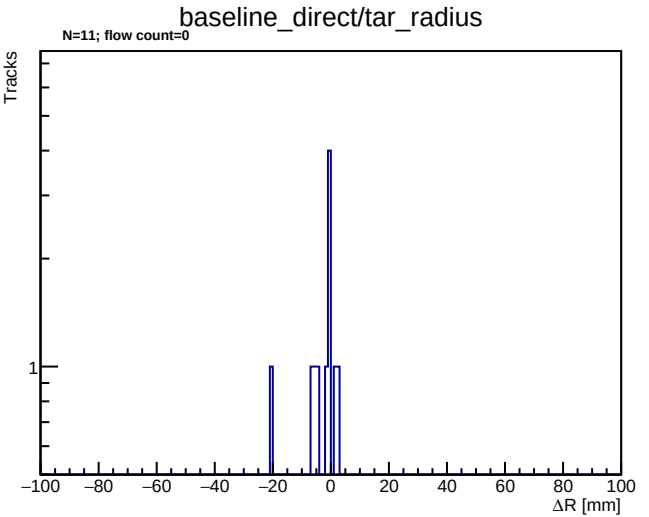

baseline_direct/tar_dx
N=11; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−600
−400
−200
0
200
400
600
Δx [mm]

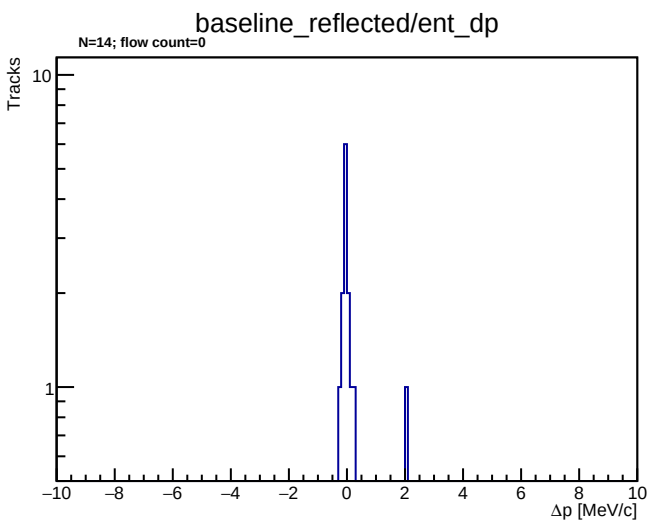

baseline_direct/tar_dy
N=11; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δy [mm]

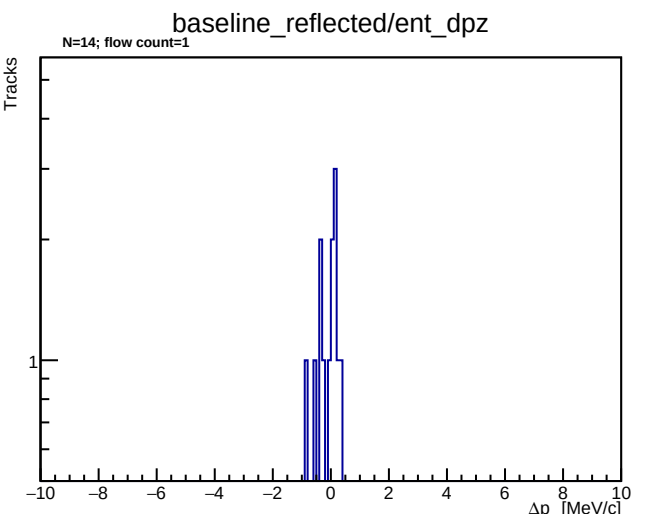

baseline_direct/tar_phase
N=11; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−150
−100
−50
0
50
100
150
Δϕ [degrees]

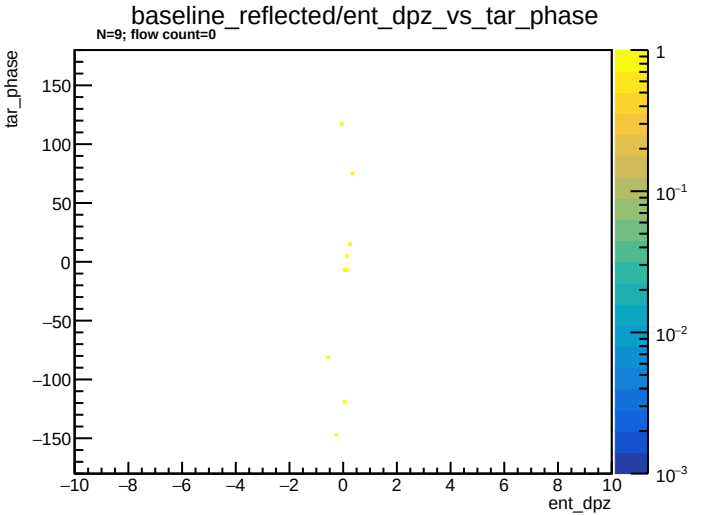

baseline_direct/tar_phase_vs_tar_dr
N=11; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
$10^{-1}$
$10^{-2}$
$10^{-3}$

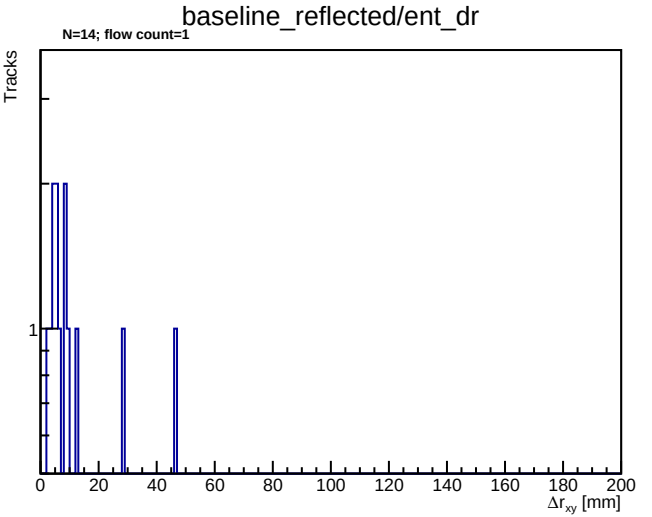

baseline_direct/tar_pitch
N=11; flow count=0
Tracks
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

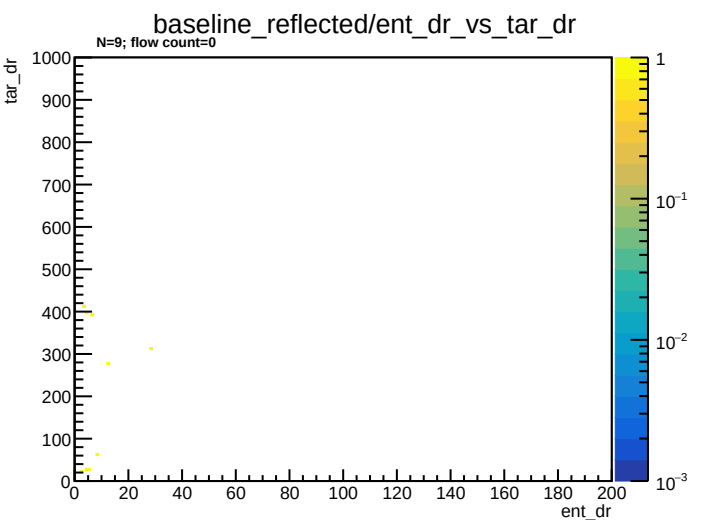

baseline_direct/tar_radius
N=11; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

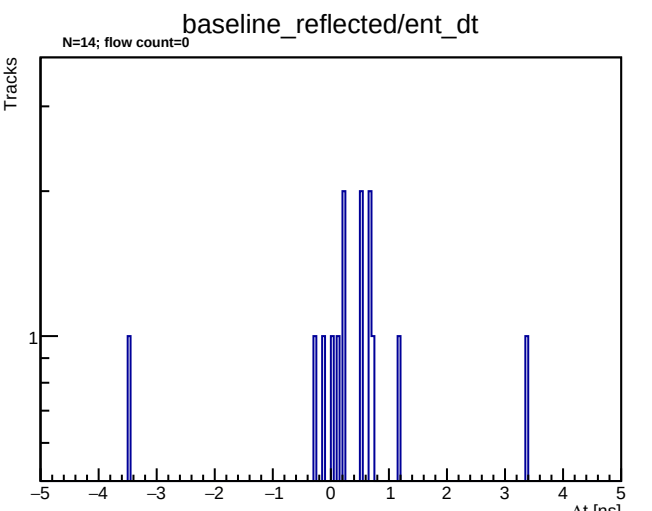

baseline_reflected/ent_dp
N=14; flow count=0
Tracks
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

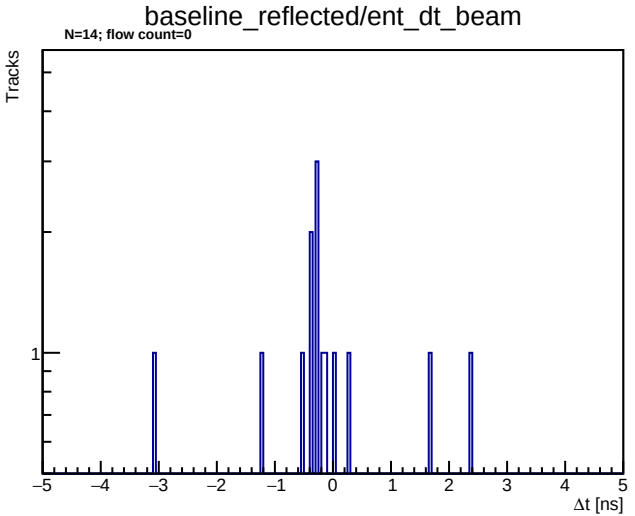

baseline_reflected/ent_dpz
N=14; flow count=1
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

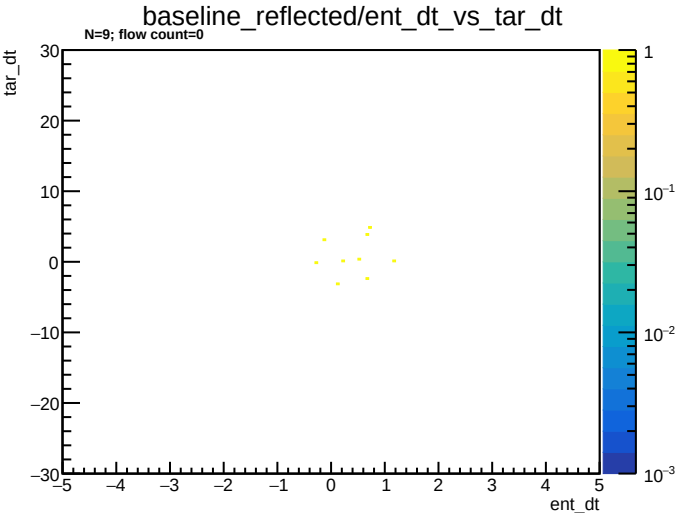

baseline_reflected/ent_dpz_vs_tar_phase
N=9; flow count=0
tar_phase
ent_dpz

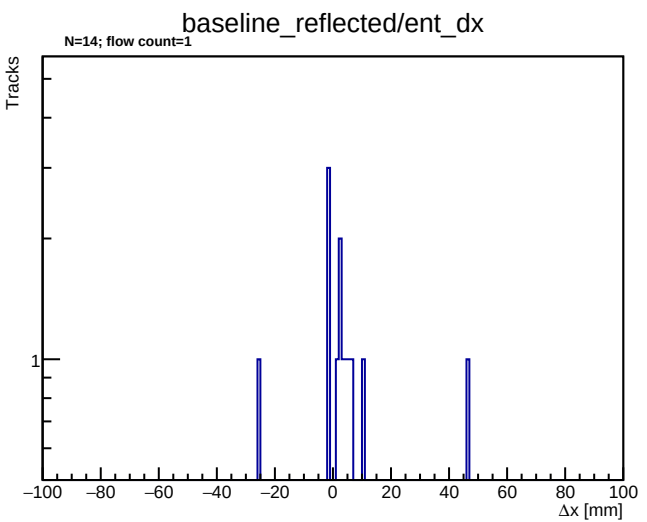

baseline_reflected/ent_dr
N=14; flow count=1
Tracks
Δr_xy [mm]

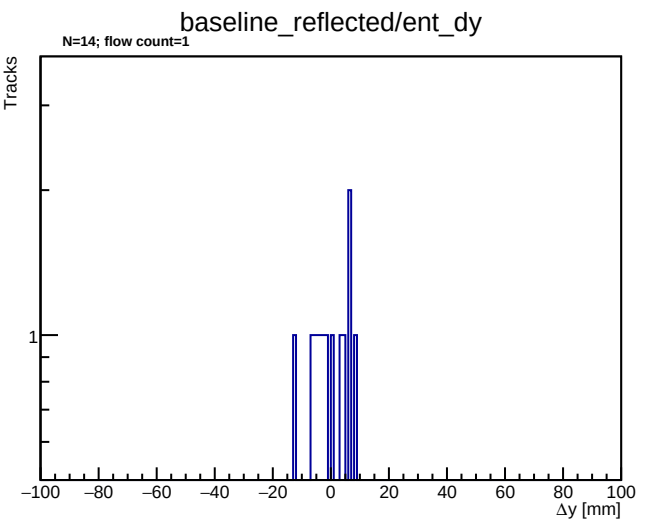

baseline_reflected/ent_dr_vs_tar_dr
N=9; flow count=0
tar_dr
ent_dr

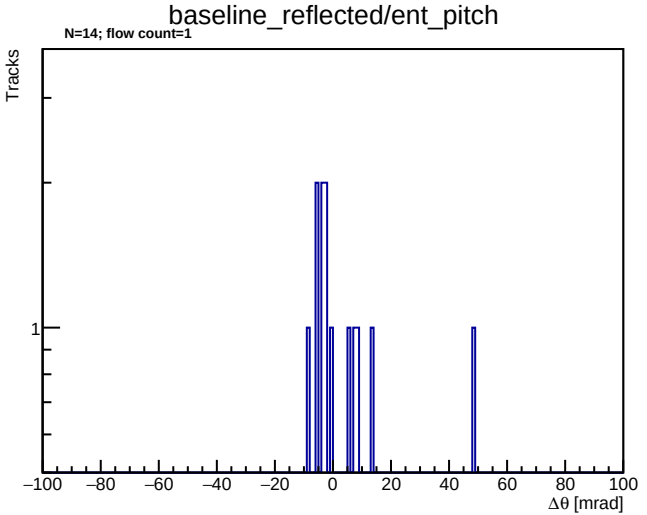

baseline_reflected/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

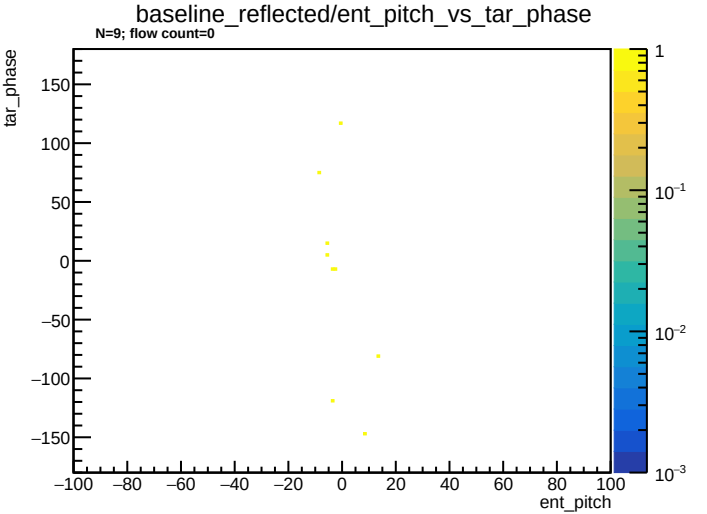

baseline_reflected/ent_dt_beam
N=14; flow count=0
Tracks
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

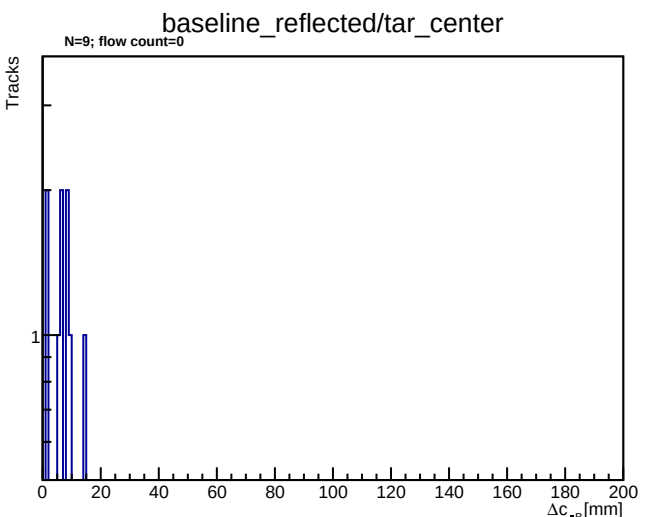

baseline_reflected/ent_dt_vs_tar_dt
N=9; flow count=0
tar_dt
30 20 10 0 −10 −20 −30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^-1
10^-2
10^-3

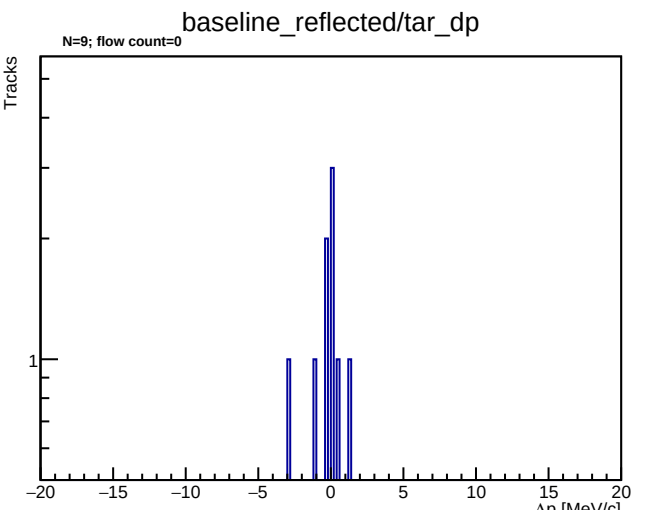

baseline_reflected/ent_dx
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

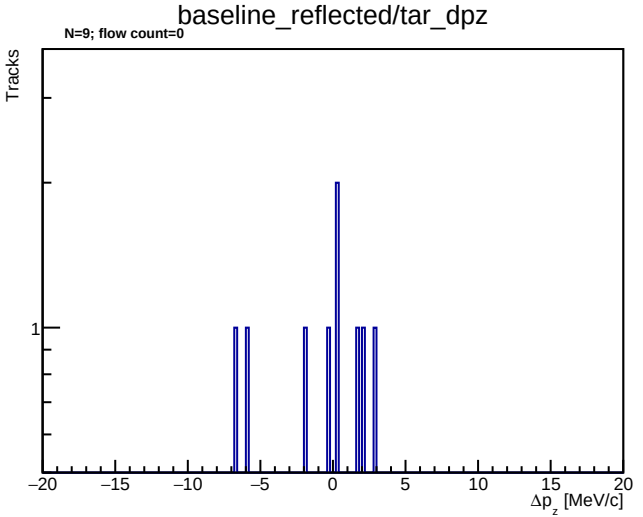

baseline_reflected/ent_dy
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

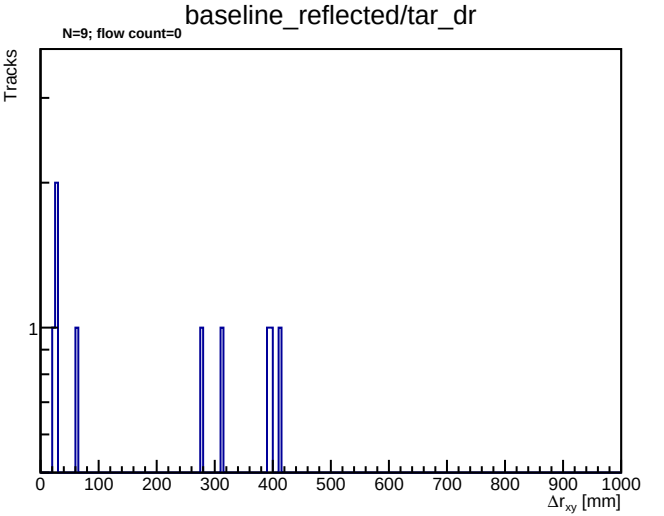

baseline_reflected/ent_pitch
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

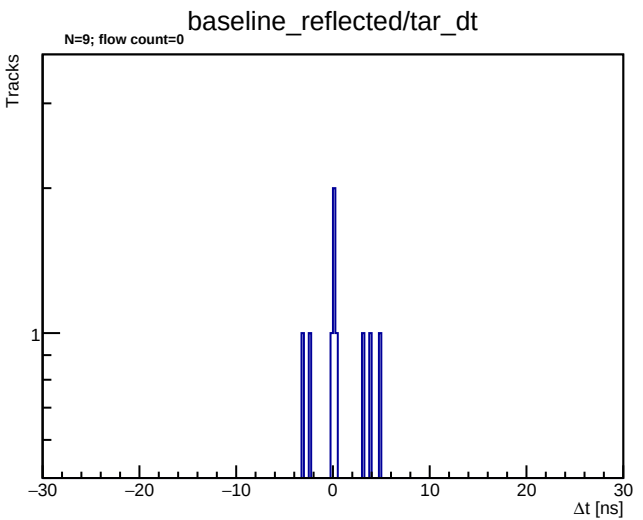

baseline_reflected/ent_pitch_vs_tar_phase
N=9; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10^-1
10^-2
10^-3

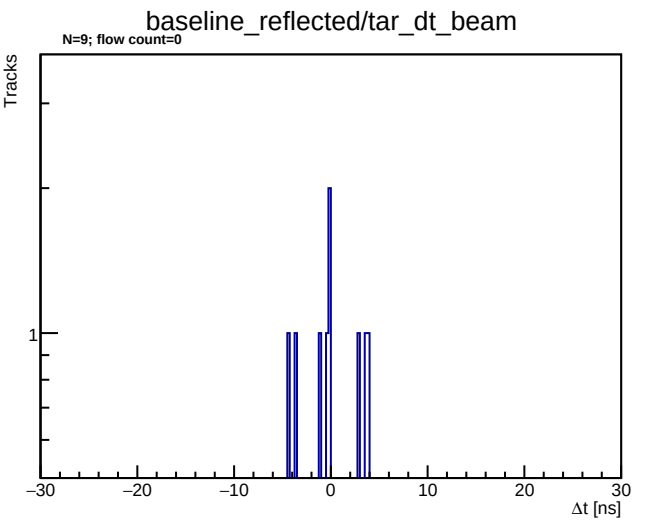

baseline_reflected/tar_center
N=9; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

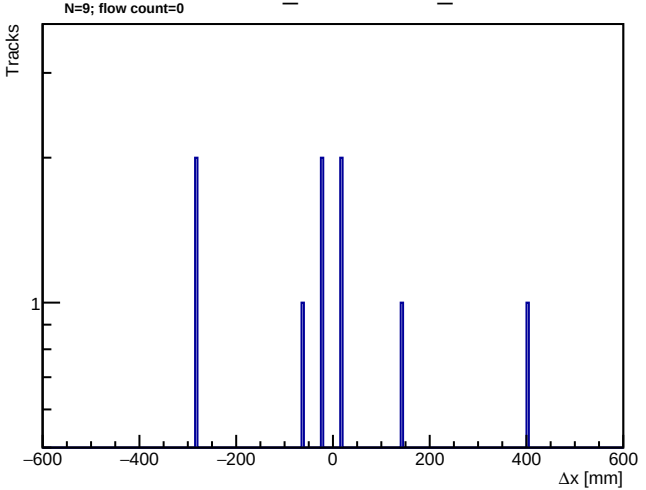

baseline_reflected/tar_dp
N=9; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

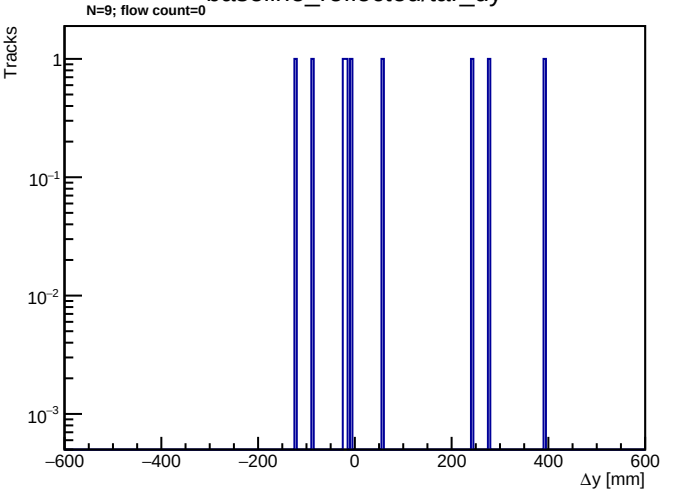

baseline_reflected/tar_dpz
N=9; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

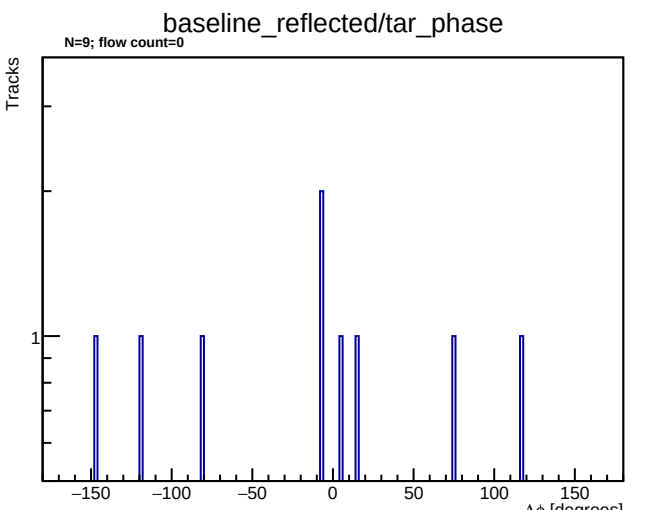

baseline_reflected/tar_dr
N=9; flow count=0
Tracks
1
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

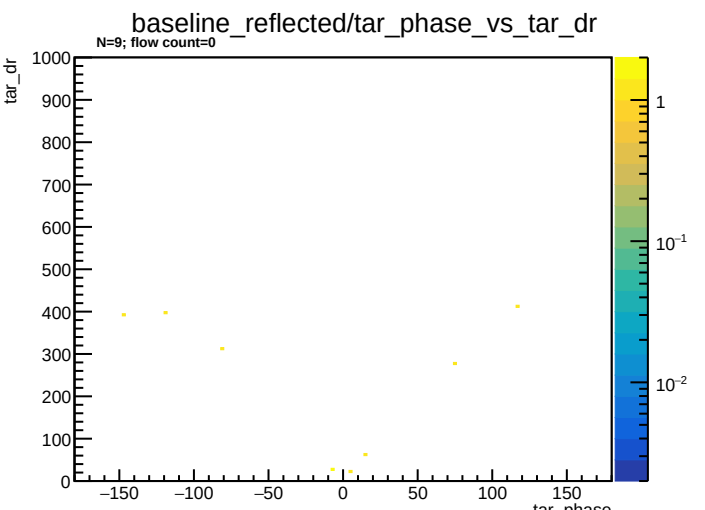

baseline_reflected/tar_dt
N=9; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

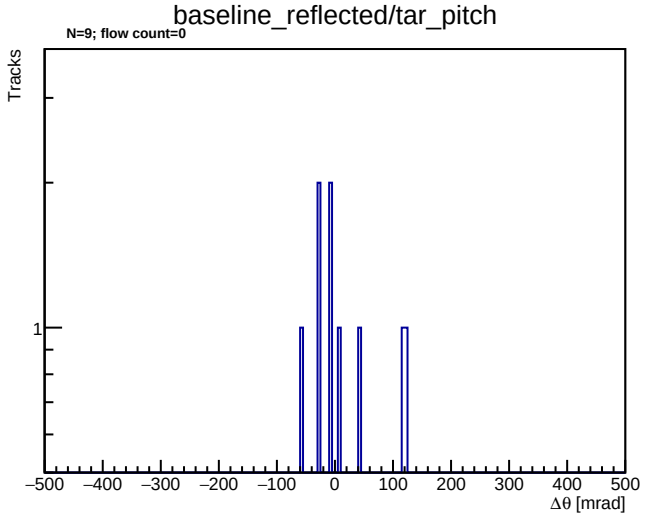

baseline_reflected/tar_dt_beam
N=9; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

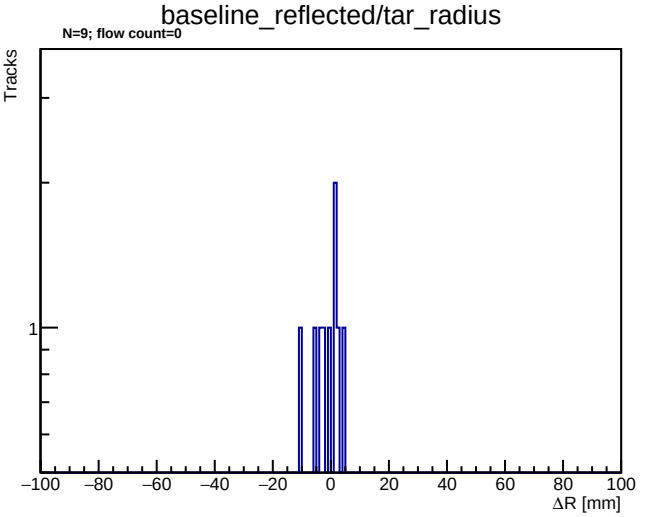

baseline_reflected/tar_dx
N=9; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δx [mm]

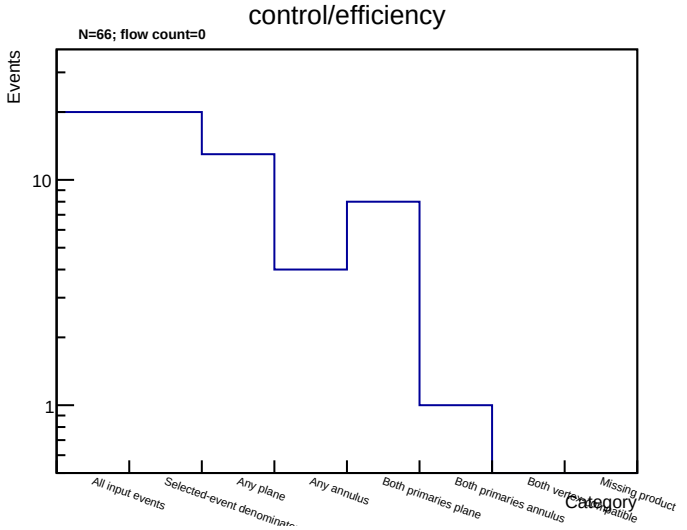

baseline_reflected/tar_dy
N=9; flow count=0
Tracks
1
10⁻¹
10⁻²
10⁻³
−600
−400
−200
0
200
400
600
Δy [mm]

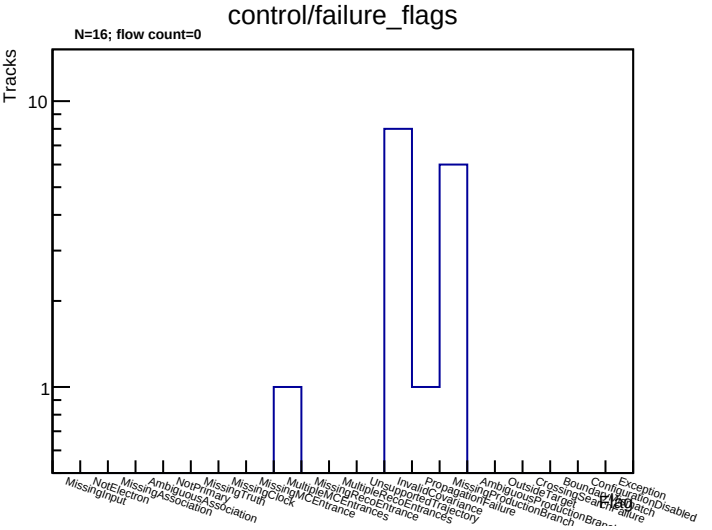

baseline_reflected/tar_phase
N=9; flow count=0
Tracks
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

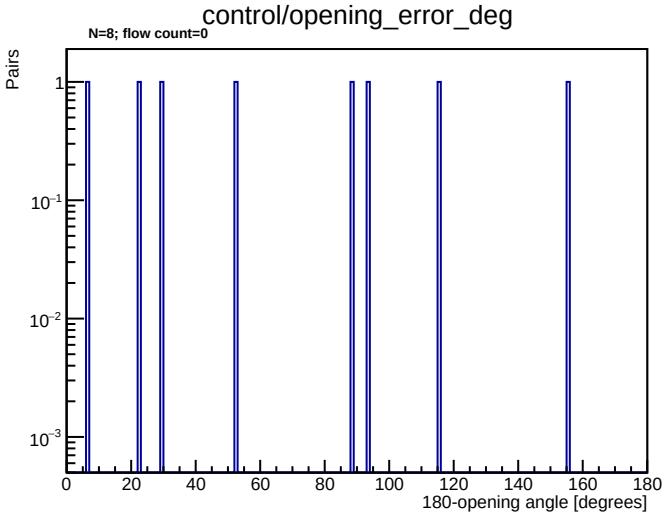

baseline_reflected/tar_phase_vs_tar_dr
N=9; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²

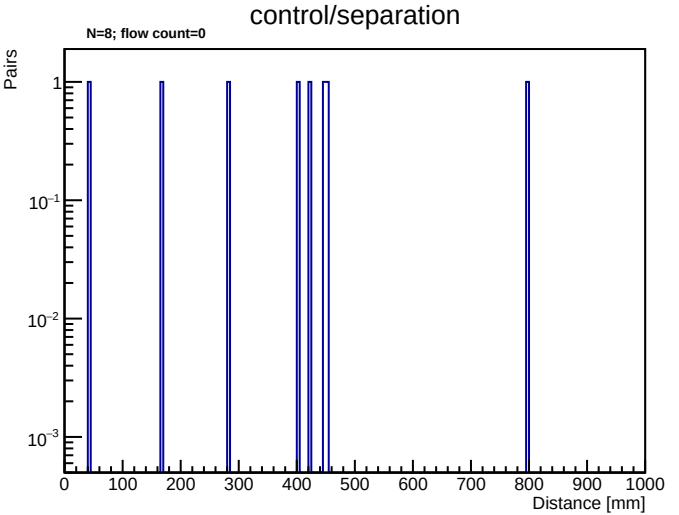

baseline_reflected/tar_pitch
N=9; flow count=0
Tracks
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

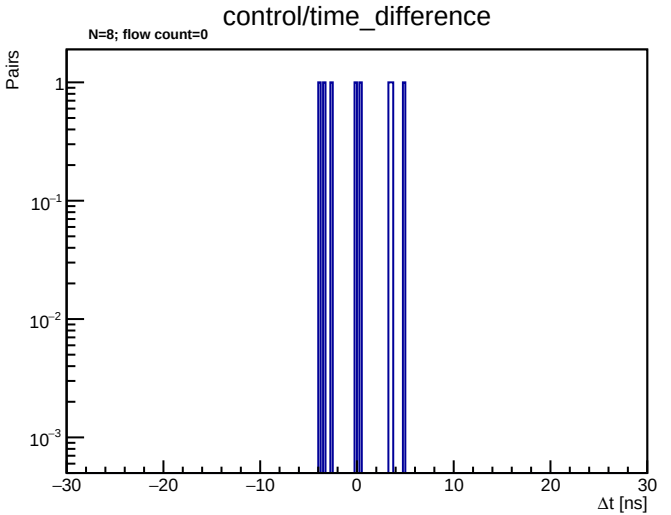

baseline_reflected/tar_radius
N=9; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

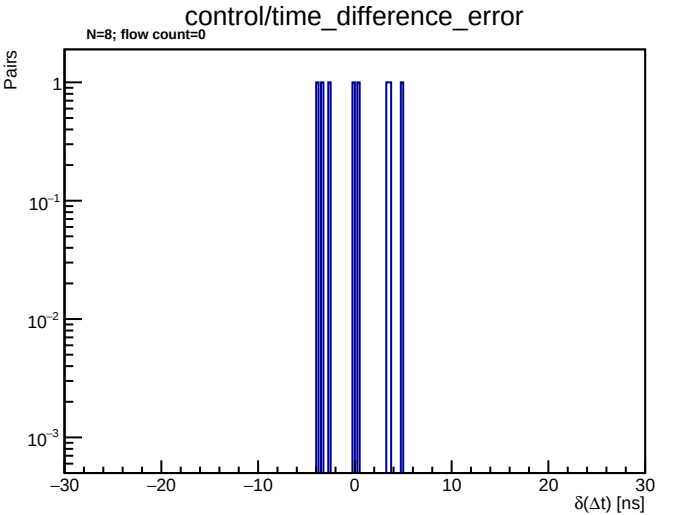

control/efficiency
N=66; flow count=0
Events
10
1
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Missing product
Category

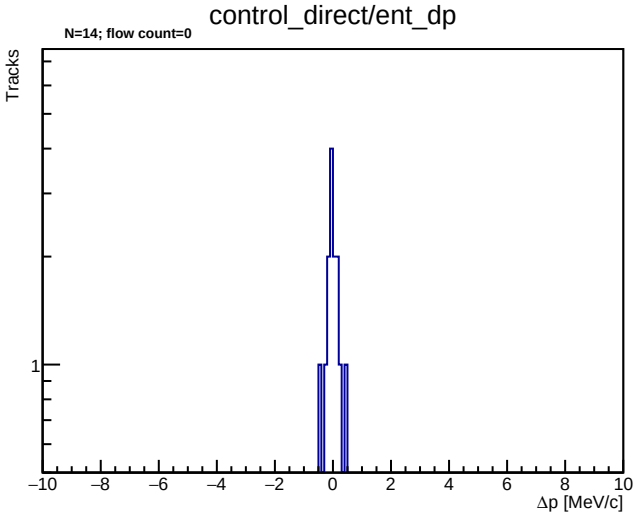

control/failure_flags
N=16; flow count=0
Tracks
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimary
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleReferenceEntrances
MissingReferenceEntrance
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProductionBranch
AmbiguousProductionBranch
OutsideTarget
CrossingSearchPlane
BoundaryProximity
ConfigurationBranch
ConfigurationDisabled
Exception
Flag

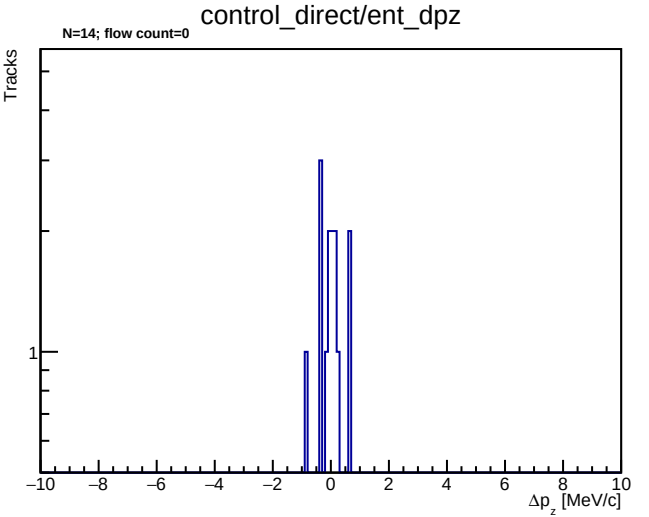

control/opening_error_deg
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

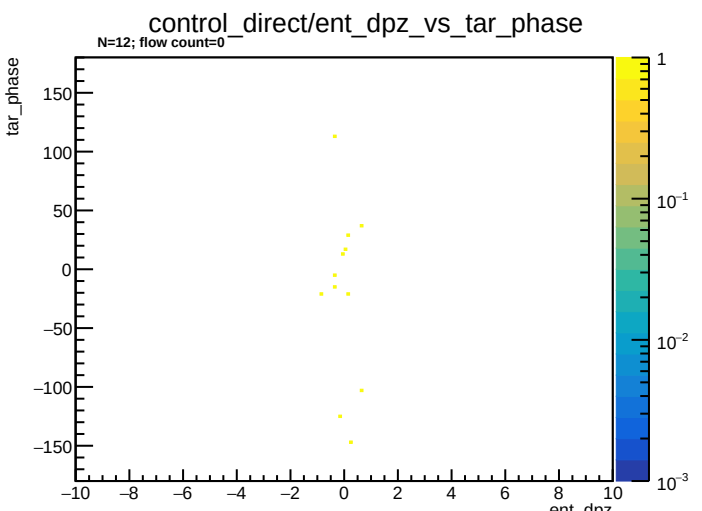

control/separation
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

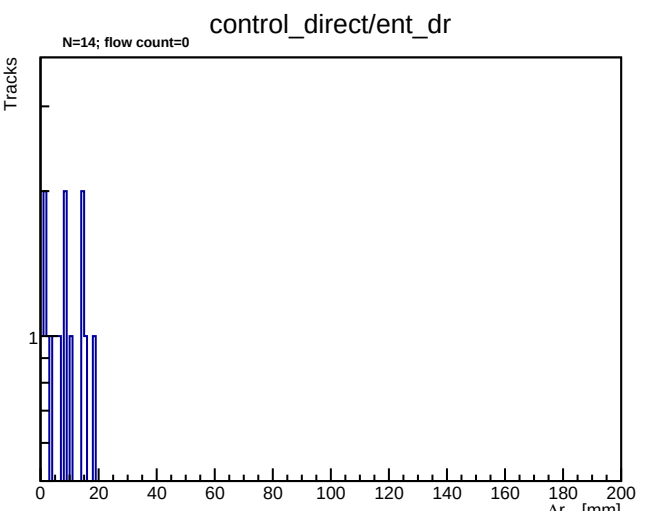

control/time_difference
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
–30
–20
–10
0
10
20
30
Δt [ns]

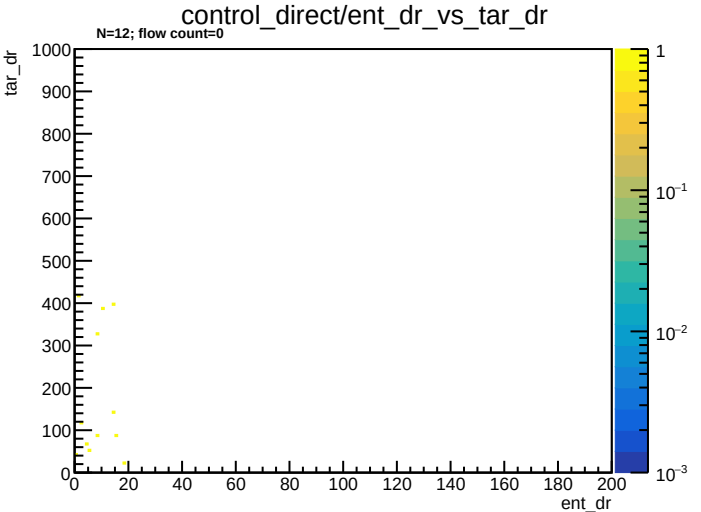

control/time_difference_error
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
–30
–20
–10
0
10
20
30
δ(Δt) [ns]

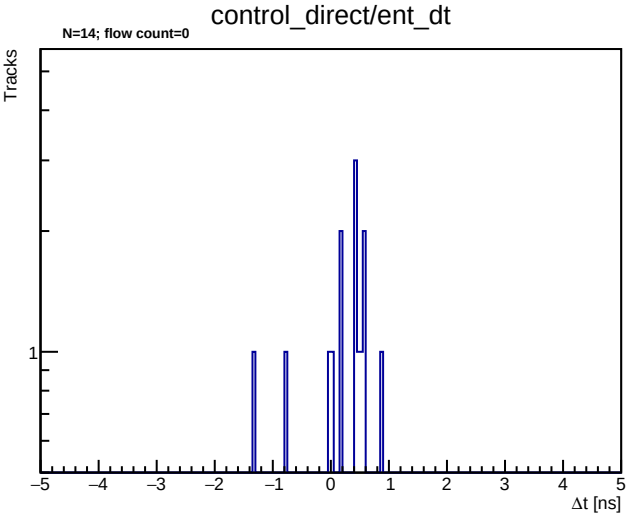

control_direct/ent_dp
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

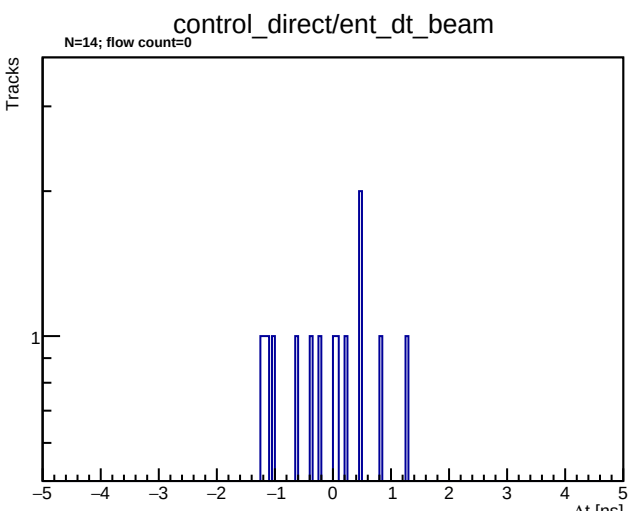

control_direct/ent_dpz
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

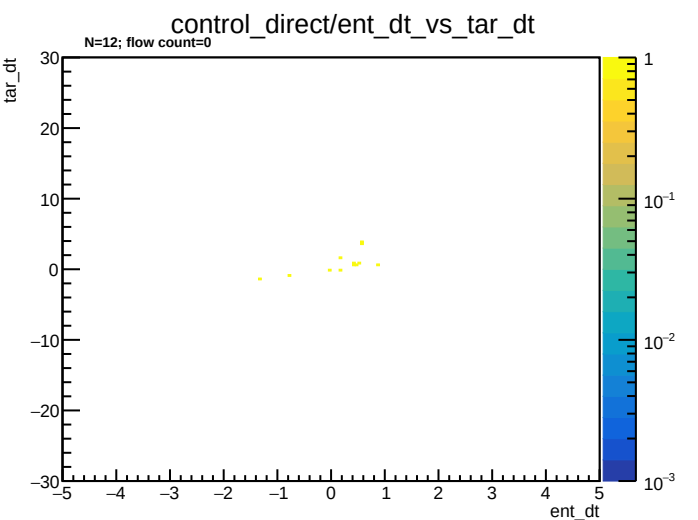

control_direct/ent_dpz_vs_tar_phase
N=12; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−10 −8 −6 −4 −2 0 2 4 6 8 10
ent_dpz
1
10^−1
10^−2
10^−3

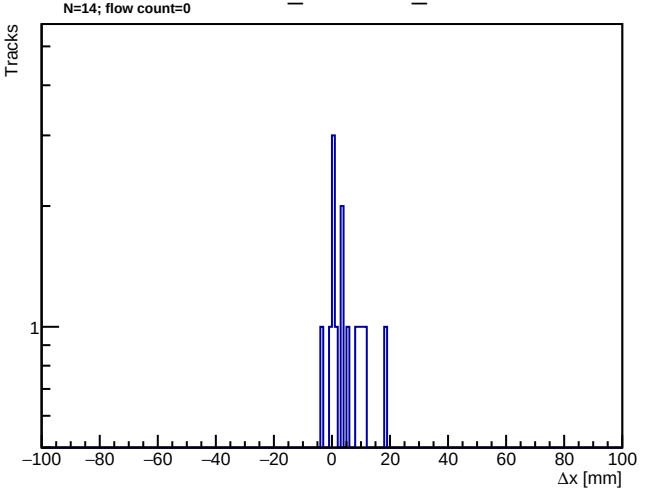

control_direct/ent_dr
N=14; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δr_xy [mm]

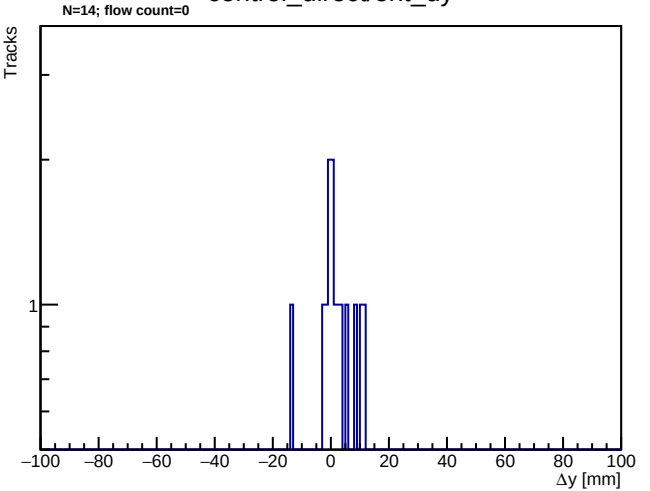

control_direct/ent_dr_vs_tar_dr
N=12; flow count=0
tar_dr
ent_dr

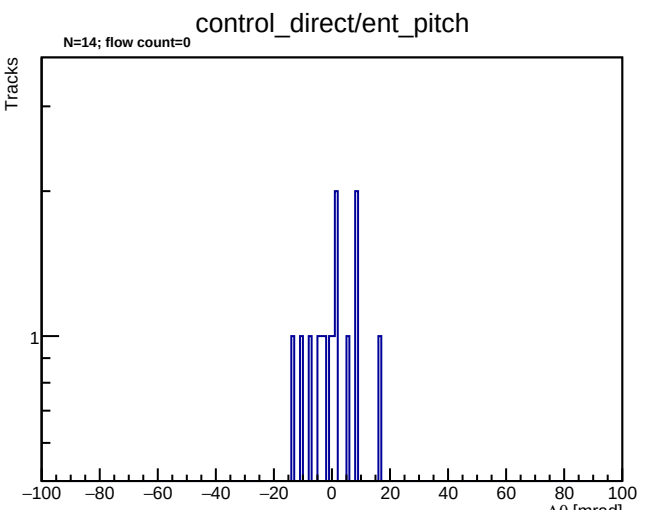

control_direct/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

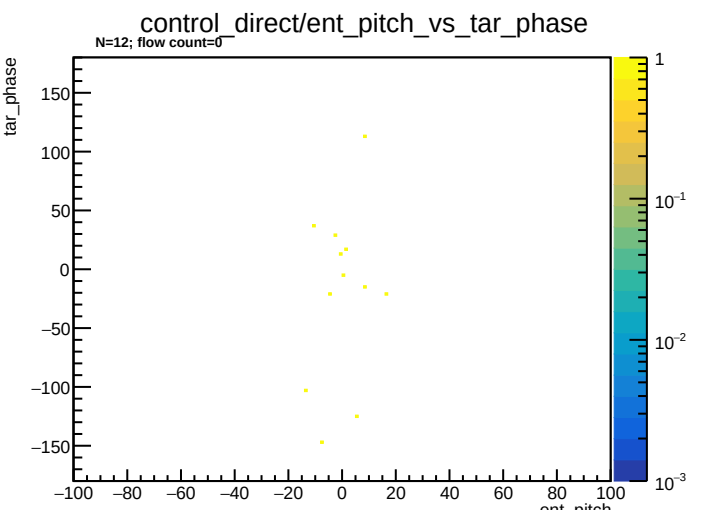

control_direct/ent_dt_beam
N=14; flow count=0
Tracks
Δt [ns]

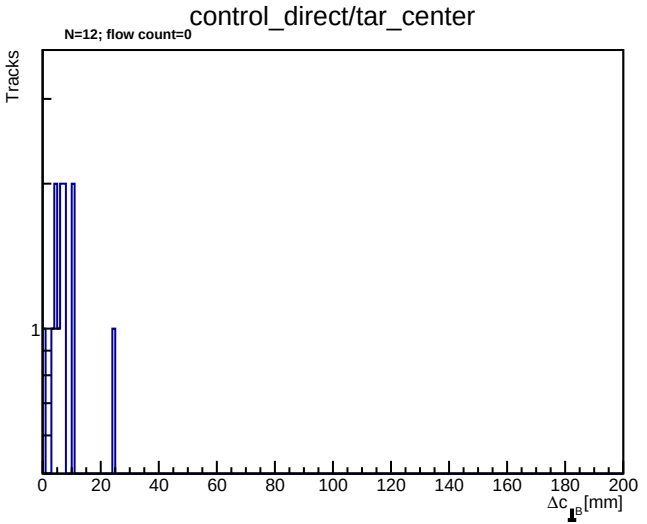

control_direct/ent_dt_vs_tar_dt
N=12; flow count=0
tar_dt
ent_dt

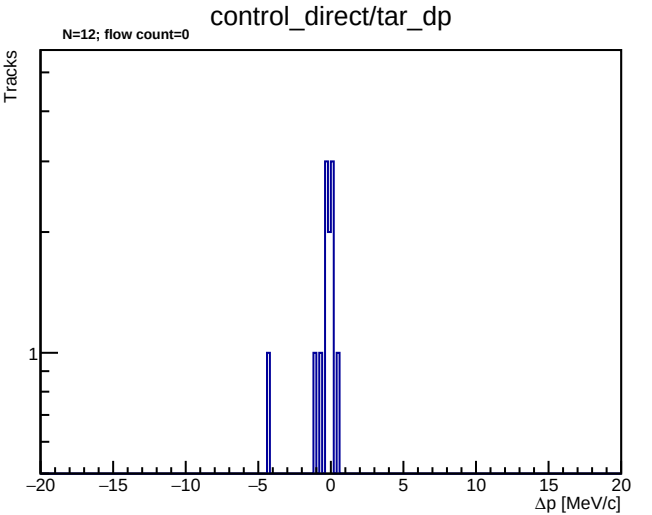

control_direct/ent_dx
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

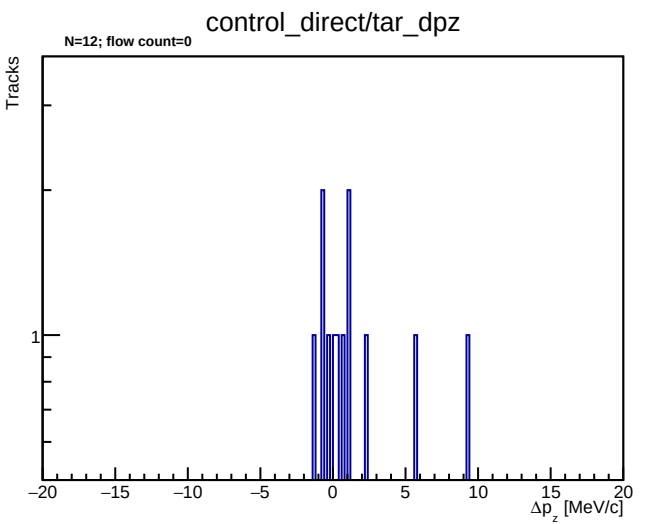

control_direct/ent_dy
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

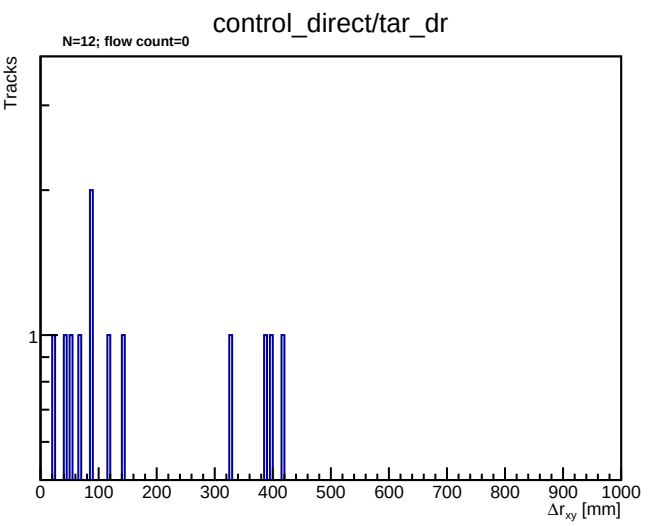

control_direct/ent_pitch
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

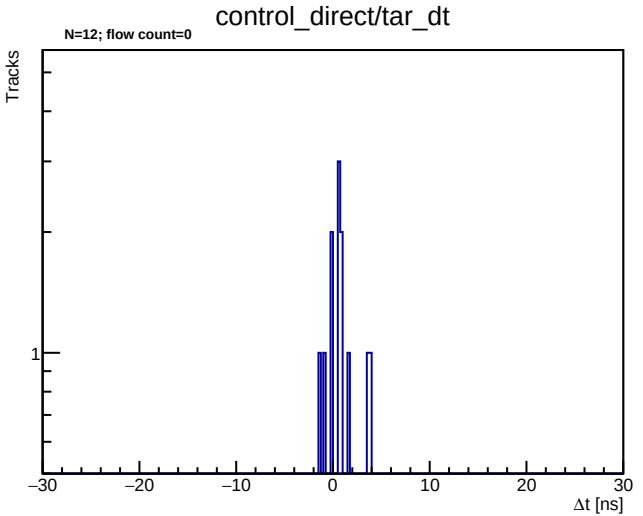

control_direct/ent_pitch_vs_tar_phase
N=12; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²
10⁻³

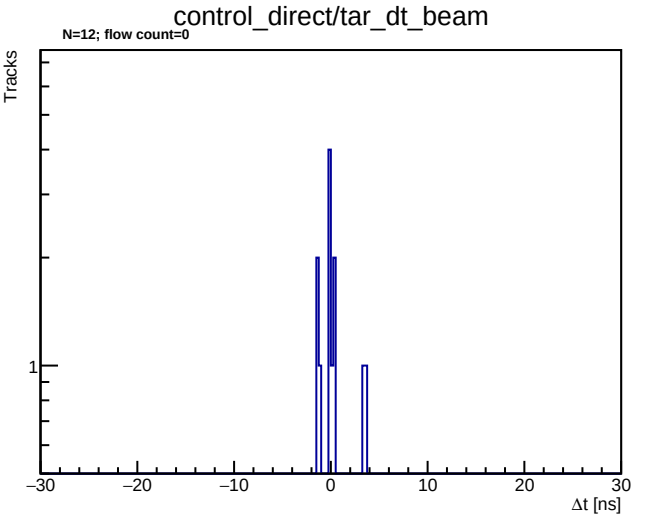

control_direct/tar_center
N=12; flow count=0
Tracks
1
0
20
40
60
80
100
120
140
160
180
200
Δc_B [mm]

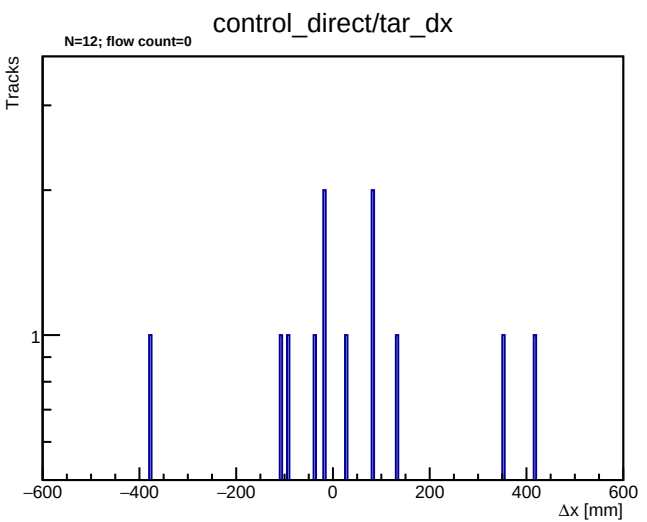

control_direct/tar_dp
N=12; flow count=0
Tracks
1
−20
−15
−10
−5
0
5
10
15
20
Δp [MeV/c]

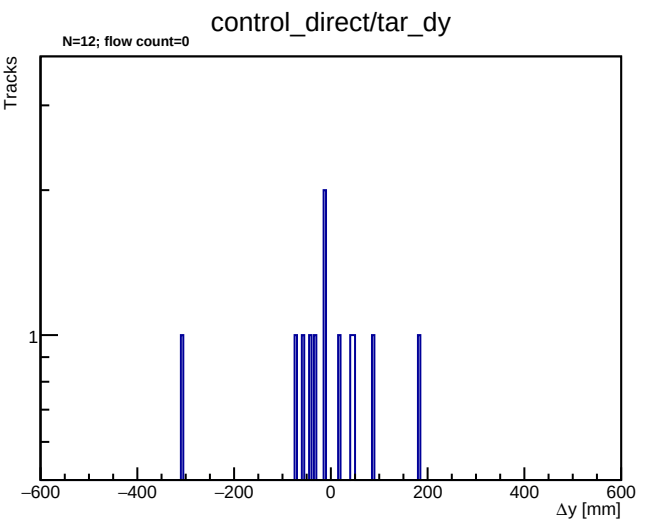

control_direct/tar_dpz
N=12; flow count=0
Tracks
1
−20
−15
−10
−5
0
5
10
15
20
Δp_z [MeV/c]

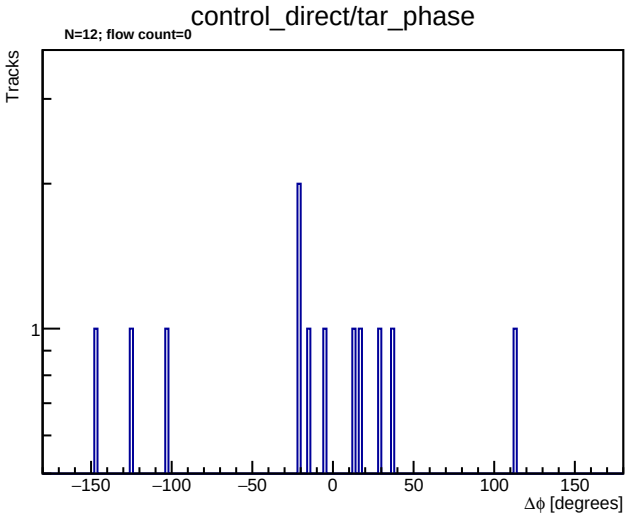

control_direct/tar_dr
N=12; flow count=0
Tracks
1
0
100
200
300
400
500
600
700
800
900
1000
Δr_xy [mm]

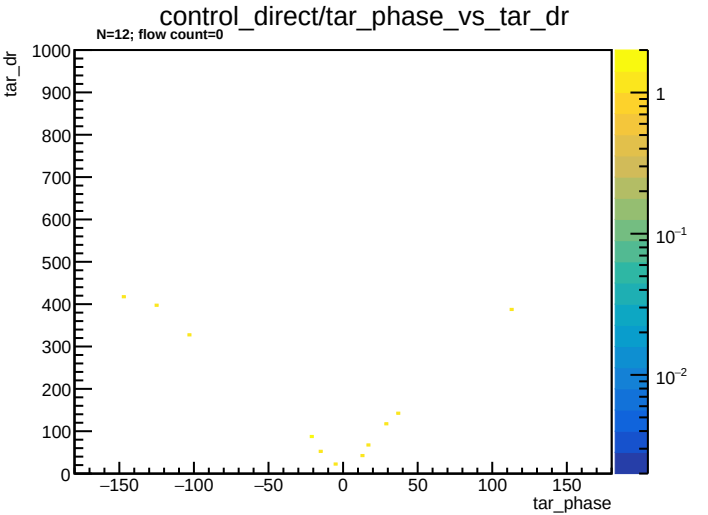

control_direct/tar_dt
N=12; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

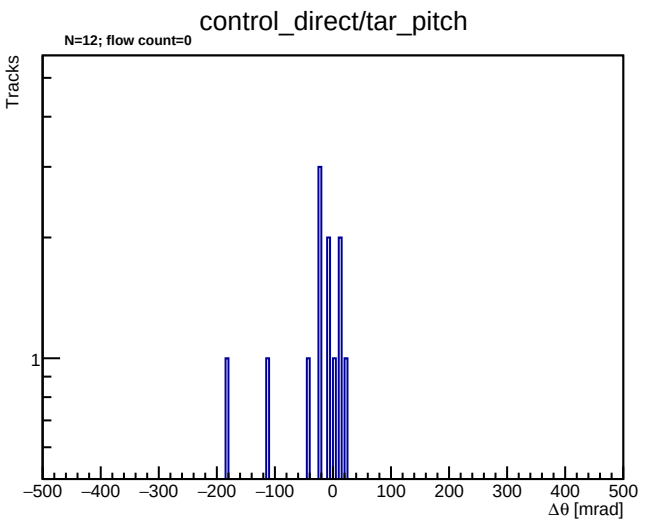

control_direct/tar_dt_beam
N=12; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

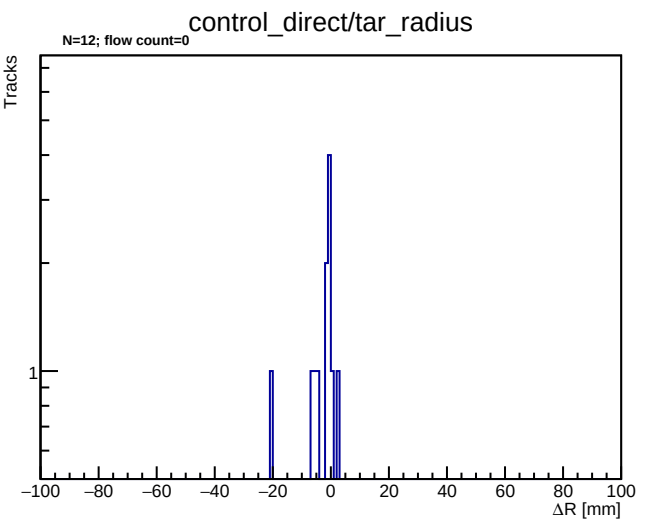

control_direct/tar_dx
N=12; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δx [mm]

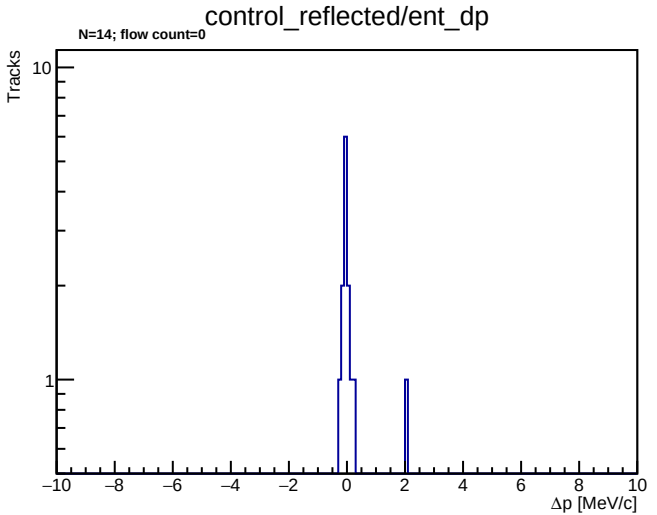

control_direct/tar_dy
N=12; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δy [mm]

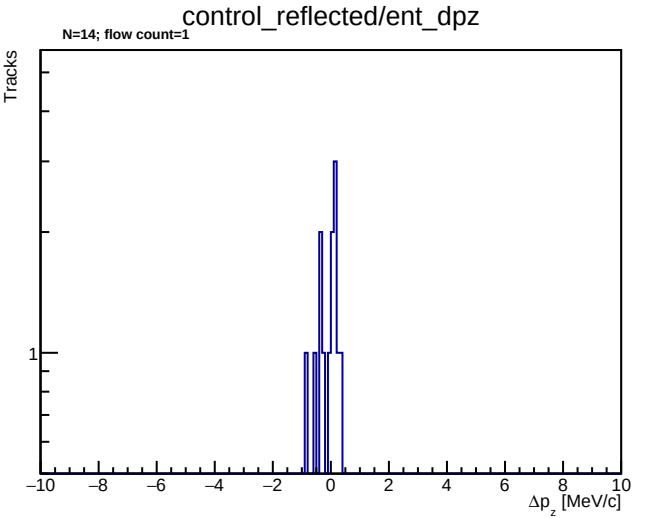

control_direct/tar_phase
N=12; flow count=0
Tracks
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

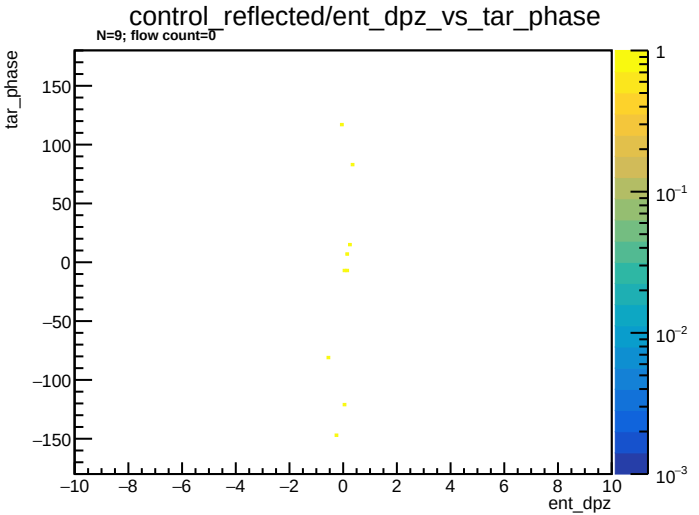

control_direct/tar_phase_vs_tar_dr
N=12; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²

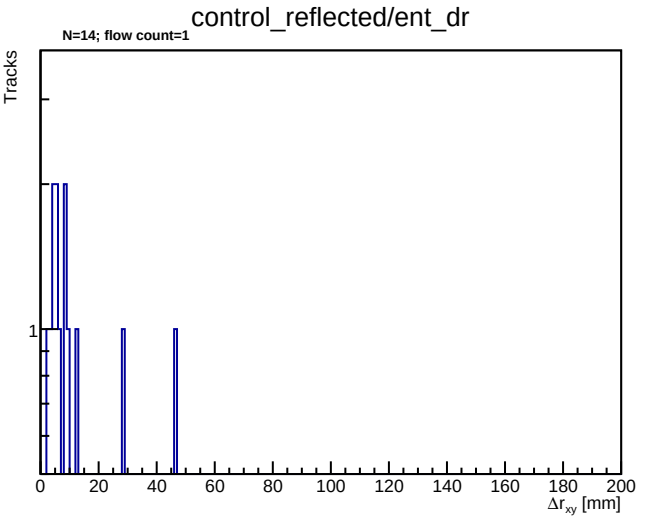

control_direct/tar_pitch
N=12; flow count=0
Tracks
1
−500
−400
−300
−200
−100
0
100
200
300
400
500
Δθ [mrad]

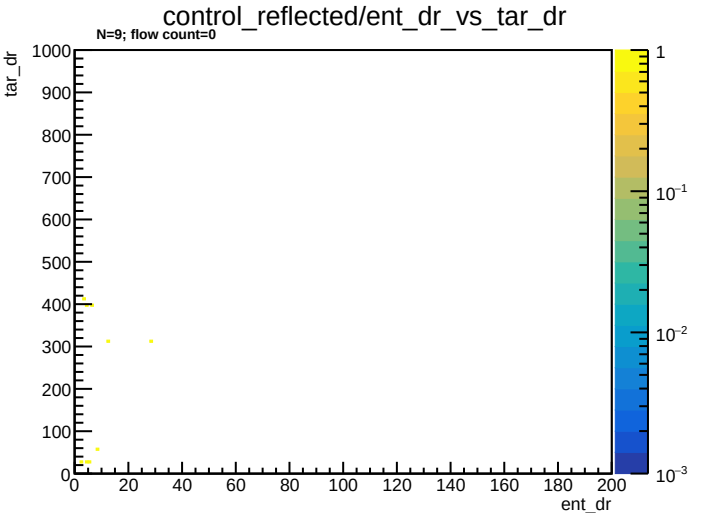

control_direct/tar_radius
N=12; flow count=0
Tracks
1
−100
−80
−60
−40
−20
0
20
40
60
80
100
ΔR [mm]

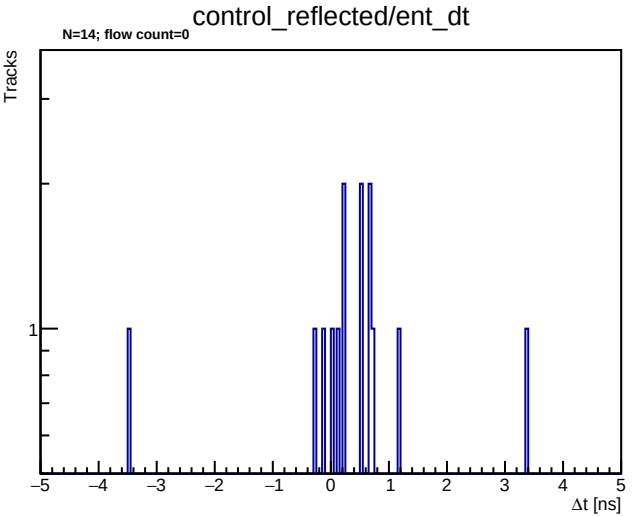

control_reflected/ent_dp
N=14; flow count=0
Tracks
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

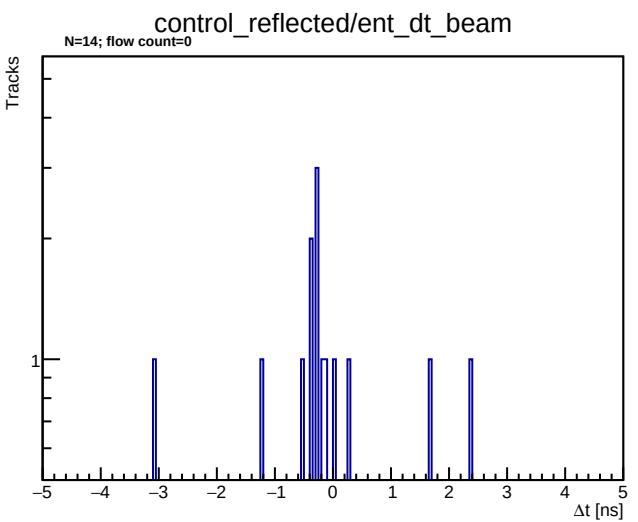

control_reflected/ent_dpz
N=14; flow count=1
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

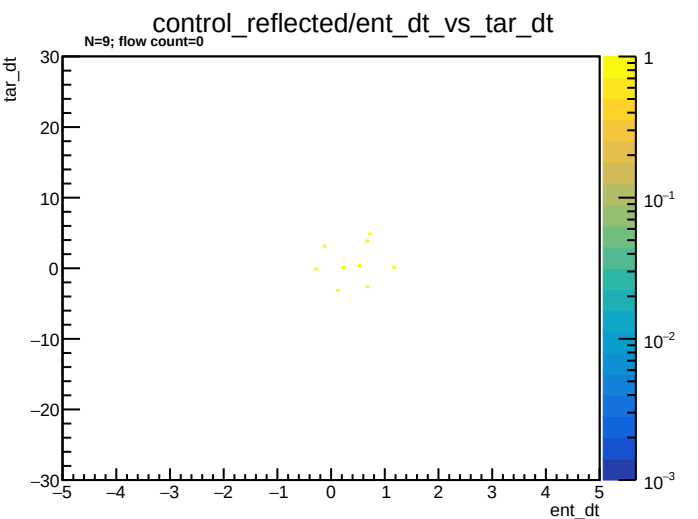

control_reflected/ent_dpz_vs_tar_phase
N=9; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−10 −8 −6 −4 −2 0 2 4 6 8 10
ent_dpz
1
10^−1
10^−2
10^−3

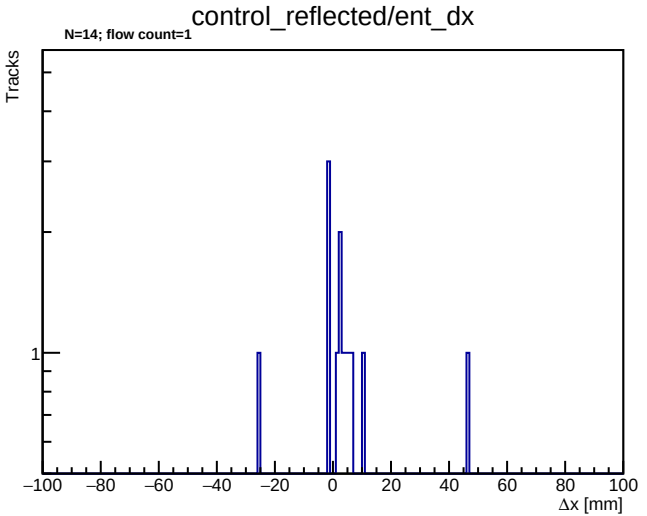

control_reflected/ent_dr
N=14; flow count=1
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δr_xy [mm]

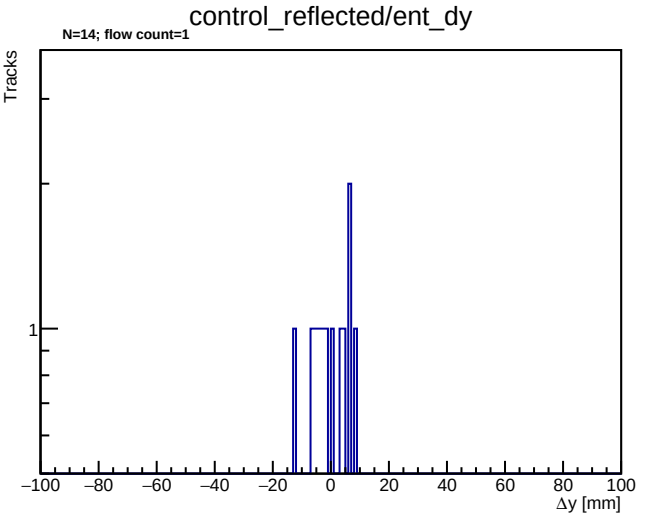

control_reflected/ent_dr_vs_tar_dr
N=9; flow count=0
tar_dr
ent_dr

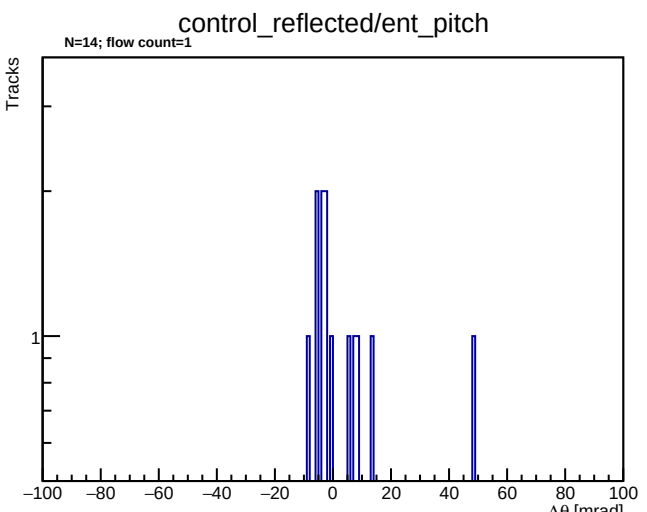

control_reflected/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

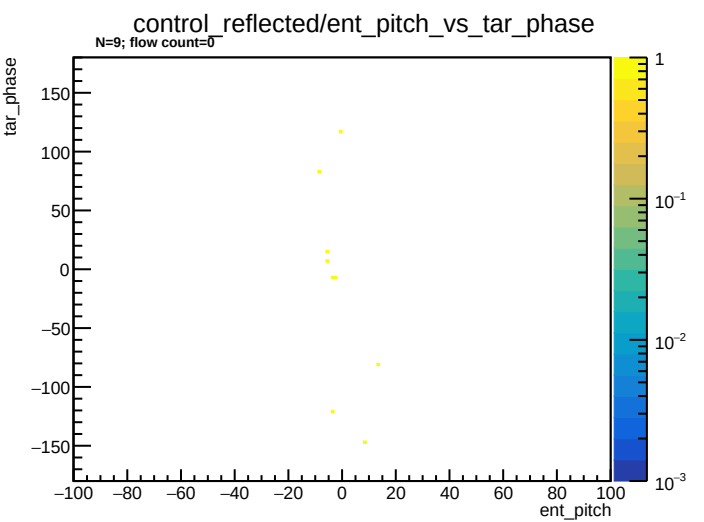

control_reflected/ent_dt_beam
N=14; flow count=0
Tracks
Δt [ns]

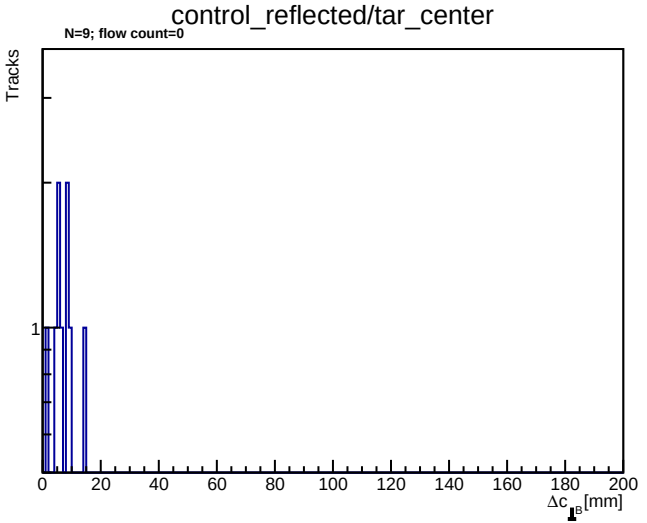

control_reflected/ent_dt_vs_tar_dt
N=9; flow count=0
tar_dt
ent_dt

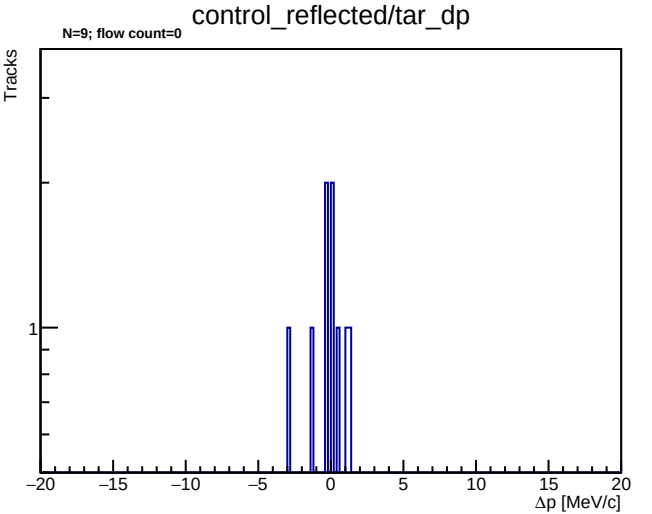

control_reflected/ent_dx
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

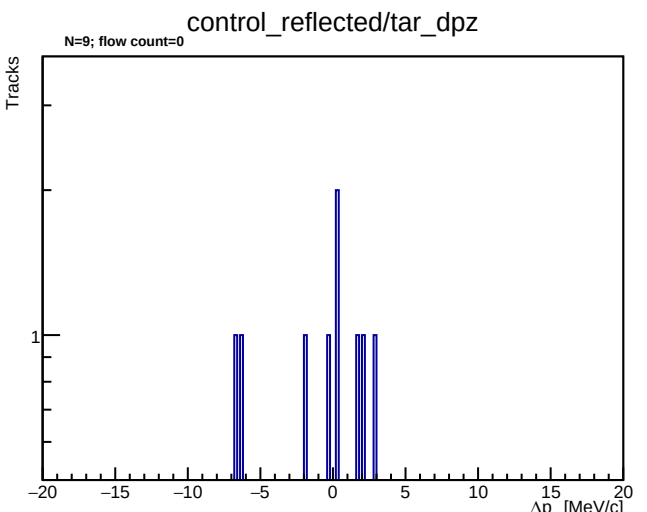

control_reflected/ent_dy
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

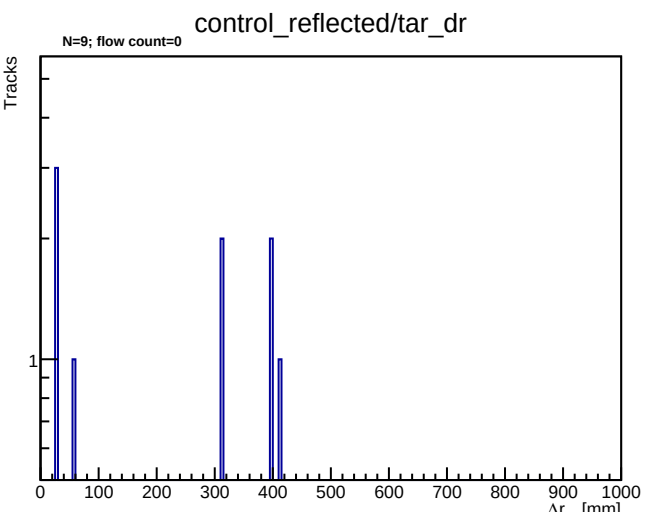

control_reflected/ent_pitch
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

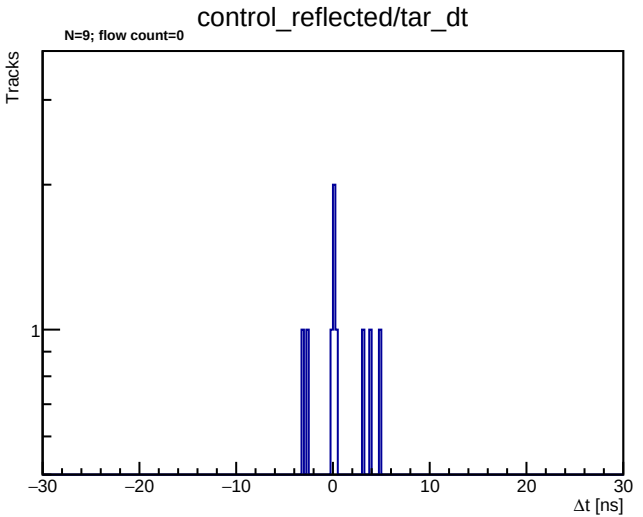

control_reflected/ent_pitch_vs_tar_phase
N=9; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²
10⁻³

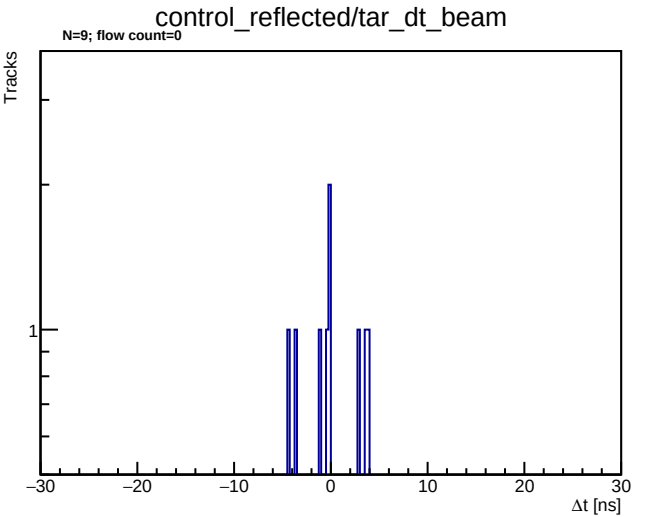

control_reflected/tar_center
N=9; flow count=0
Tracks
1
0
20
40
60
80
100
120
140
160
180
200
Δc_B [mm]

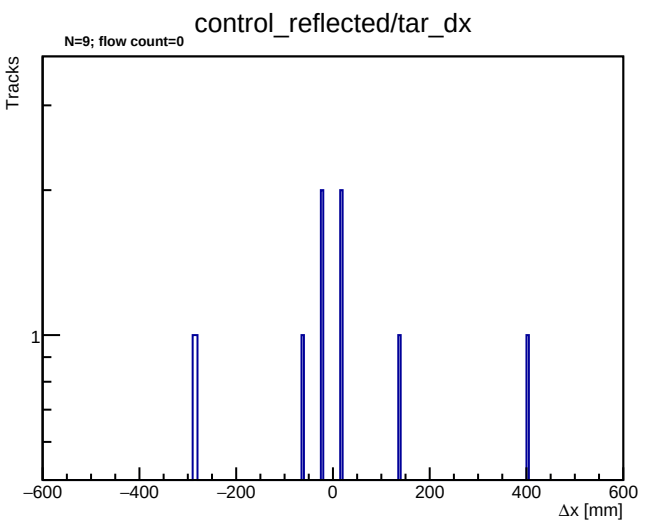

control_reflected/tar_dp
N=9; flow count=0
Tracks
1
−20
−15
−10
−5
0
5
10
15
20
Δp [MeV/c]

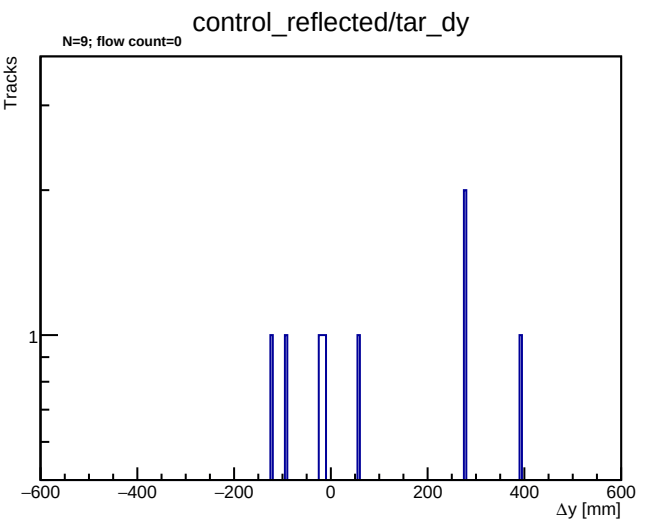

control_reflected/tar_dpz
N=9; flow count=0
Tracks
1
−20
−15
−10
−5
0
5
10
15
20
Δp_z [MeV/c]

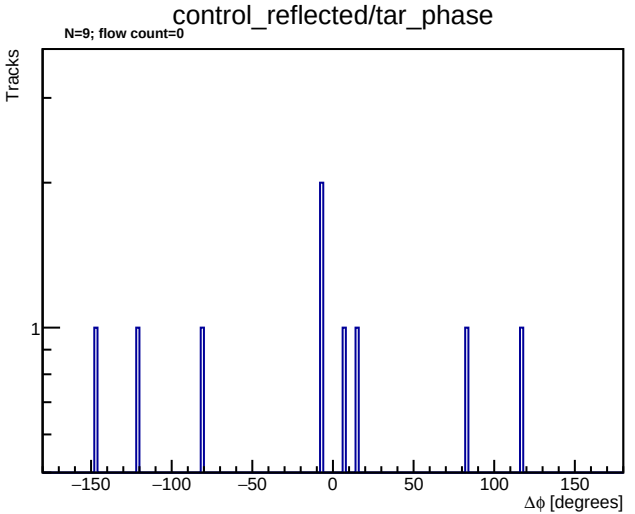

control_reflected/tar_dr
N=9; flow count=0
Tracks
1
0
100
200
300
400
500
600
700
800
900
1000
Δr_xy [mm]

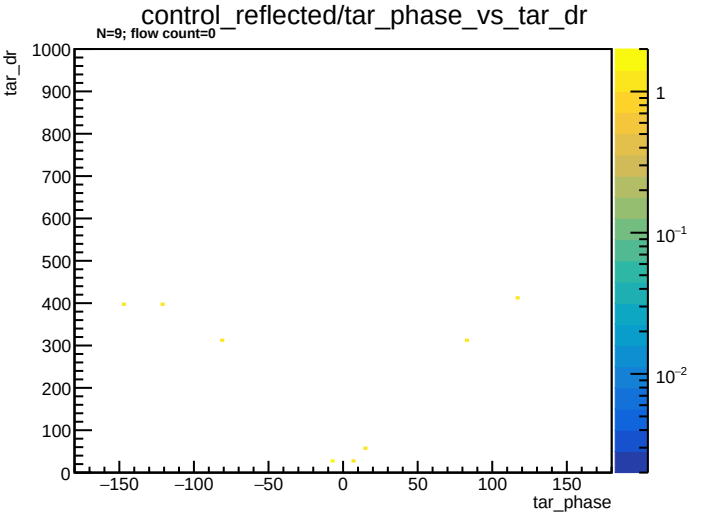

control_reflected/tar_dt
N=9; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

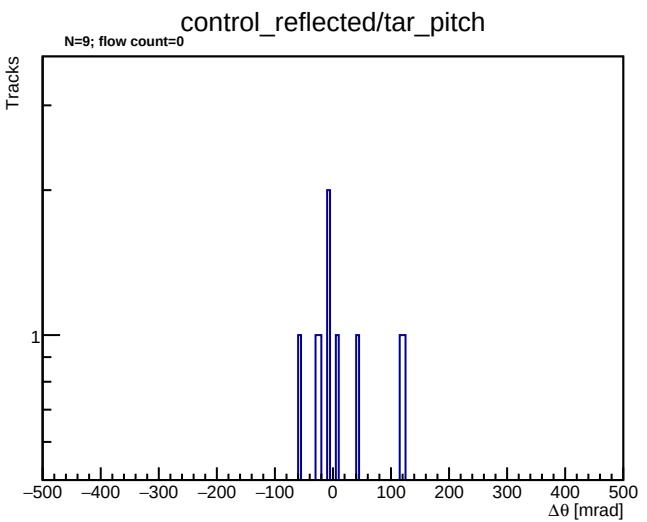

control_reflected/tar_dt_beam
N=9; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

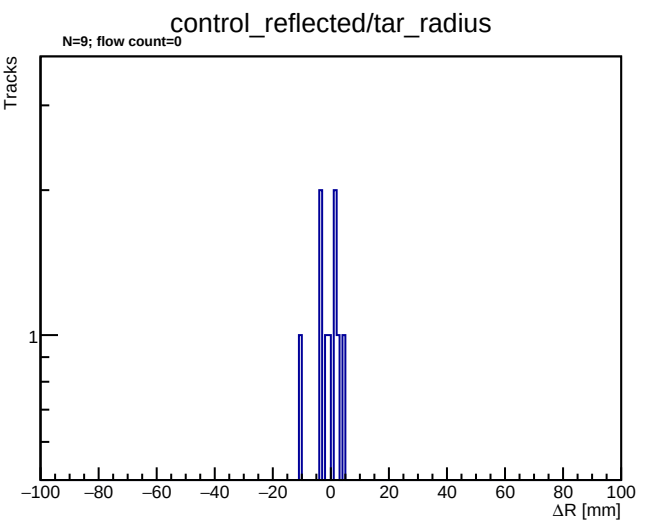

control_reflected/tar_dx
N=9; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δx [mm]

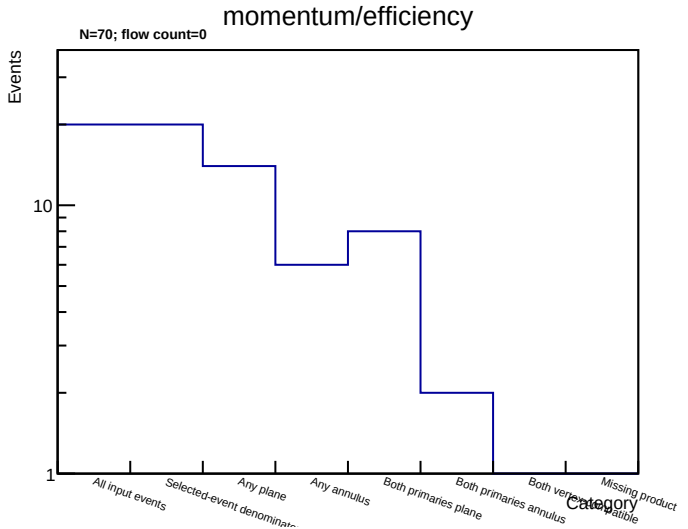

control_reflected/tar_dy
N=9; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δy [mm]

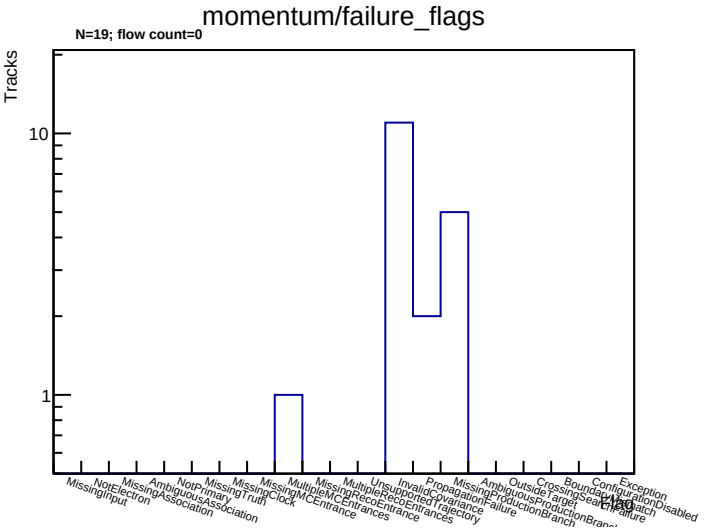

control_reflected/tar_phase
N=9; flow count=0
Tracks
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

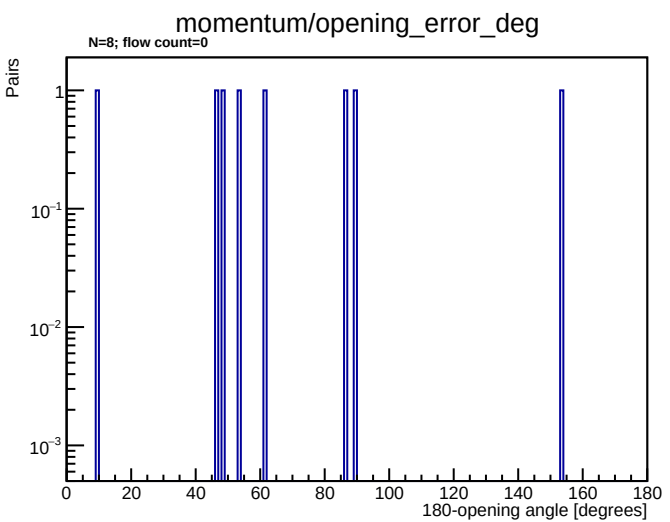

control_reflected/tar_phase_vs_tar_dr
N=9; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²

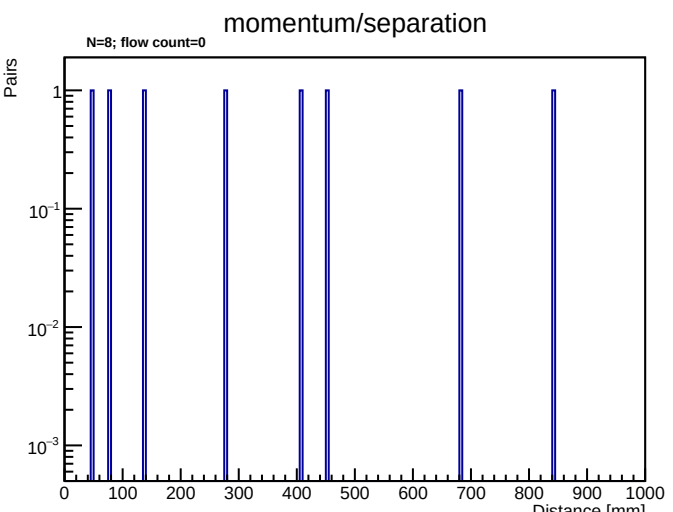

control_reflected/tar_pitch
N=9; flow count=0
Tracks
1
−500
−400
−300
−200
−100
0
100
200
300
400
500
Δθ [mrad]

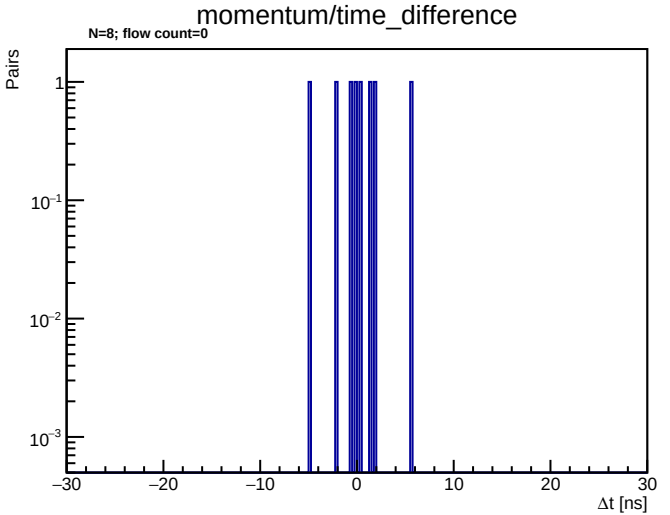

control_reflected/tar_radius
N=9; flow count=0
Tracks
1
−100
−80
−60
−40
−20
0
20
40
60
80
100
ΔR [mm]

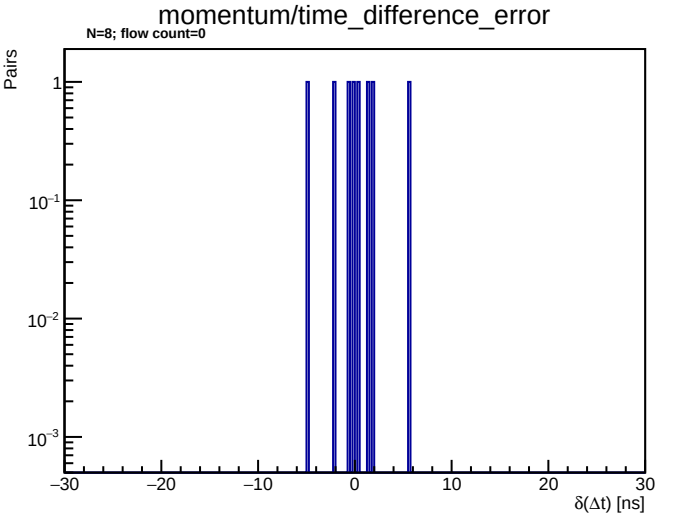

momentum/efficiency
N=70; flow count=0
Events
10
1
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both ver
Missing product
Category

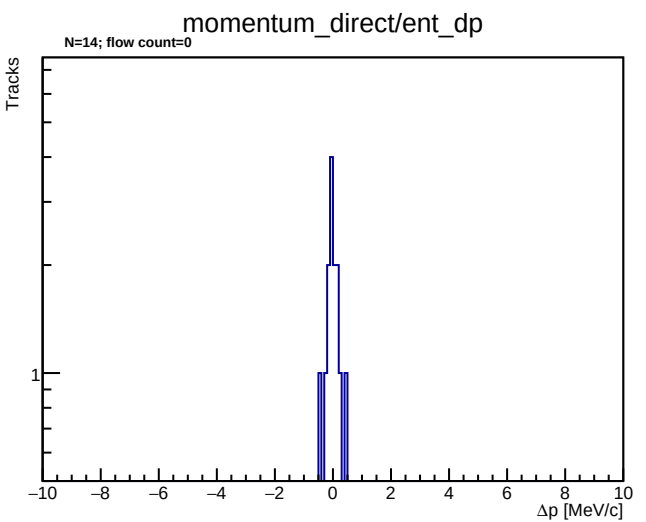

momentum/failure_flags
N=19; flow count=0
Tracks
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimary
MissingTruth
MissingClock
MultipleMCEntrance
MissingRecoEntrances
MultipleRecoEntrance
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeed
BoundaryBranch
ConfigurationDisabled
Exception
Failed

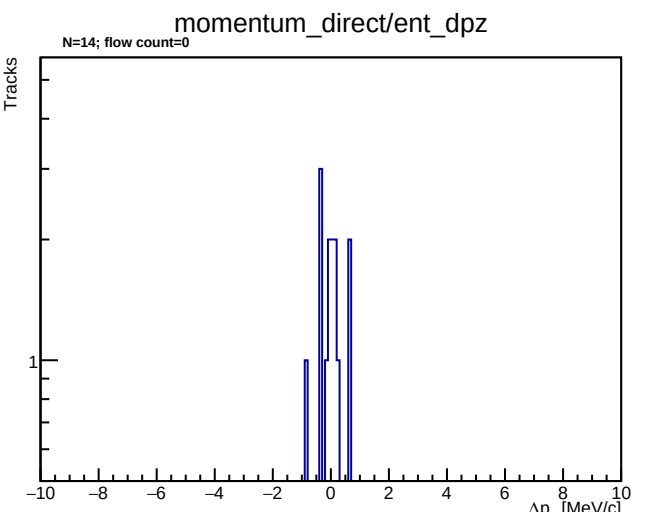

momentum/opening_error_deg
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

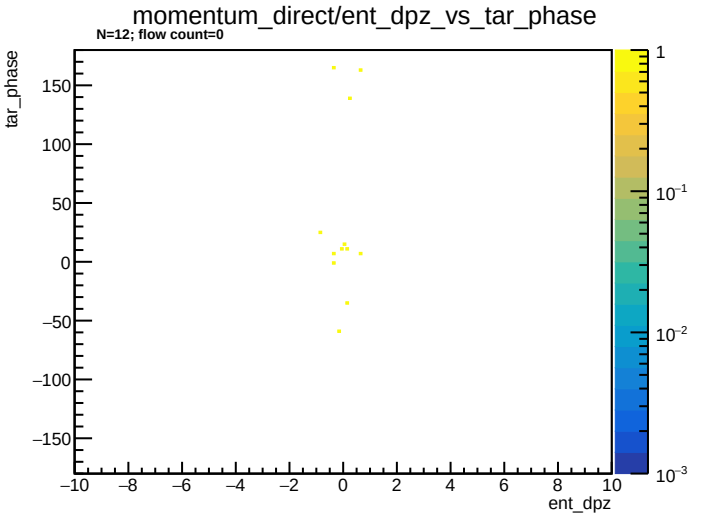

momentum/separation
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

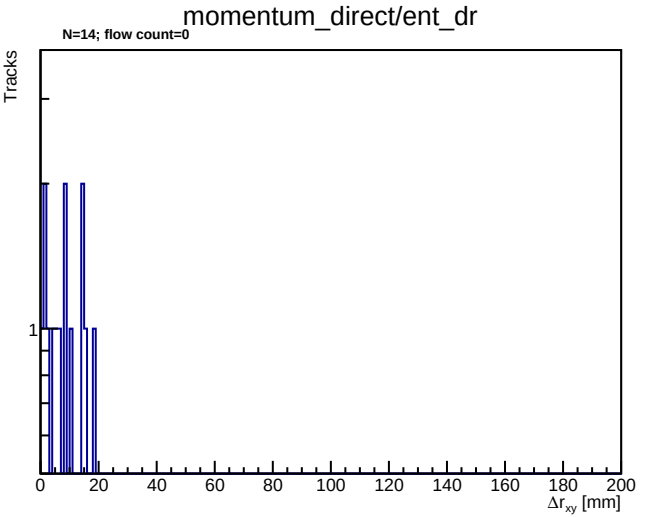

momentum/time_difference
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−30 −20 −10 0 10 20 30
Δt [ns]

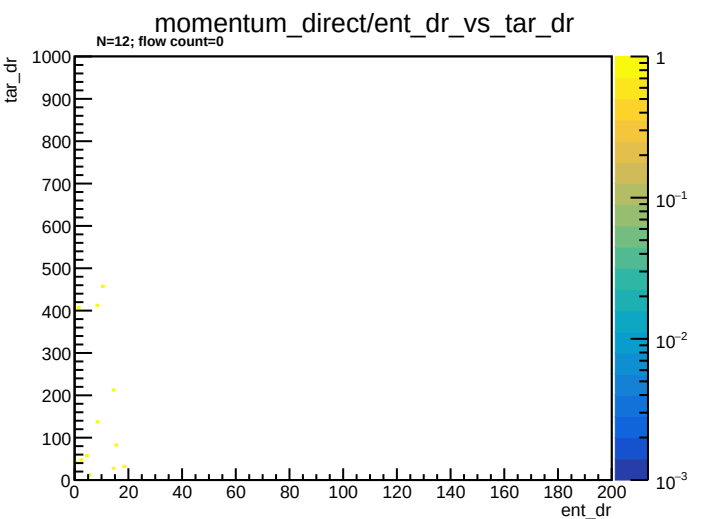

momentum/time_difference_error
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−30 −20 −10 0 10 20 30
δ(Δt) [ns]

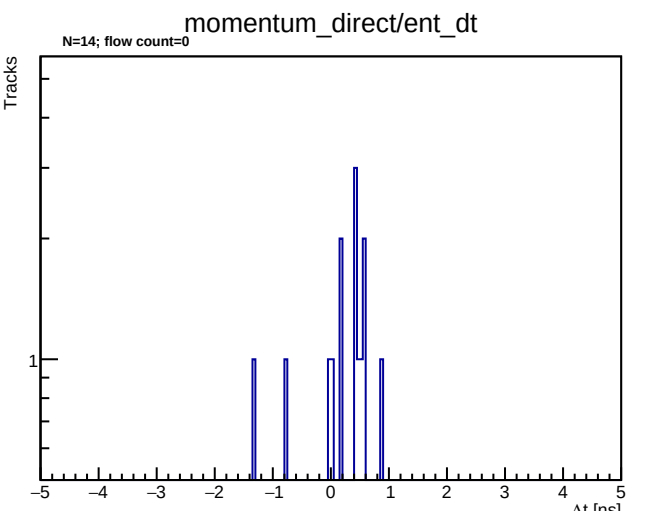

momentum_direct/ent_dp
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

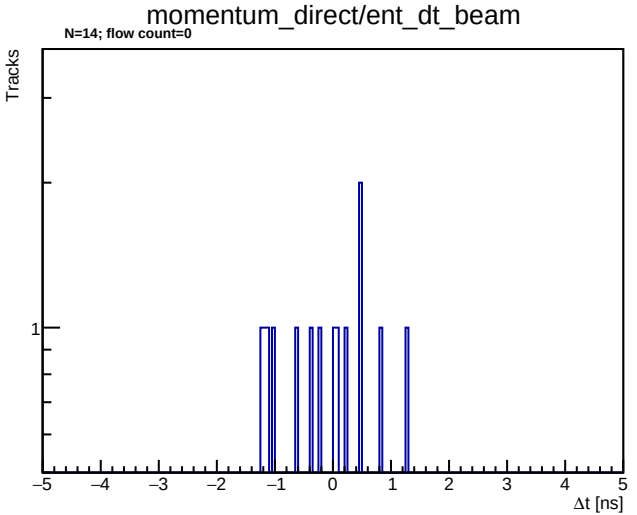

momentum_direct/ent_dpz
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp$_z$ [MeV/c]

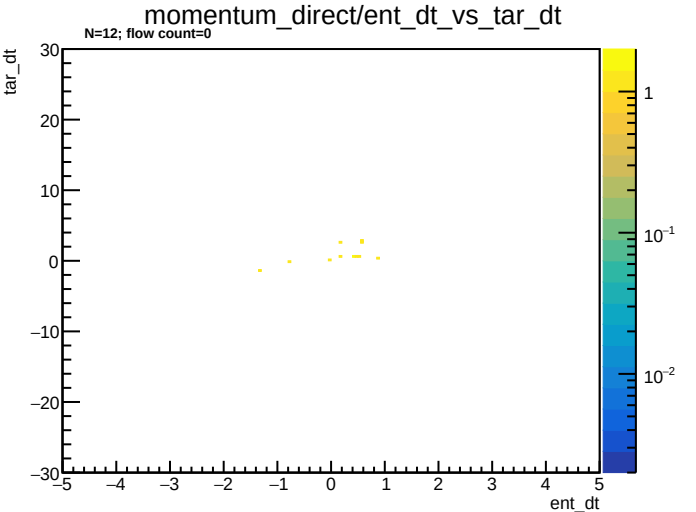

momentum_direct/ent_dpz_vs_tar_phase
N=12; flow count=0
tar_phase
ent_dpz

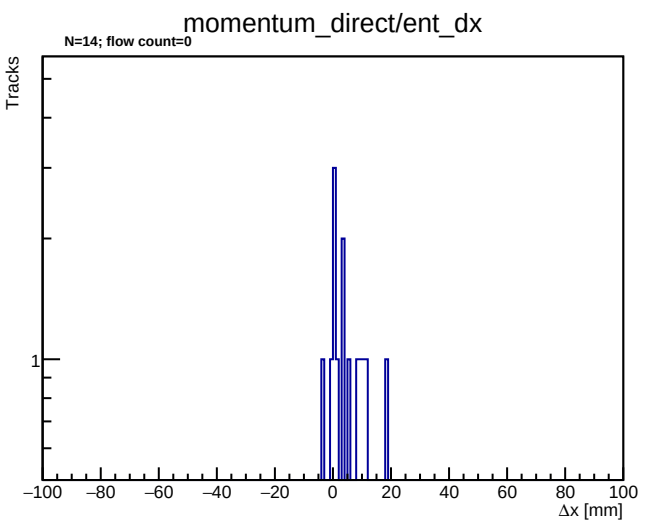

momentum_direct/ent_dr
N=14; flow count=0
Tracks
Δr_xy [mm]

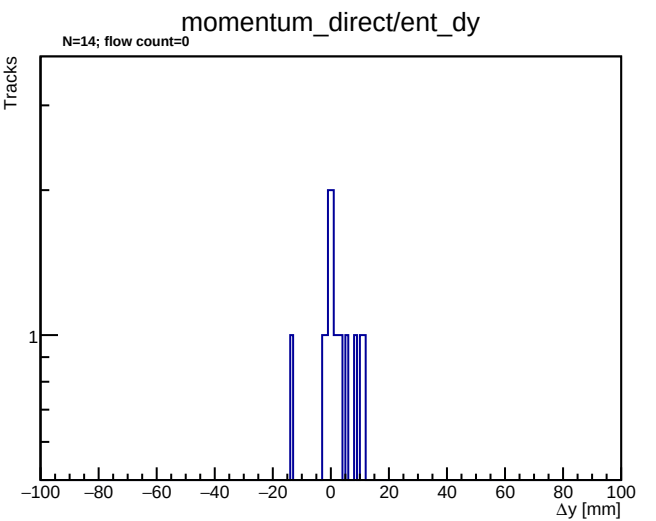

momentum_direct/ent_dr_vs_tar_dr
N=12; flow count=0
tar_dr
ent_dr

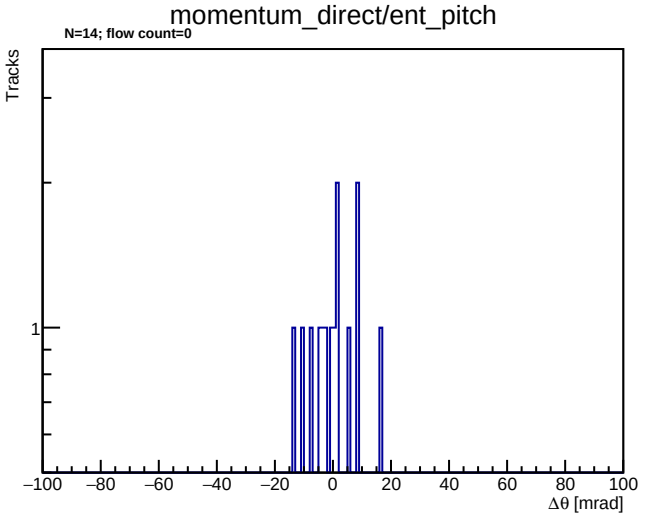

momentum_direct/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

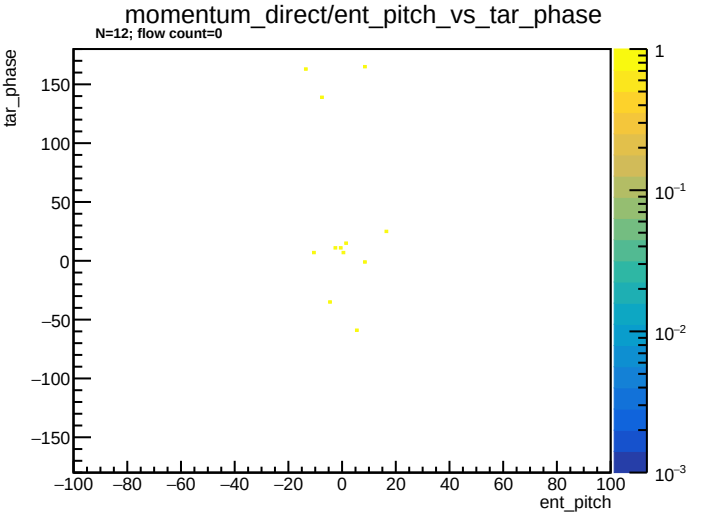

momentum_direct/ent_dt_beam
N=14; flow count=0
Tracks
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

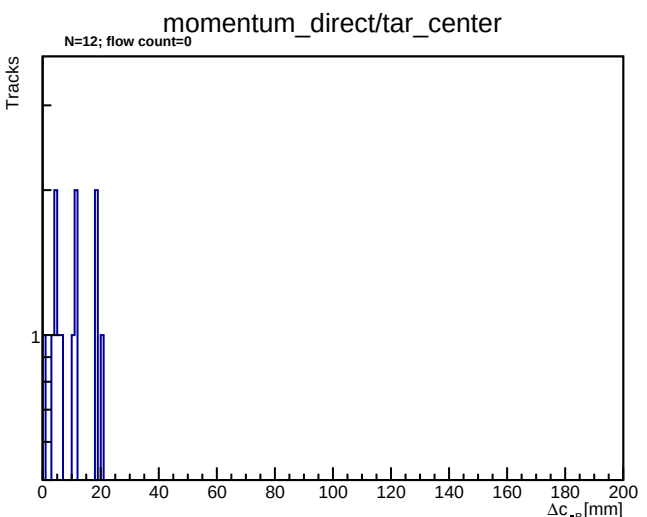

momentum_direct/ent_dt_vs_tar_dt
N=12; flow count=0
tar_dt
30 20 10 0 −10 −20 −30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10⁻¹
10⁻²

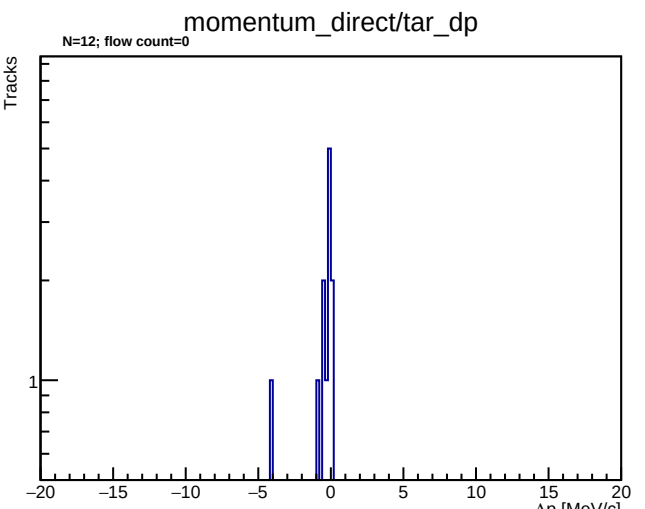

momentum_direct/ent_dx
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

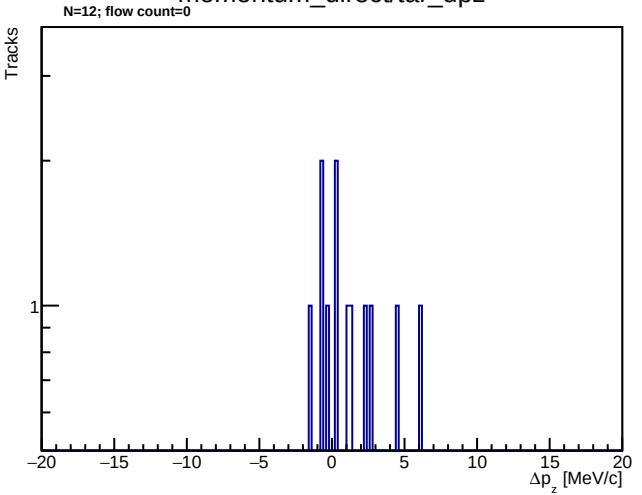

momentum_direct/ent_dy
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

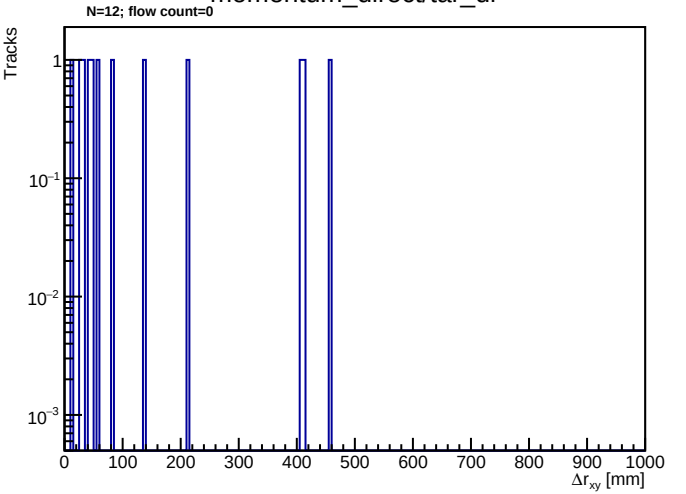

momentum_direct/ent_pitch
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

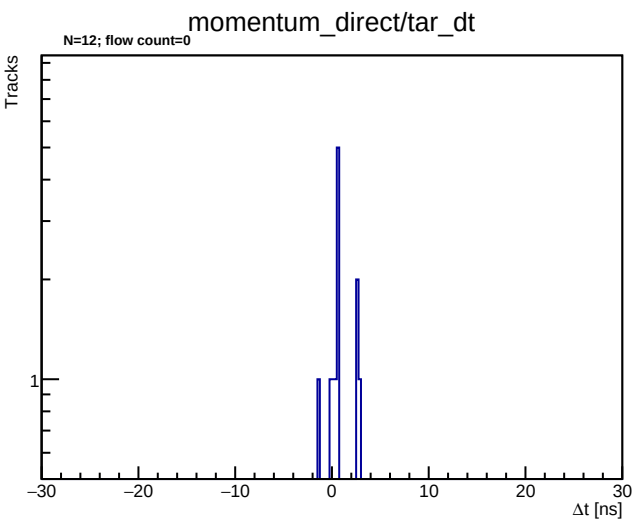

momentum_direct/ent_pitch_vs_tar_phase
N=12; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
$10^{-1}$
$10^{-2}$
$10^{-3}$

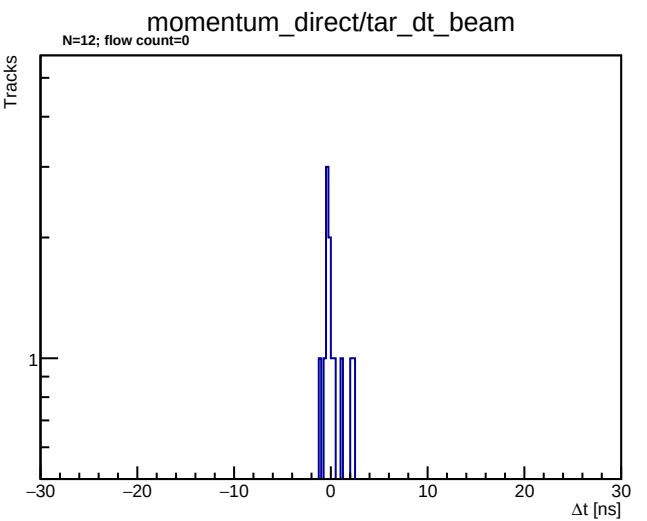

momentum_direct/tar_center
N=12; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

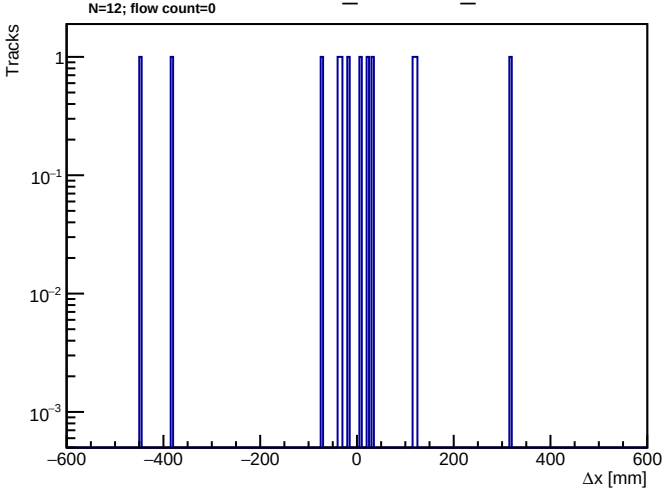

momentum_direct/tar_dp
N=12; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

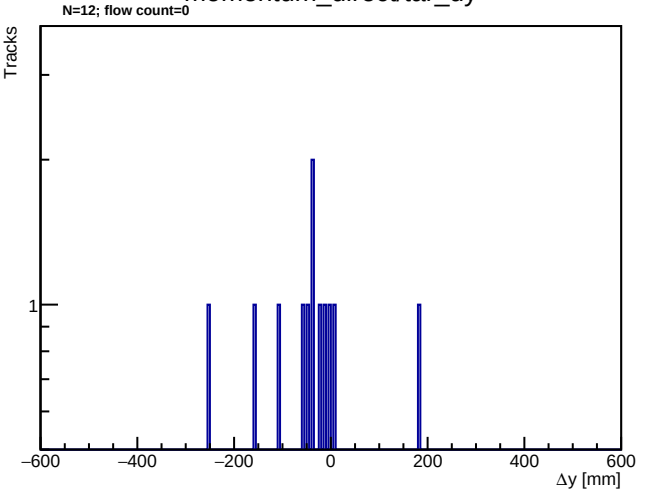

momentum_direct/tar_dpz
N=12; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

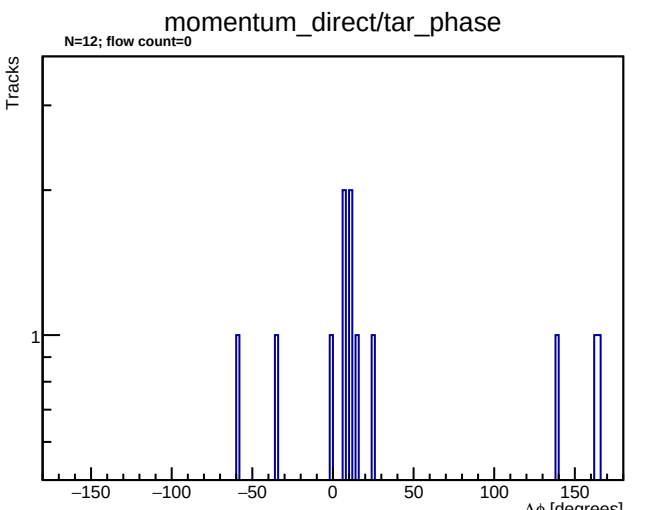

momentum_direct/tar_dr
N=12; flow count=0
Tracks
1
10^-1
10^-2
10^-3
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

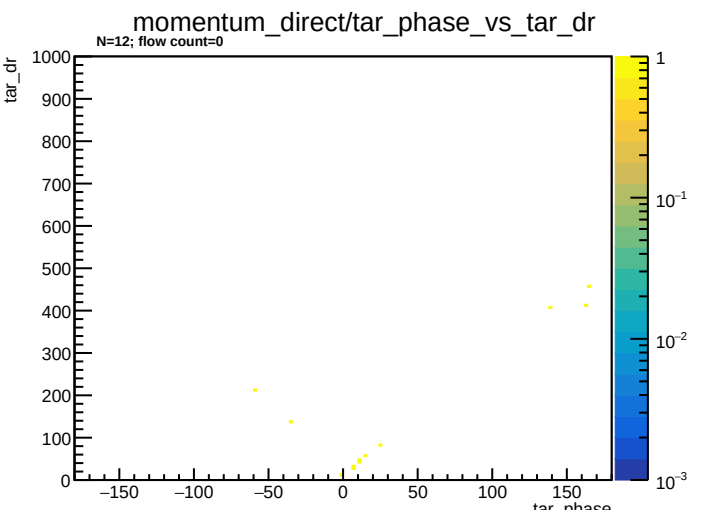

momentum_direct/tar_dt
N=12; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

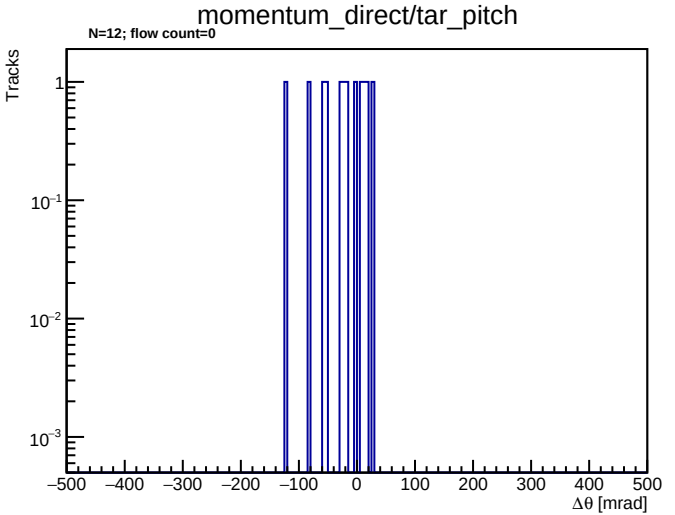

momentum_direct/tar_dt_beam
N=12; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

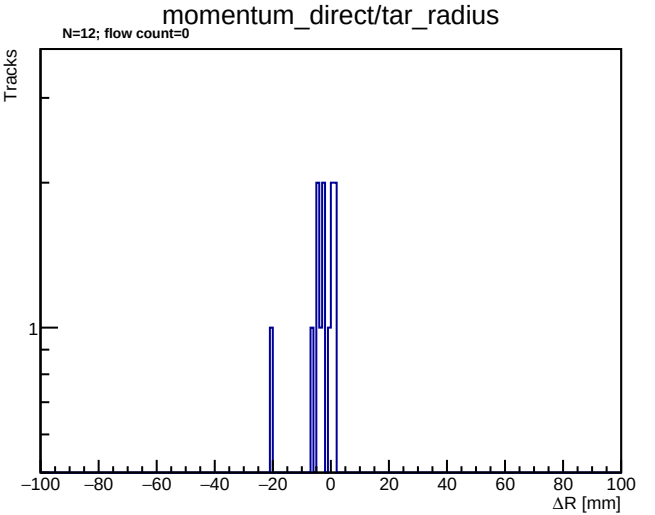

momentum_direct/tar_dx
N=12; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−600
−400
−200
0
200
400
600
Δx [mm]

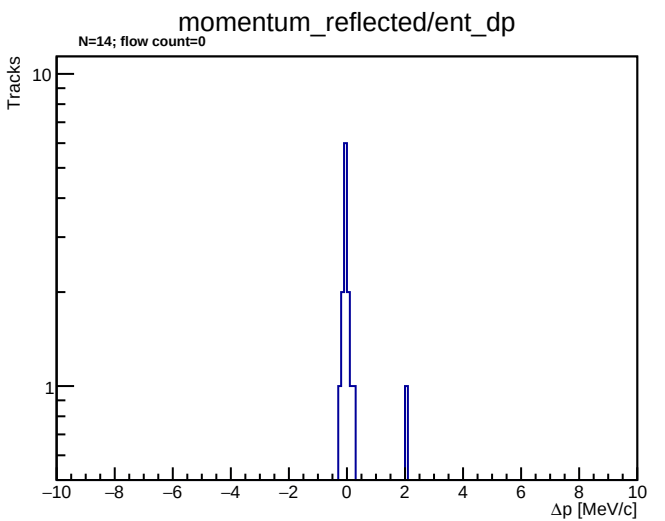

momentum_direct/tar_dy
N=12; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δy [mm]

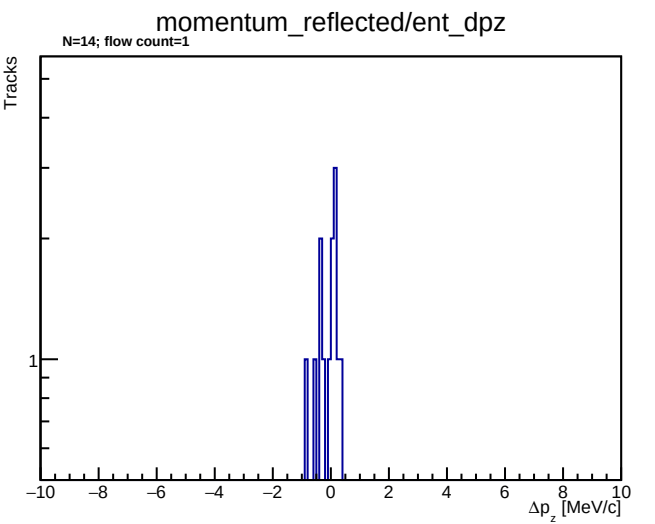

momentum_direct/tar_phase
N=12; flow count=0
Tracks
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

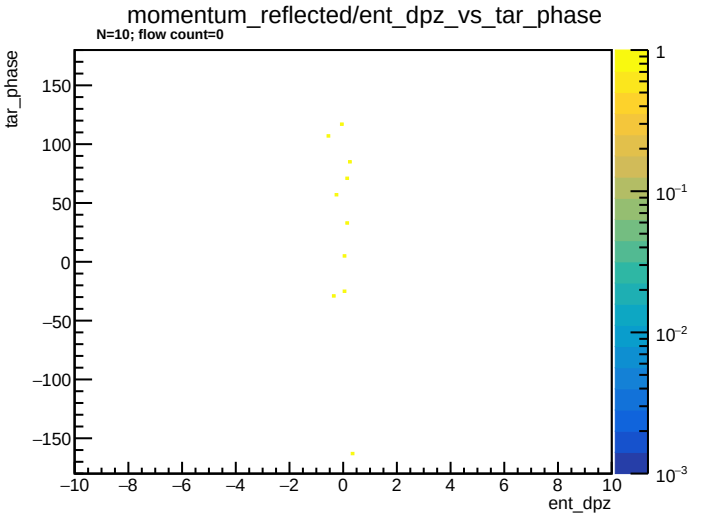

momentum_direct/tar_phase_vs_tar_dr
N=12; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
$10^{-1}$
$10^{-2}$
$10^{-3}$

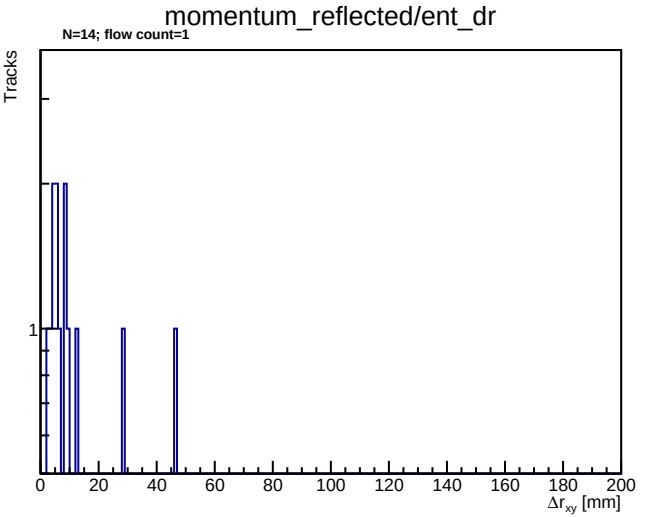

momentum_direct/tar_pitch
N=12; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

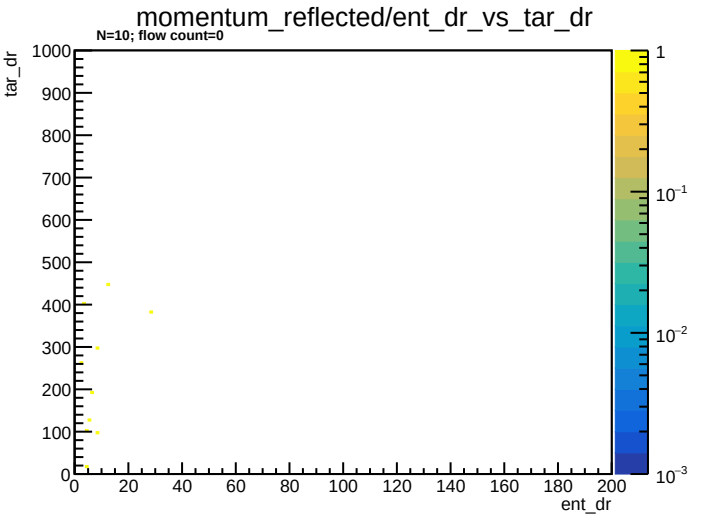

momentum_direct/tar_radius
N=12; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

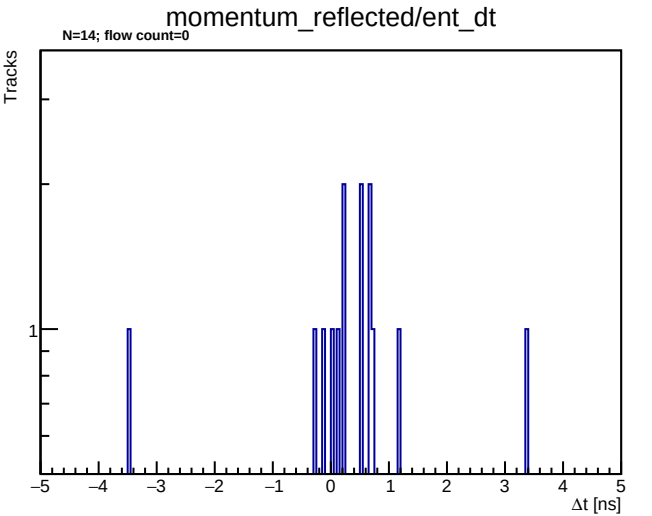

momentum_reflected/ent_dp
N=14; flow count=0
Tracks
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

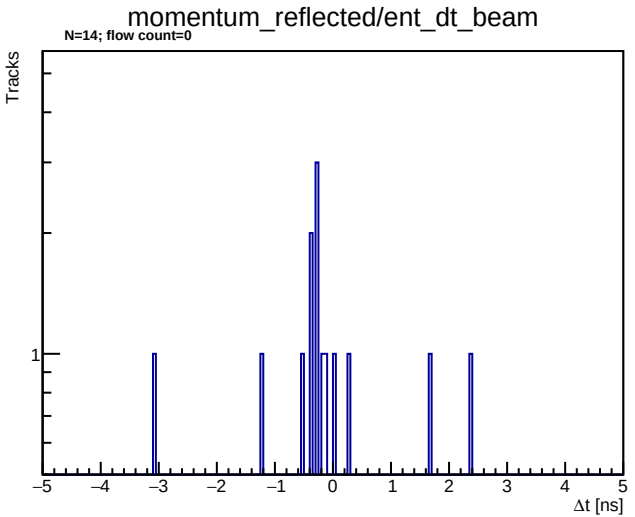

momentum_reflected/ent_dpz
N=14; flow count=1
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp$_z$ [MeV/c]

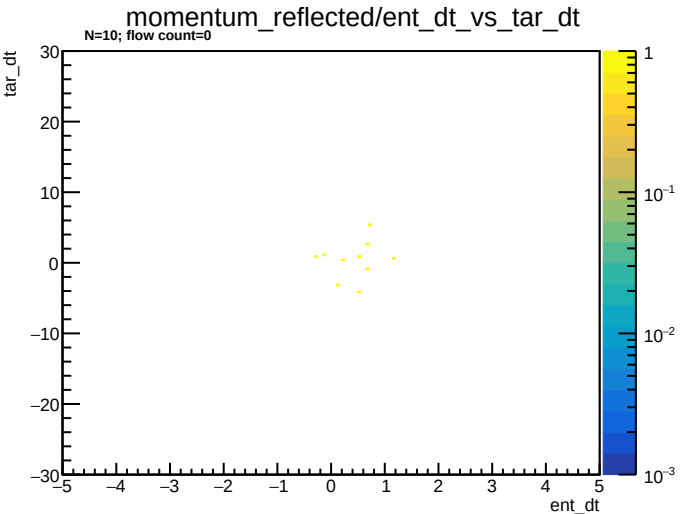

momentum_reflected/ent_dpz_vs_tar_phase
N=10; flow count=0
tar_phase
ent_dpz

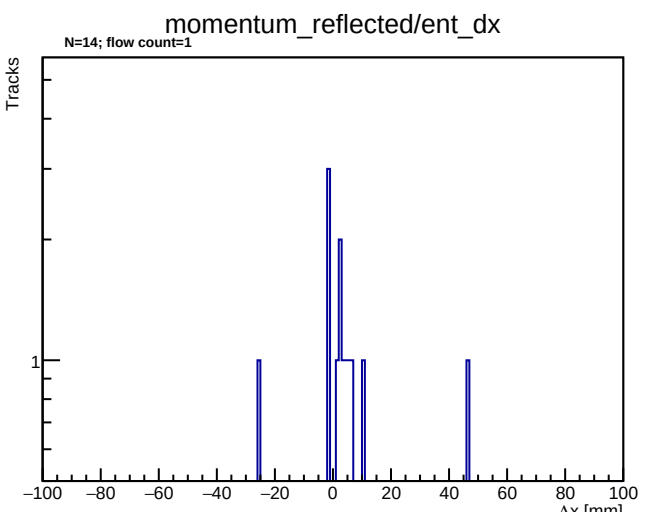

momentum_reflected/ent_dr
N=14; flow count=1
Tracks
Δr_xy [mm]

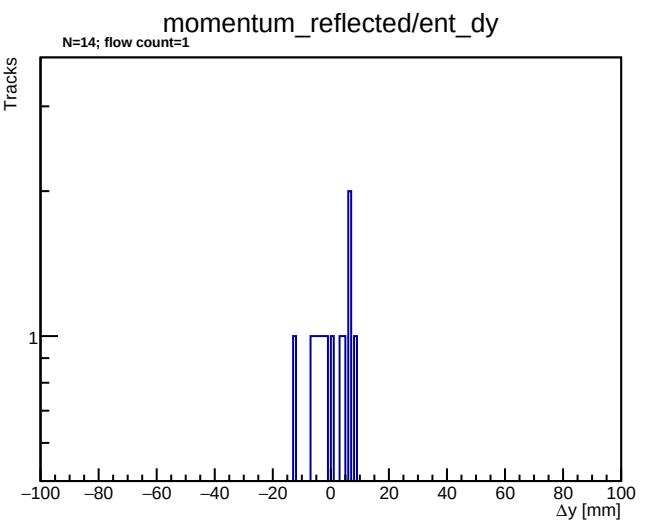

momentum_reflected/ent_dr_vs_tar_dr
N=10; flow count=0
tar_dr
ent_dr

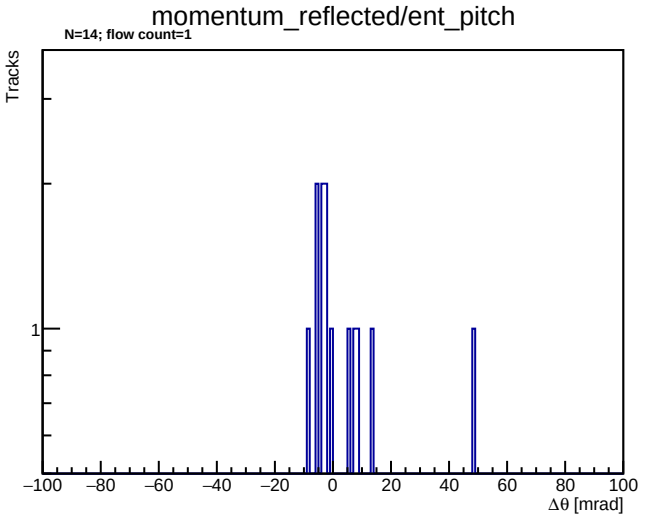

momentum_reflected/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

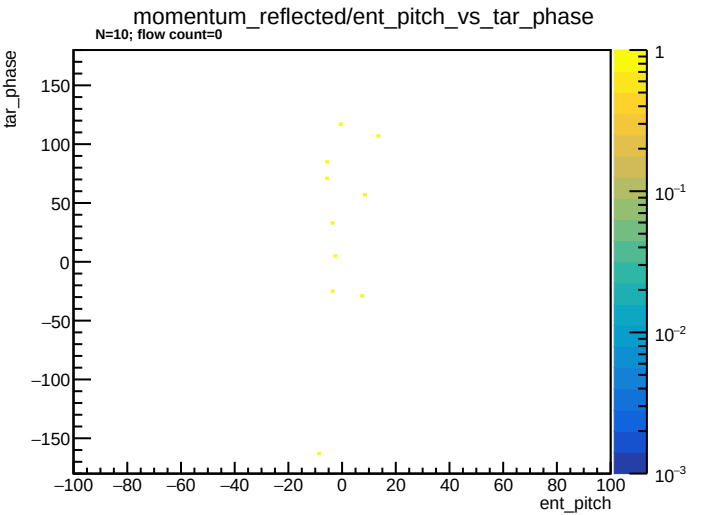

momentum_reflected/ent_dt_beam
N=14; flow count=0
Tracks
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

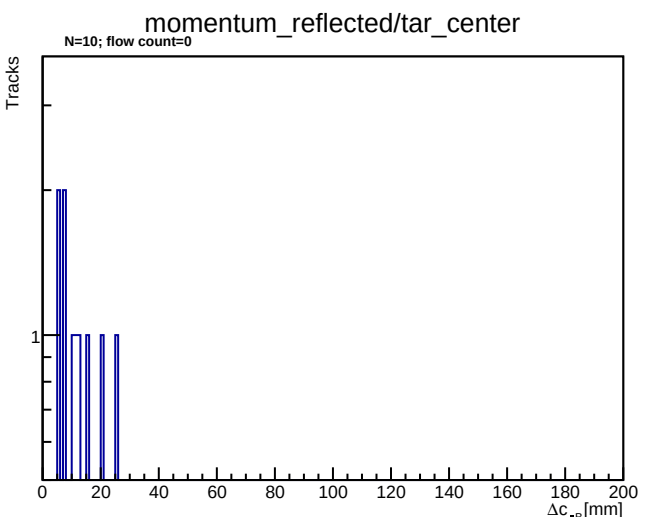

momentum_reflected/ent_dt_vs_tar_dt
N=10; flow count=0
tar_dt
30 20 10 0 −10 −20 −30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^-1
10^-2
10^-3

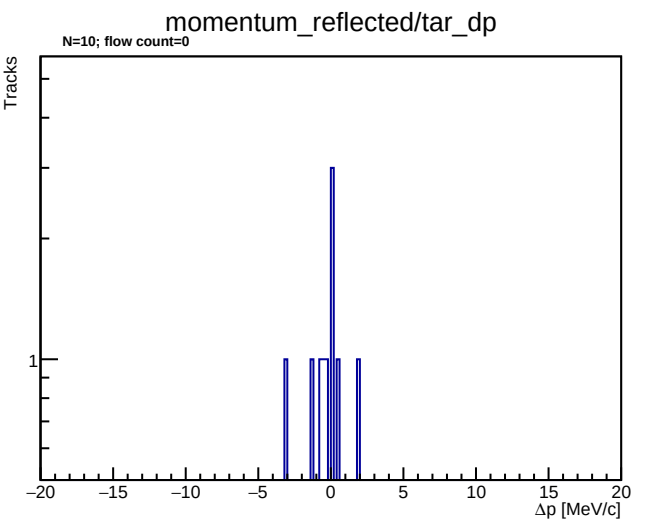

momentum_reflected/ent_dx
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

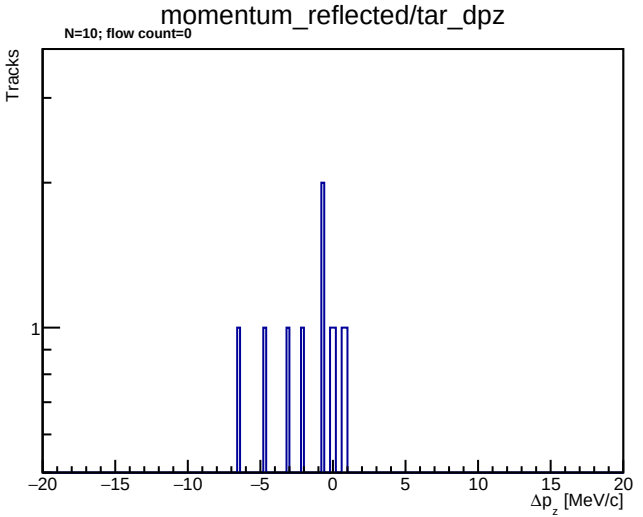

momentum_reflected/ent_dy
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

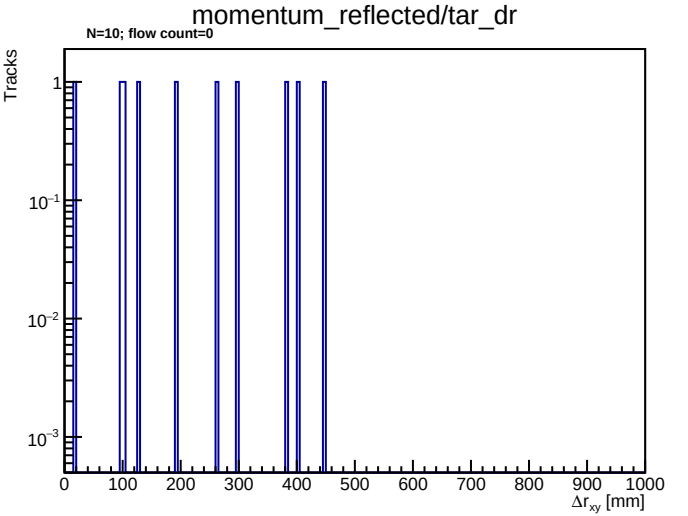

momentum_reflected/ent_pitch
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

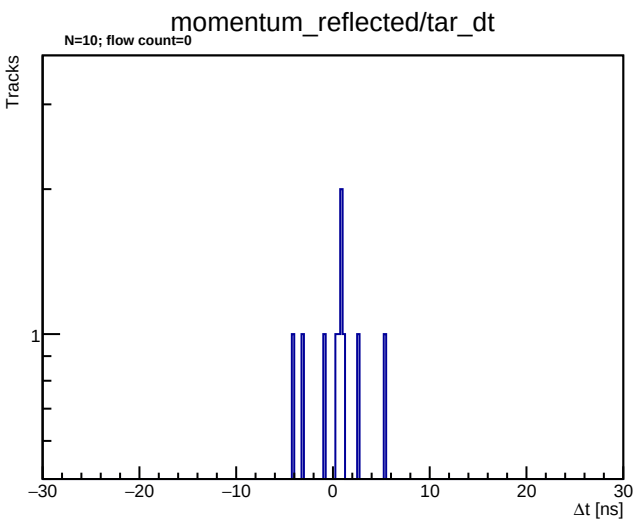

momentum_reflected/ent_pitch_vs_tar_phase
N=10; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²
10⁻³

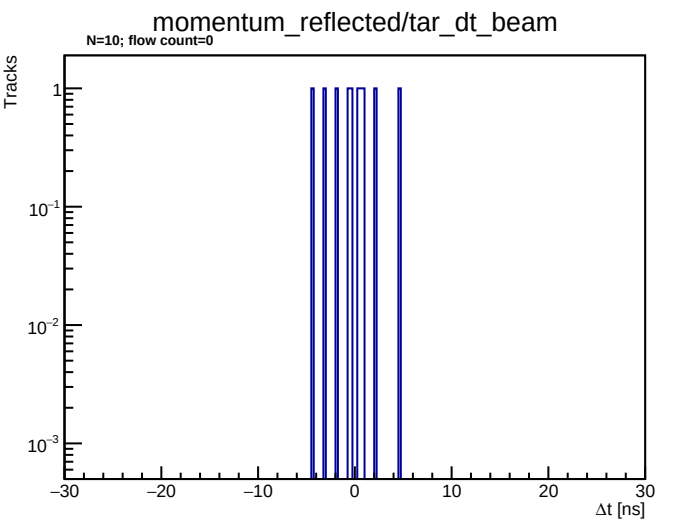

momentum_reflected/tar_center
N=10; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

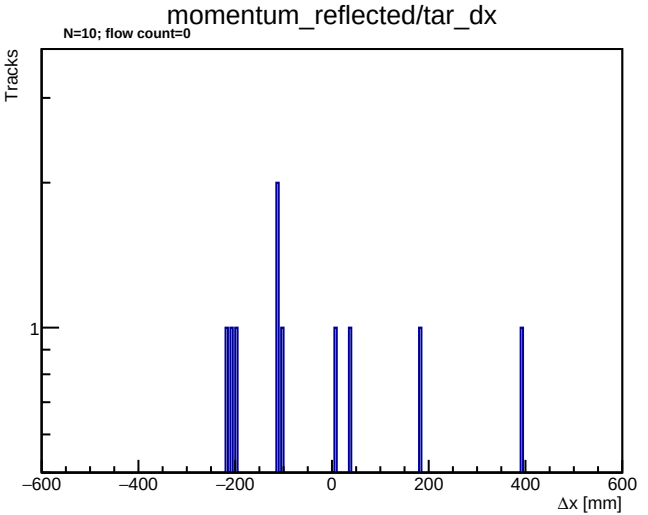

momentum_reflected/tar_dp
N=10; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

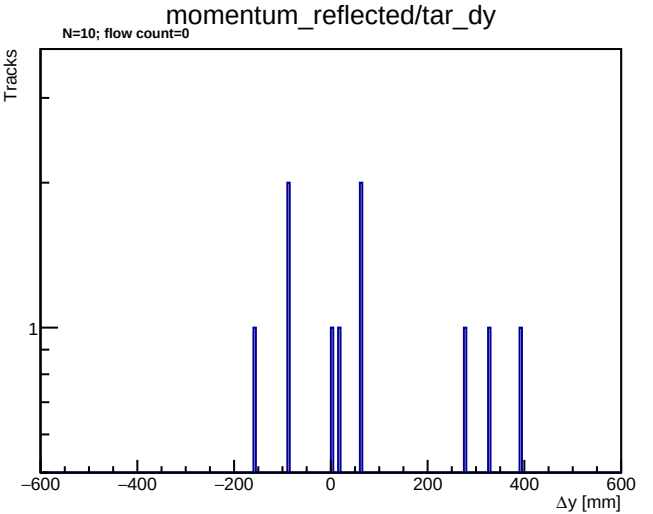

momentum_reflected/tar_dpz
N=10; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

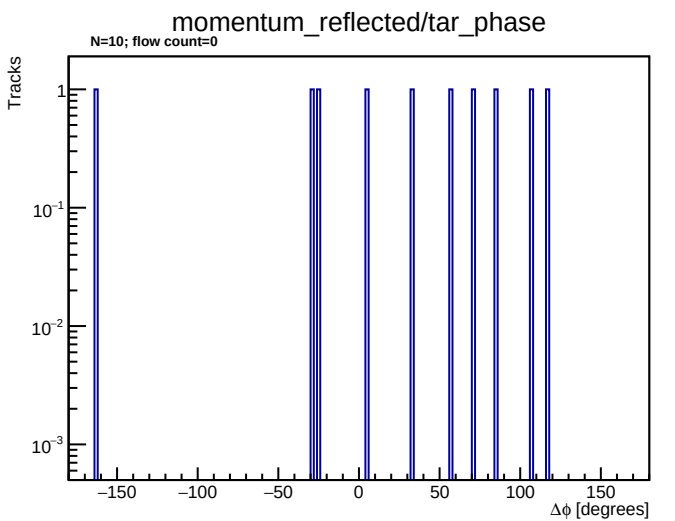

momentum_reflected/tar_dr
N=10; flow count=0
Tracks
1
10^-1
10^-2
10^-3
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

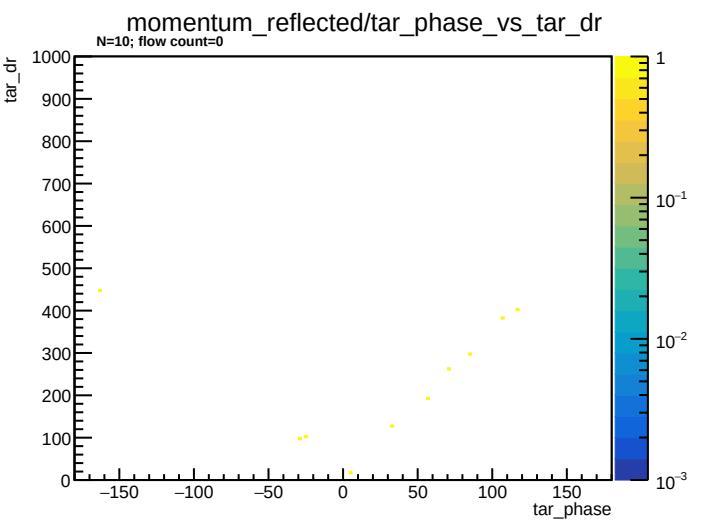

momentum_reflected/tar_dt
N=10; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

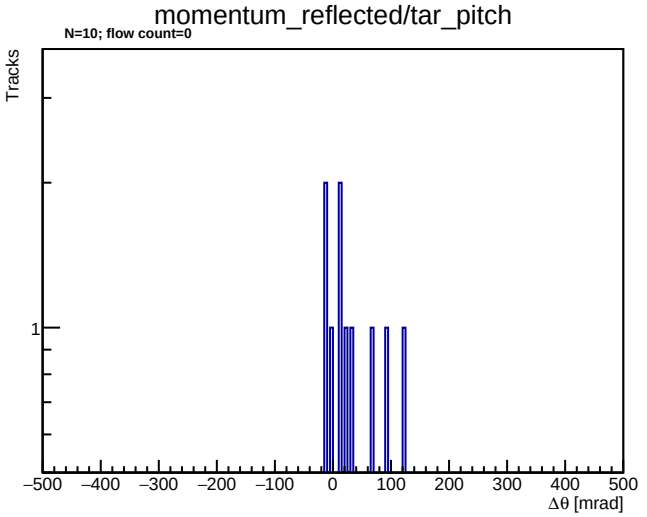

momentum_reflected/tar_dt_beam
N=10; flow count=0
Tracks
1
10^-1
10^-2
10^-3
−30 −20 −10 0 10 20 30
Δt [ns]

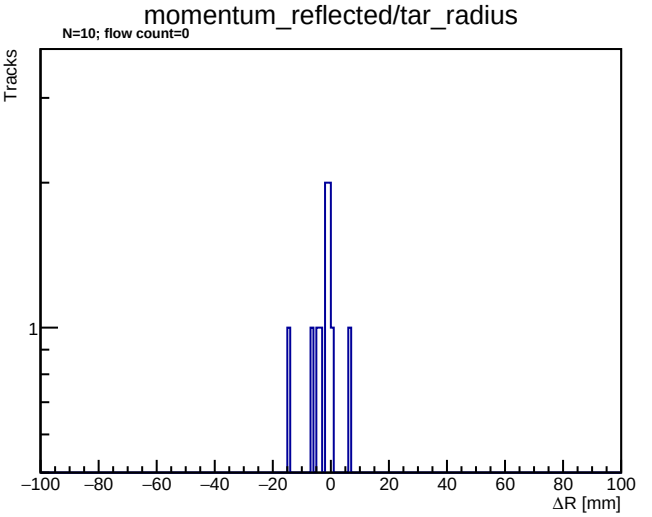

momentum_reflected/tar_dx
N=10; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δx [mm]

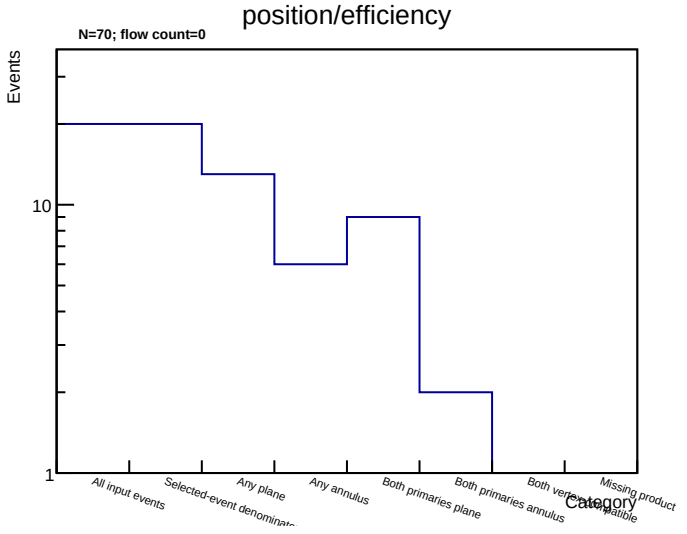

momentum_reflected/tar_dy
N=10; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δy [mm]

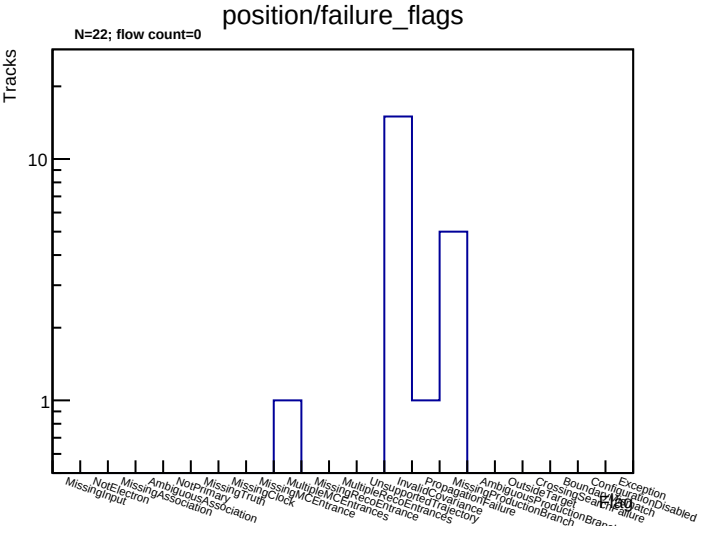

momentum_reflected/tar_phase
N=10; flow count=0
Tracks
1
10⁻¹
10⁻²
10⁻³
−150
−100
−50
0
50
100
150
Δϕ [degrees]

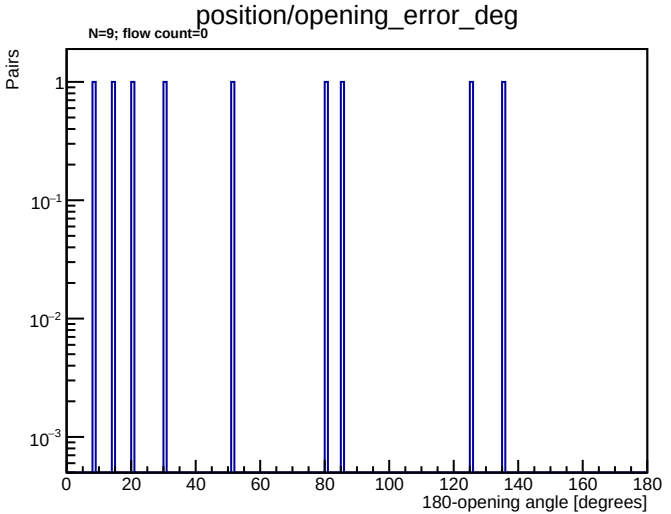

momentum_reflected/tar_phase_vs_tar_dr
N=10; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²
10⁻³

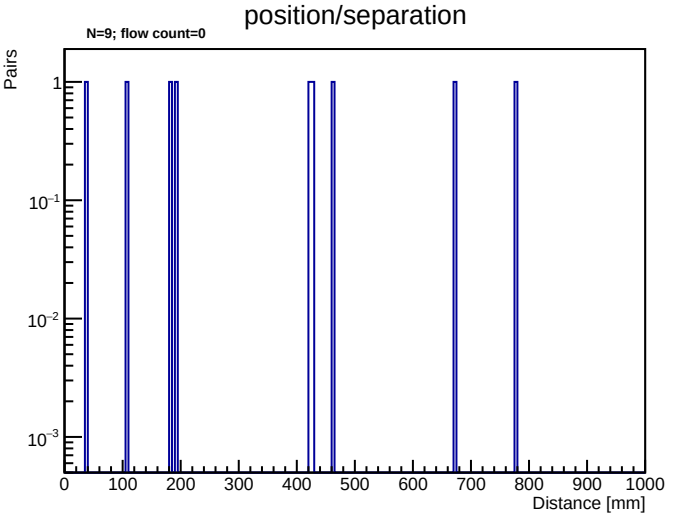

momentum_reflected/tar_pitch
N=10; flow count=0
Tracks
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

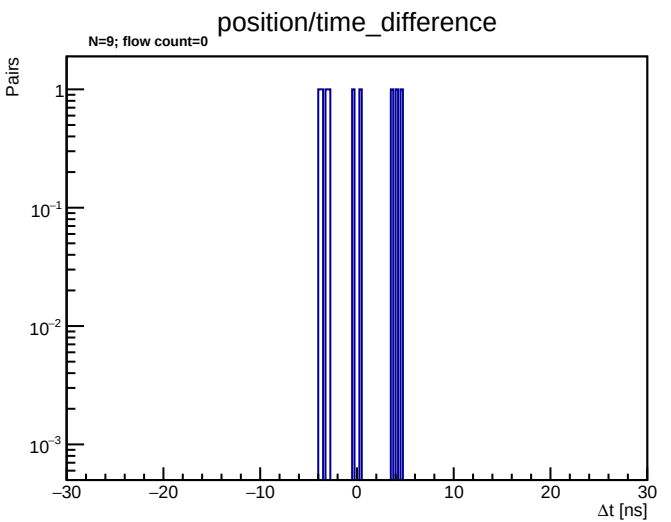

momentum_reflected/tar_radius
N=10; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

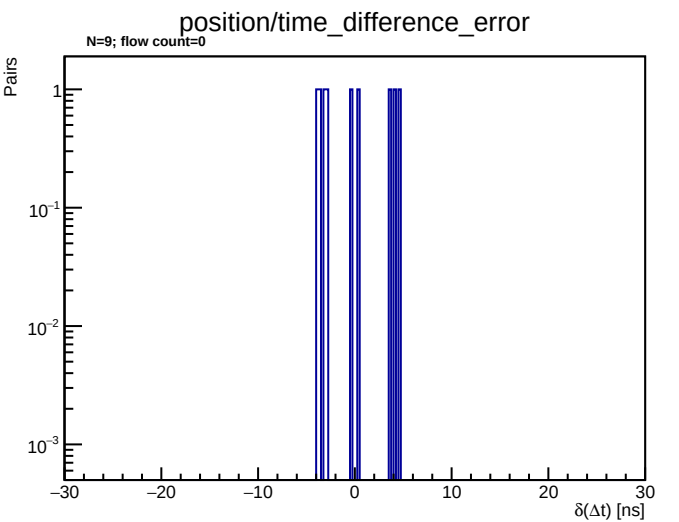

position/efficiency
N=70; flow count=0
Events
10
1
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Missing product
Category

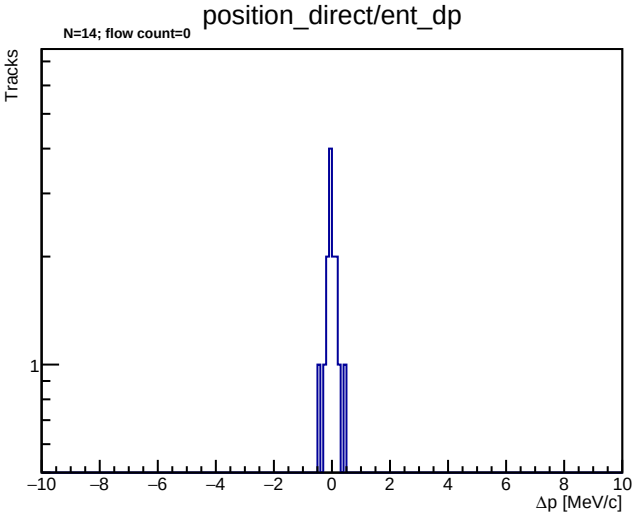

position/failure_flags
N=22; flow count=0
Tracks
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleRecoEntrances
MissingRecoEntrance
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProductionBranch
AmbiguousProduction
OutsideTarget
CrossingSeam
BoundaryProximity
ConfigurationDisabled
Exception
Flag

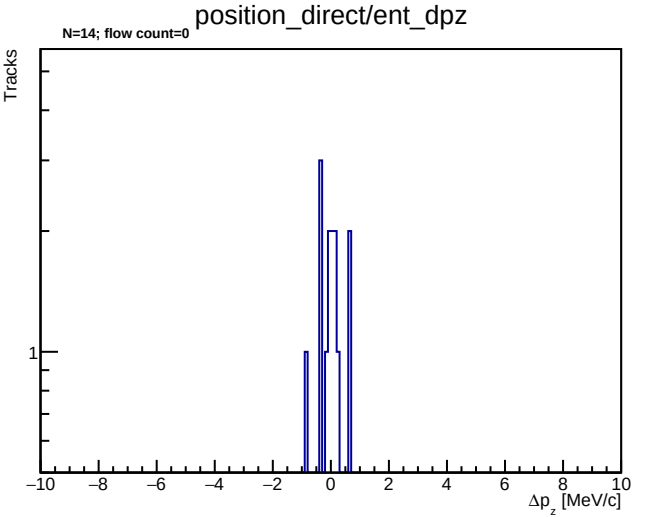

position/opening_error_deg
N=9; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

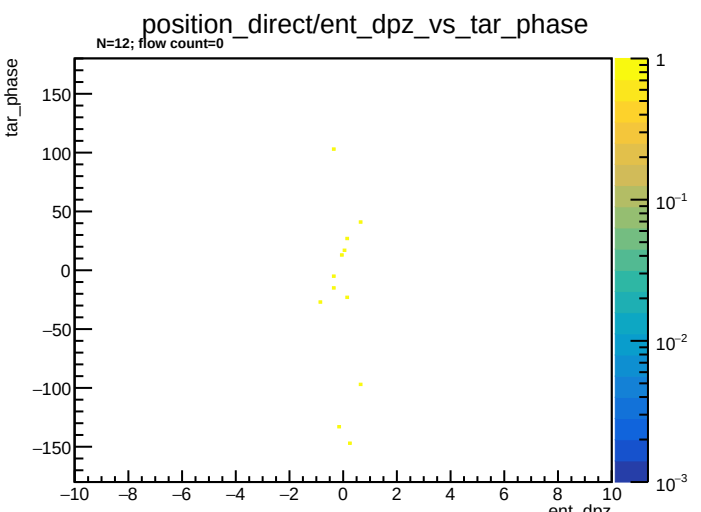

position/separation
N=9; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

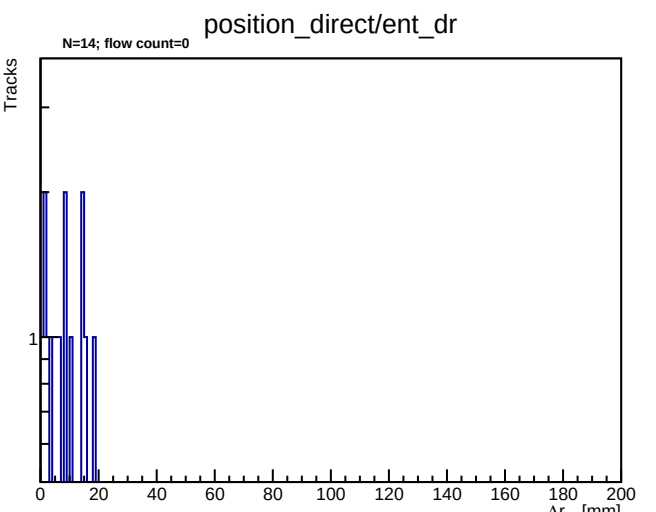

position/time_difference
N=9; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
–30
–20
–10
0
10
20
30
Δt [ns]

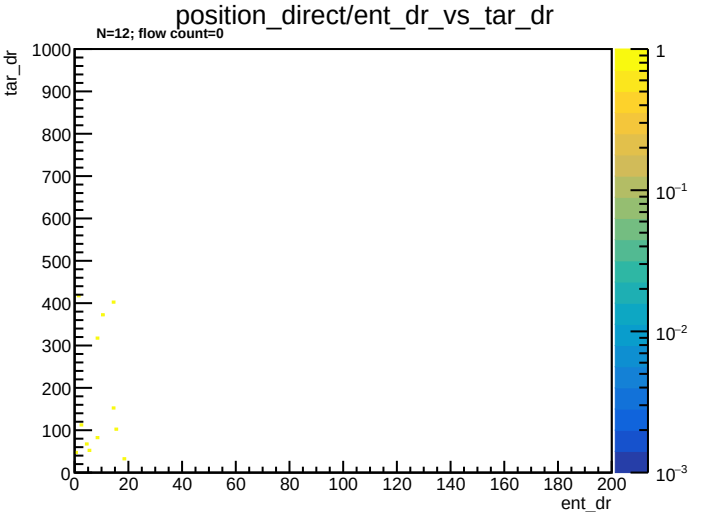

position/time_difference_error
N=9; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
–30
–20
–10
0
10
20
30
δ(Δt) [ns]

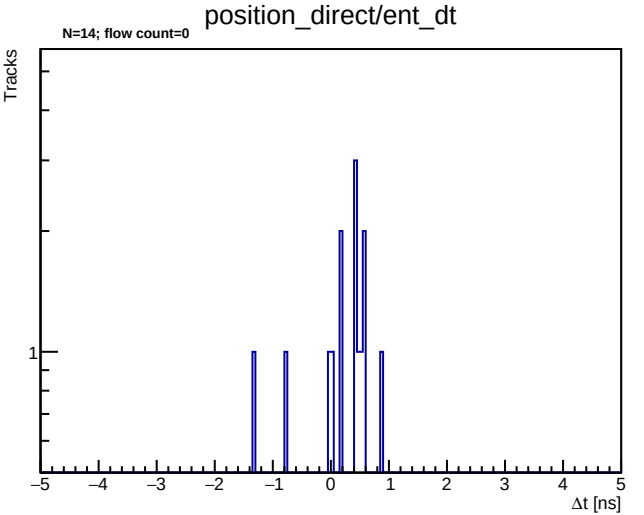

position_direct/ent_dp
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

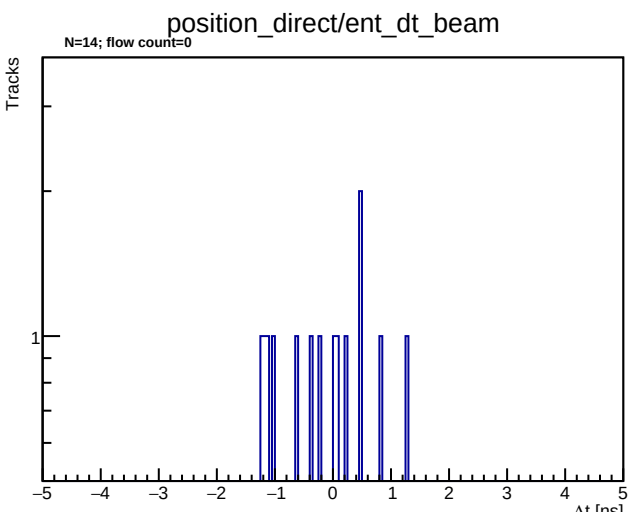

position_direct/ent_dpz
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

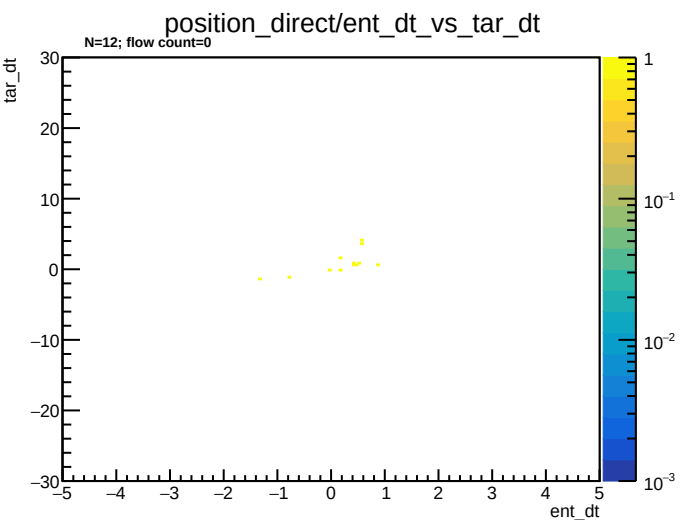

position_direct/ent_dpz_vs_tar_phase
N=12; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−10 −8 −6 −4 −2 0 2 4 6 8 10
ent_dpz
1
10^−1
10^−2
10^−3

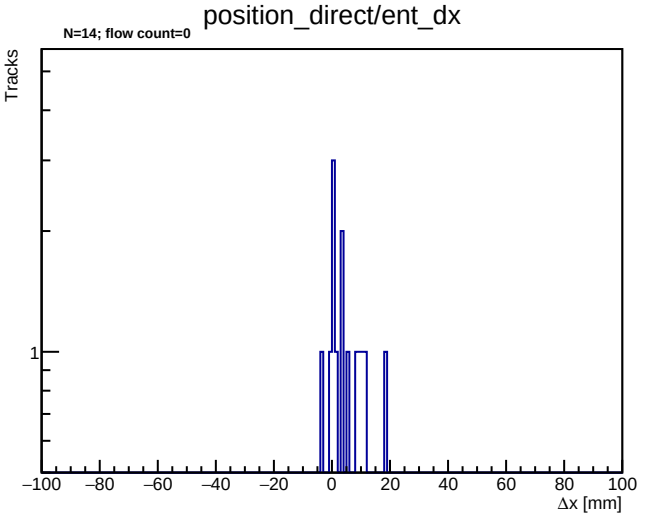

position_direct/ent_dr
N=14; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δr_xy [mm]

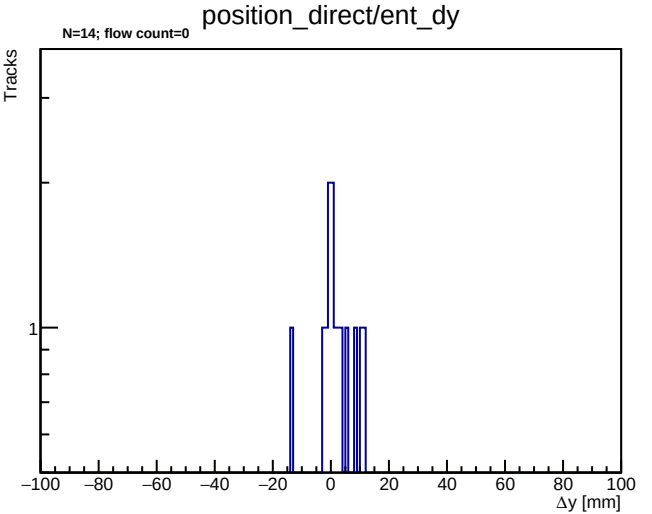

position_direct/ent_dr_vs_tar_dr
N=12; flow count=0
tar_dr
ent_dr

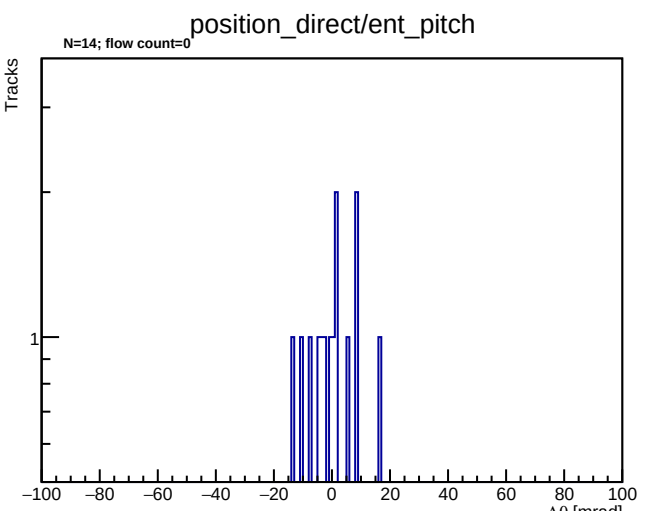

position_direct/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

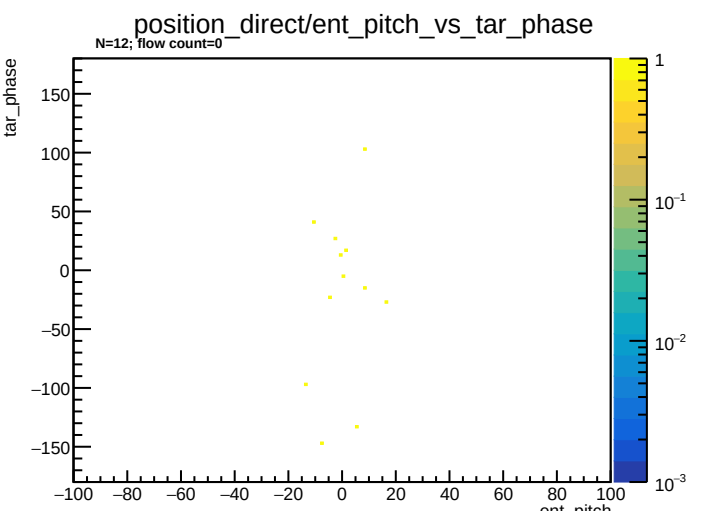

position_direct/ent_dt_beam
N=14; flow count=0
Tracks
Δt [ns]

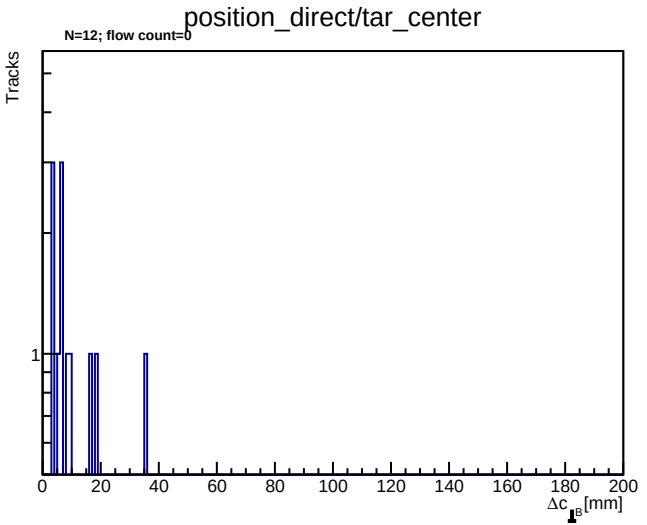

position_direct/ent_dt_vs_tar_dt
N=12; flow count=0
tar_dt
ent_dt

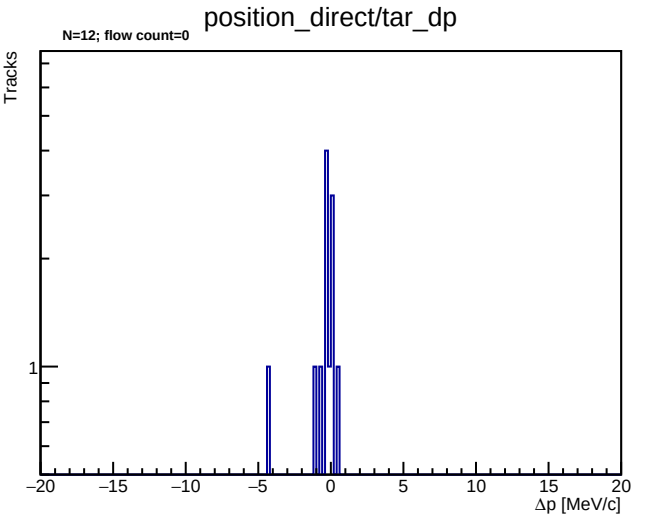

position_direct/ent_dx
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

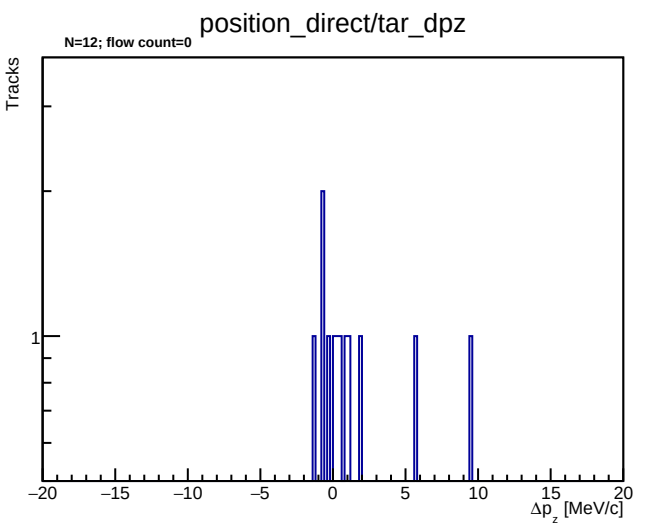

position_direct/ent_dy
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

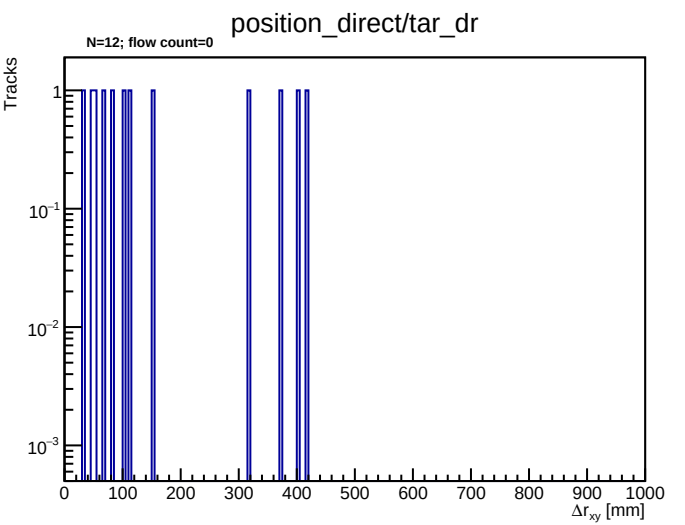

position_direct/ent_pitch
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

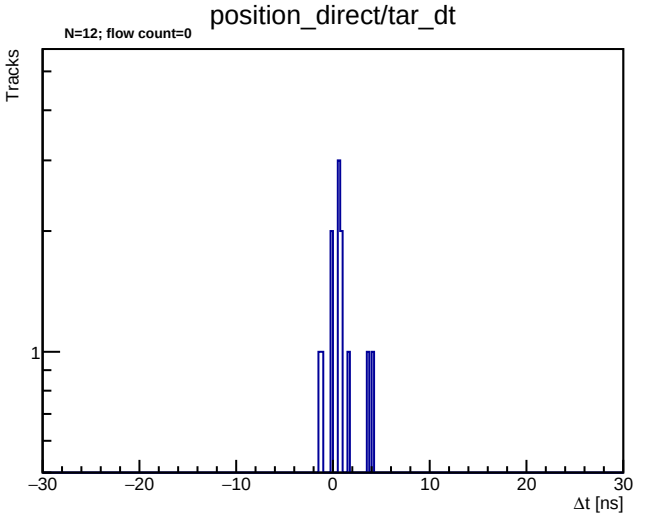

position_direct/ent_pitch_vs_tar_phase
N=12; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²
10⁻³

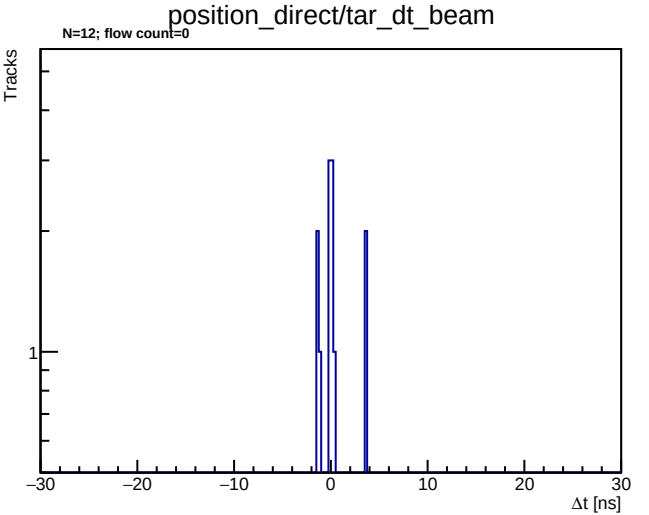

position_direct/tar_center
N=12; flow count=0
Tracks
1
0
20
40
60
80
100
120
140
160
180
200
Δc_B [mm]

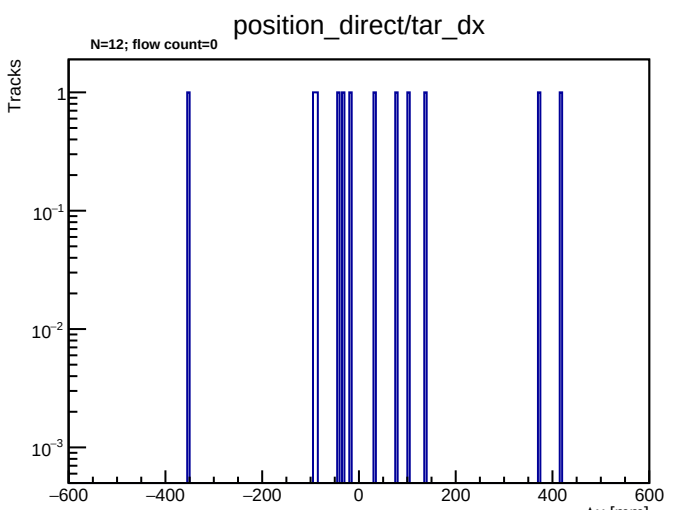

position_direct/tar_dp
N=12; flow count=0
Tracks
1
−20
−15
−10
−5
0
5
10
15
20
Δp [MeV/c]

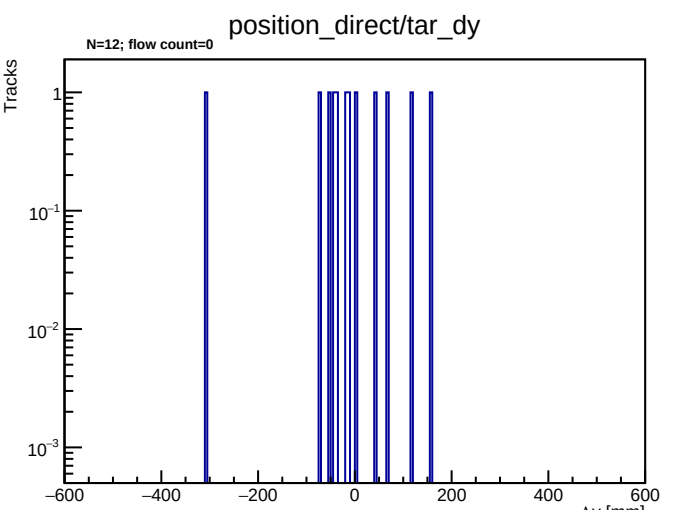

position_direct/tar_dpz
N=12; flow count=0
Tracks
1
−20
−15
−10
−5
0
5
10
15
20
Δp_z [MeV/c]

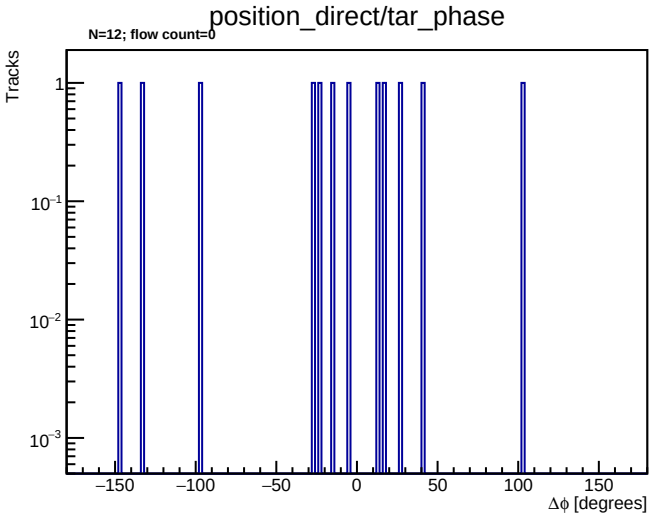

position_direct/tar_dr
N=12; flow count=0
Tracks
1
10^-1
10^-2
10^-3
0
100
200
300
400
500
600
700
800
900
1000
Δr_xy [mm]

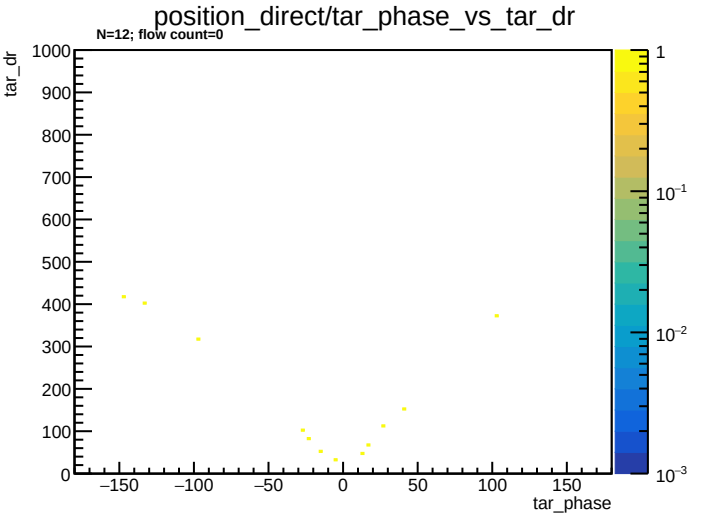

position_direct/tar_dt
N=12; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

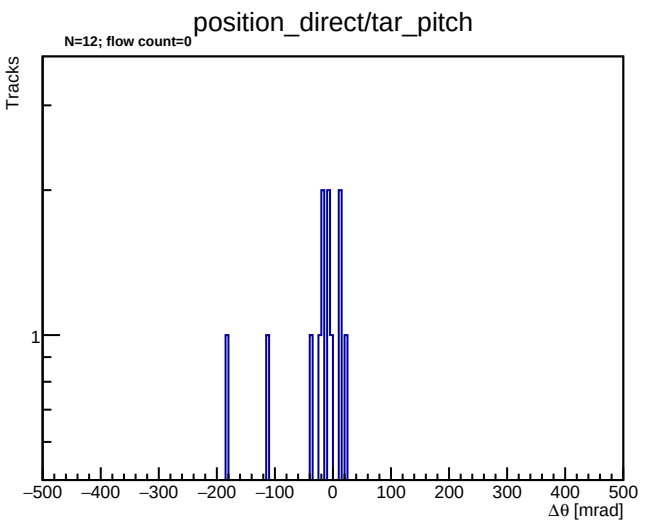

position_direct/tar_dt_beam
N=12; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

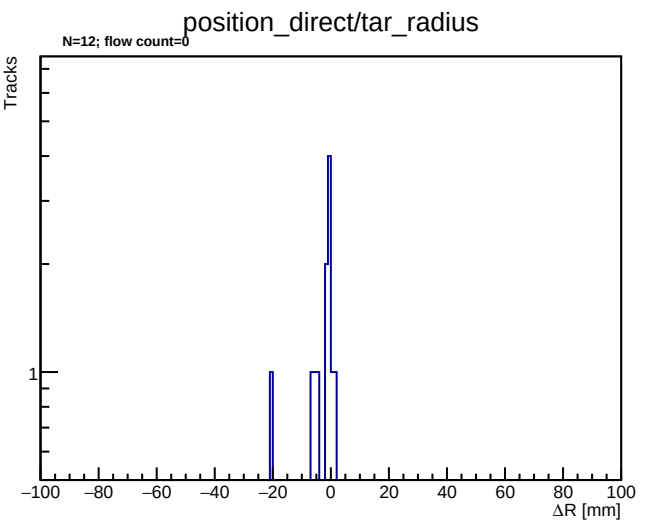

position_direct/tar_dx
N=12; flow count=0
Tracks
1
10⁻¹
10⁻²
10⁻³
−600
−400
−200
0
200
400
600
Δx [mm]

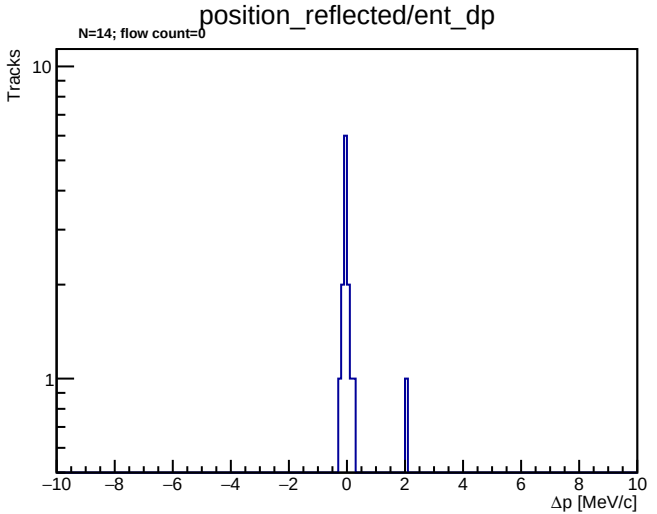

position_direct/tar_dy
N=12; flow count=0
Tracks
1
10⁻¹
10⁻²
10⁻³
−600
−400
−200
0
200
400
600
Δy [mm]

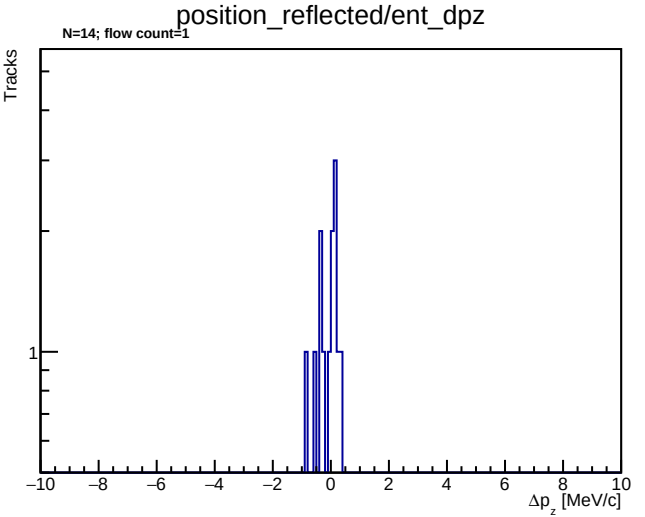

position_direct/tar_phase
N=12; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−150 −100 −50 0 50 100 150
Δϕ [degrees]

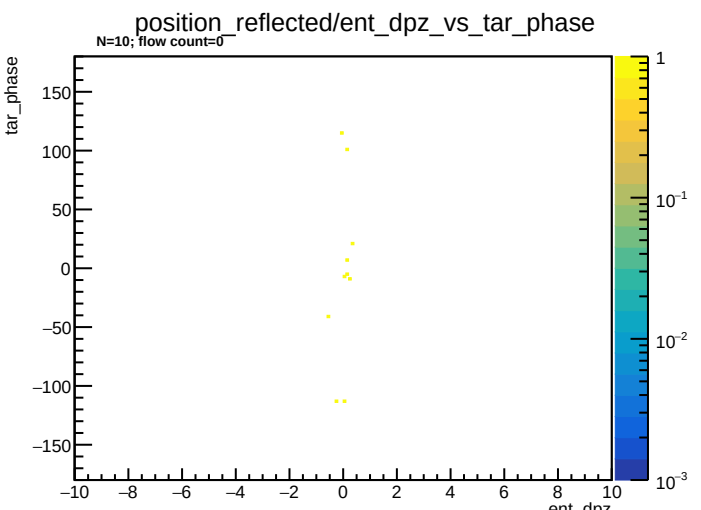

position_direct/tar_phase_vs_tar_dr
N=12; flow count=0
tar_dr
1000 900 800 700 600 500 400 300 200 100 0
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−150 −100 −50 0 50 100 150
tar_phase

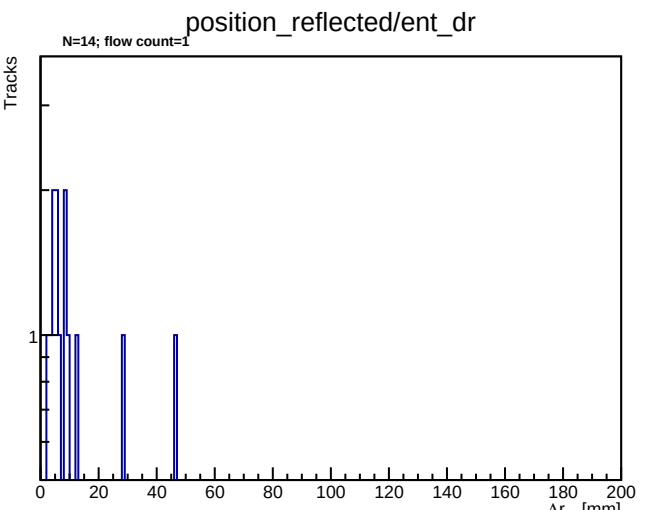

position_direct/tar_pitch
N=12; flow count=0
Tracks
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

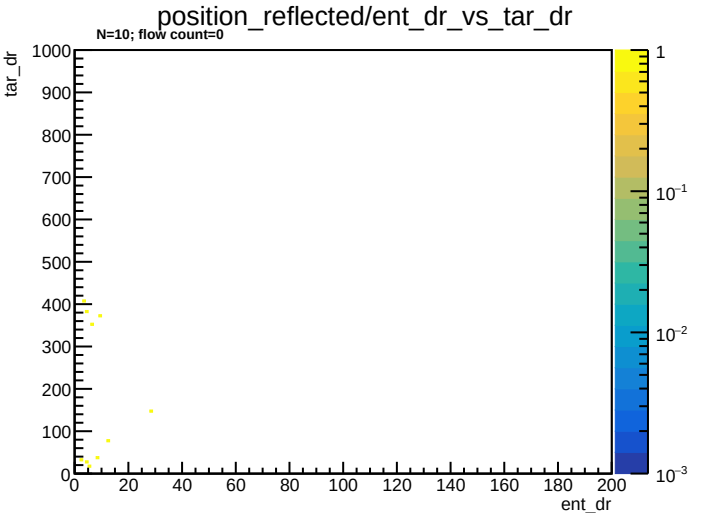

position_direct/tar_radius
N=12; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

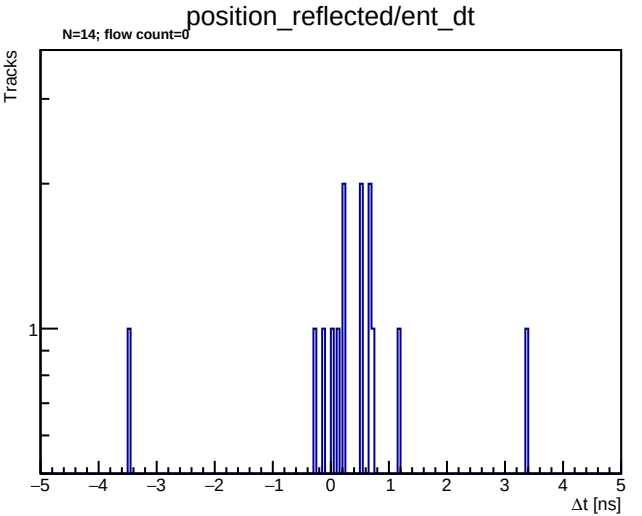

position_reflected/ent_dp
N=14; flow count=0
Tracks
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

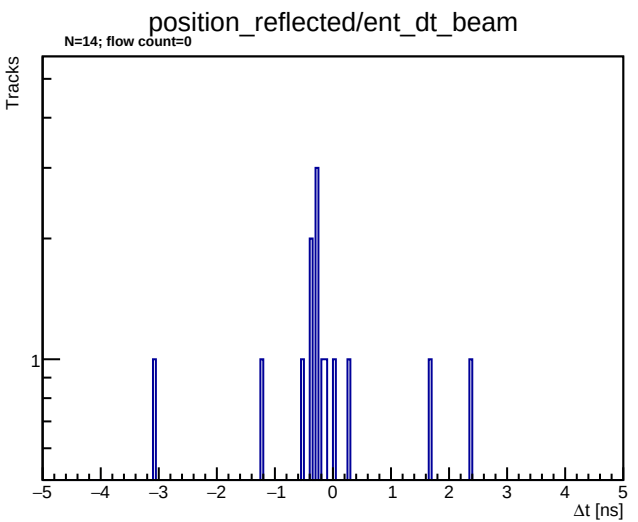

position_reflected/ent_dpz
N=14; flow count=1
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

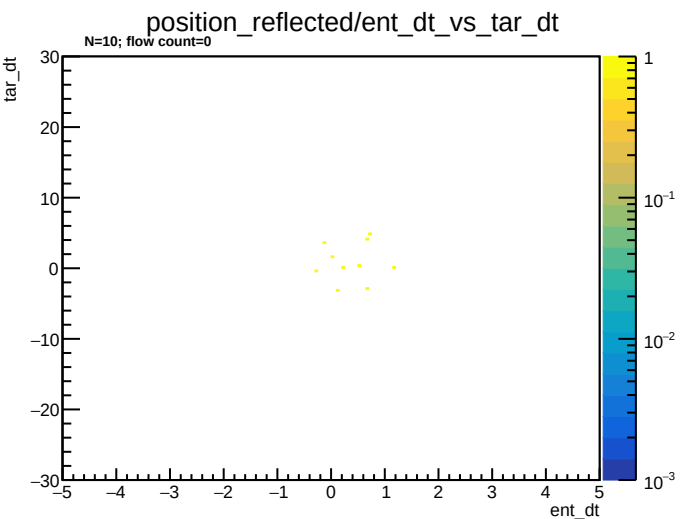

position_reflected/ent_dpz_vs_tar_phase
N=10; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−10 −8 −6 −4 −2 0 2 4 6 8 10
ent_dpz
1
10^−1
10^−2
10^−3

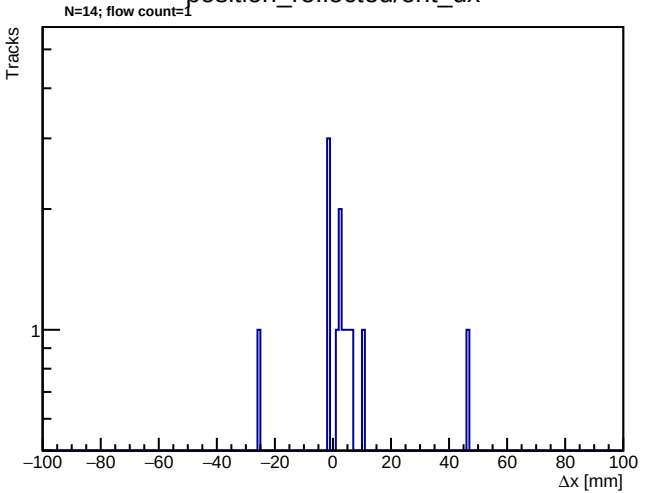

position_reflected/ent_dr
N=14; flow count=1
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δr_xy [mm]

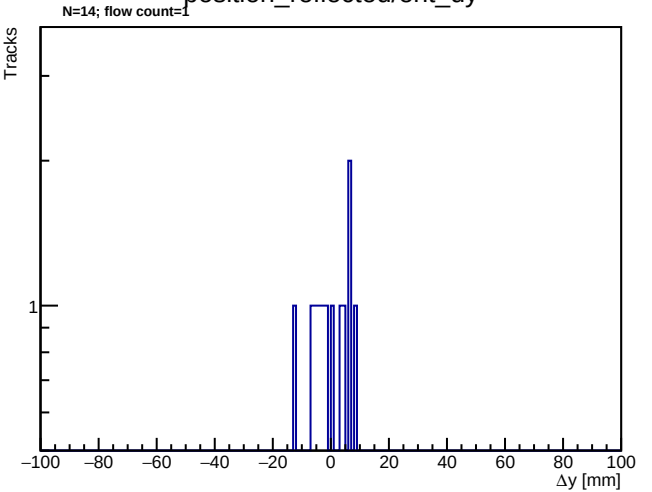

position_reflected/ent_dr_vs_tar_dr
N=10; flow count=0
tar_dr
ent_dr

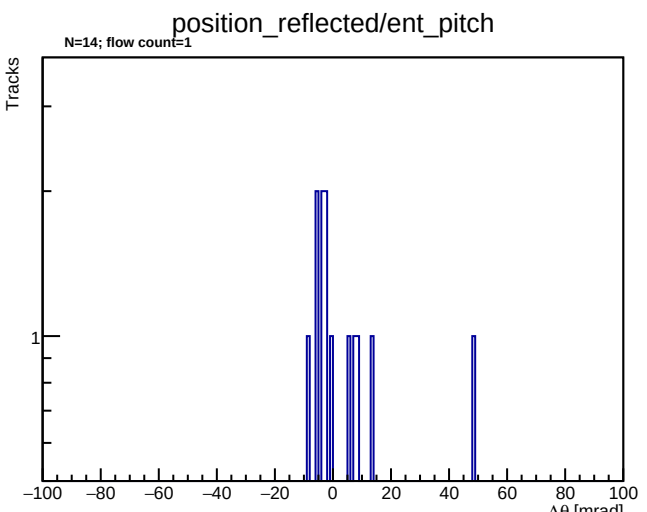

position_reflected/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

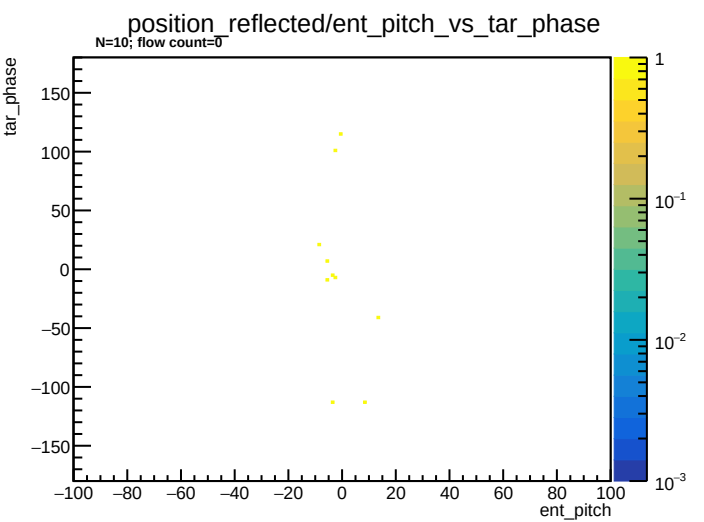

position_reflected/ent_dt_beam
N=14; flow count=0
Tracks
Δt [ns]

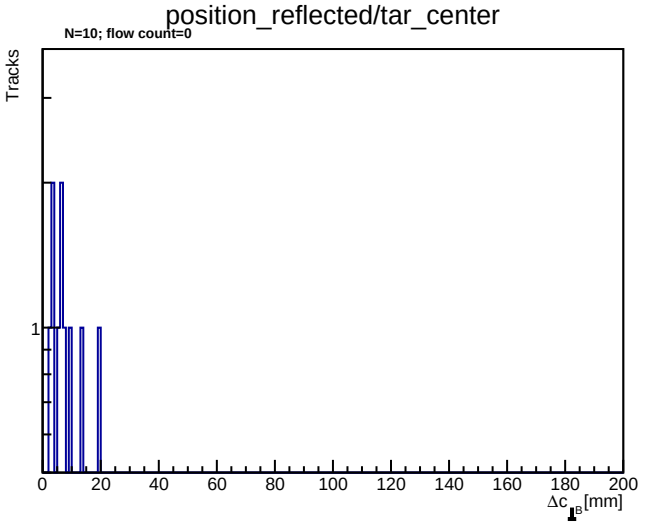

position_reflected/ent_dt_vs_tar_dt
N=10; flow count=0
tar_dt
ent_dt

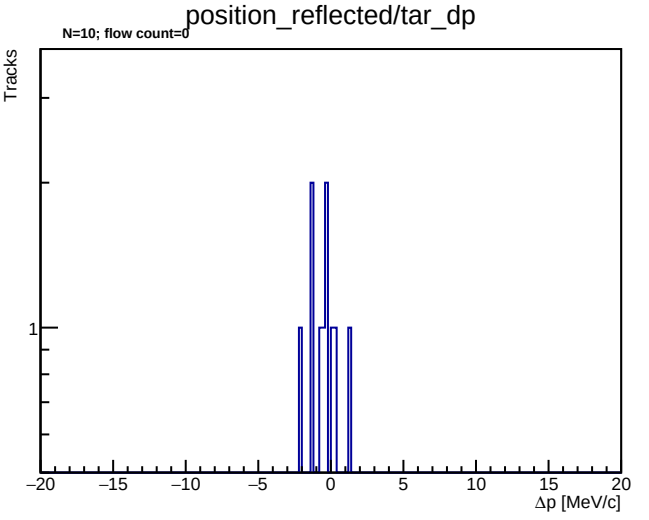

position_reflected/ent_dx
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

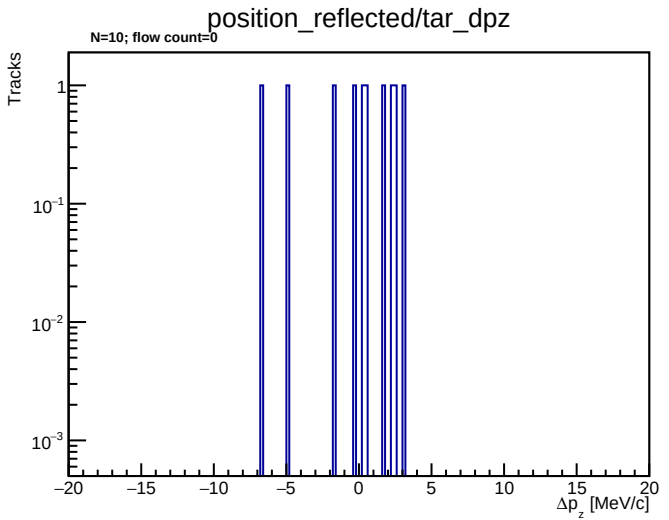

position_reflected/ent_dy
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

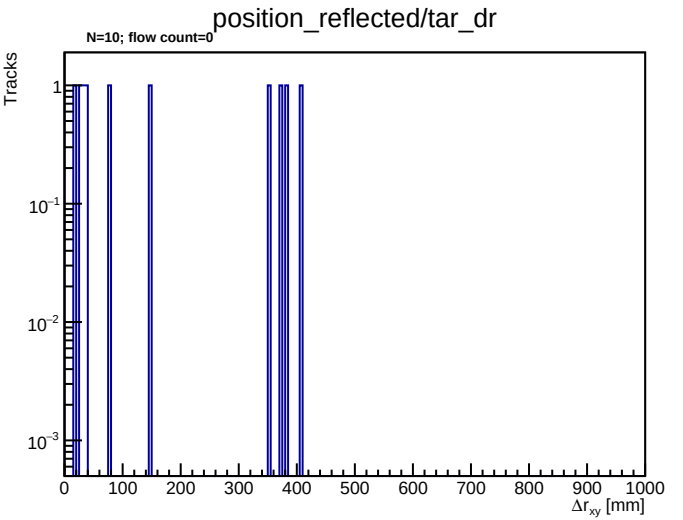

position_reflected/ent_pitch
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

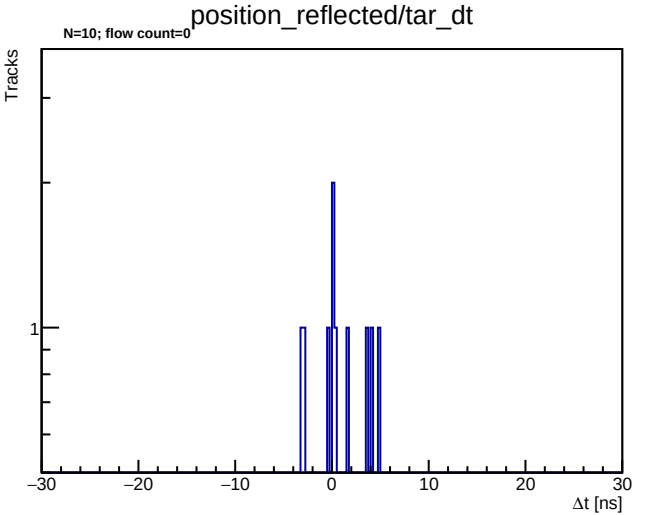

position_reflected/ent_pitch_vs_tar_phase
N=10; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
10⁻¹
10⁻²
10⁻³

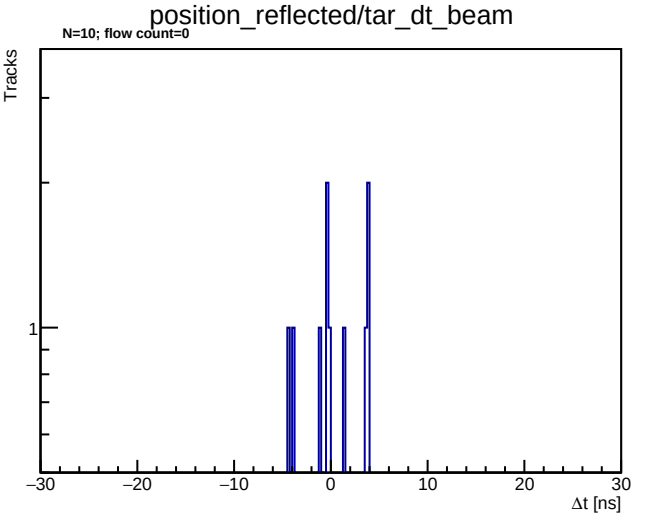

position_reflected/tar_center
N=10; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

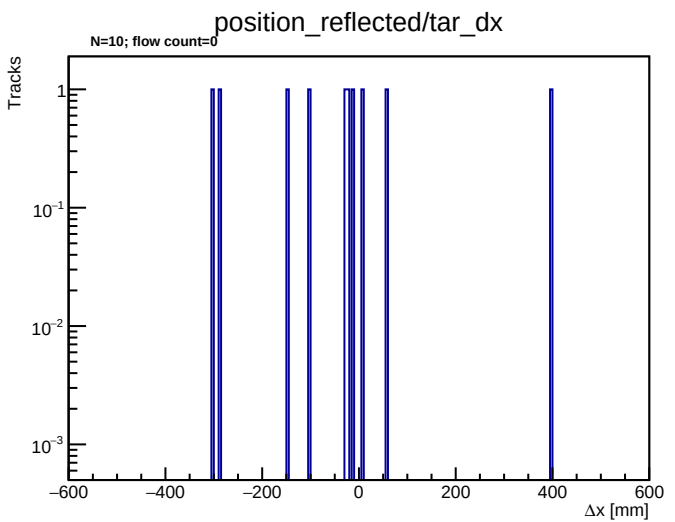

position_reflected/tar_dp
N=10; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

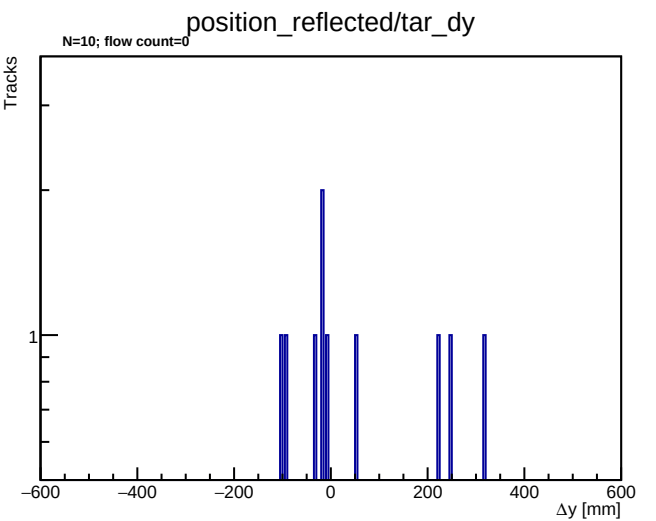

position_reflected/tar_dpz
N=10; flow count=0
Tracks
1
10^-1
10^-2
10^-3
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

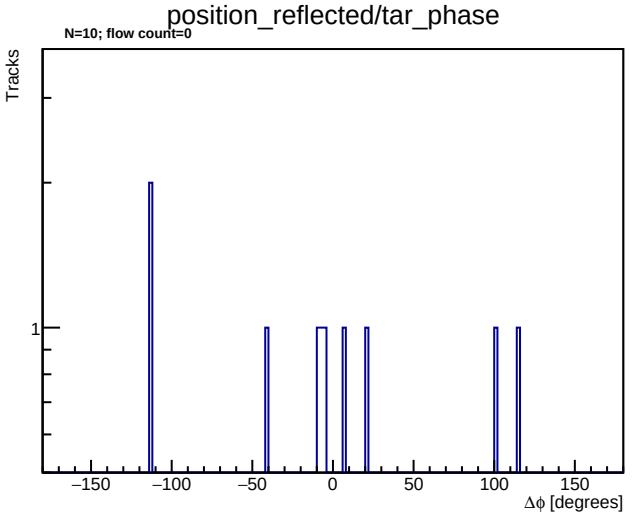

position_reflected/tar_dr
N=10; flow count=0
Tracks
1
10^-1
10^-2
10^-3
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

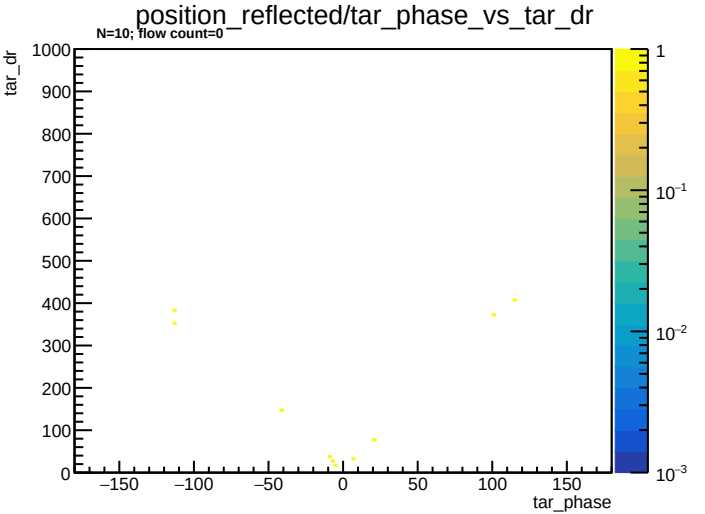

position_reflected/tar_dt
N=10; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

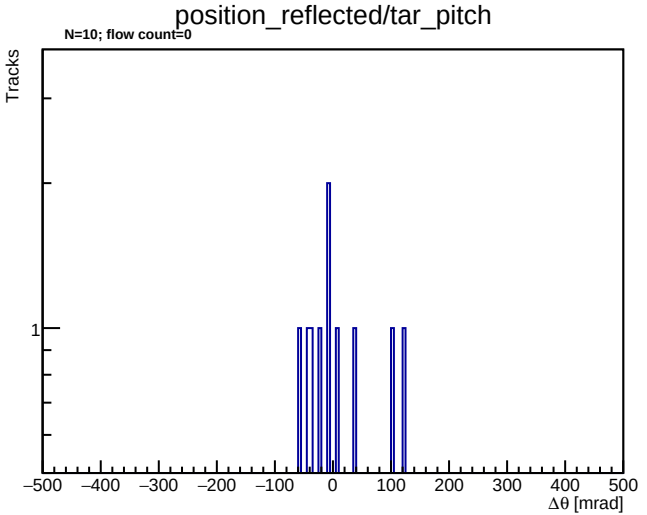

position_reflected/tar_dt_beam
N=10; flow count=0
Tracks
1
−30
−20
−10
0
10
20
30
Δt [ns]

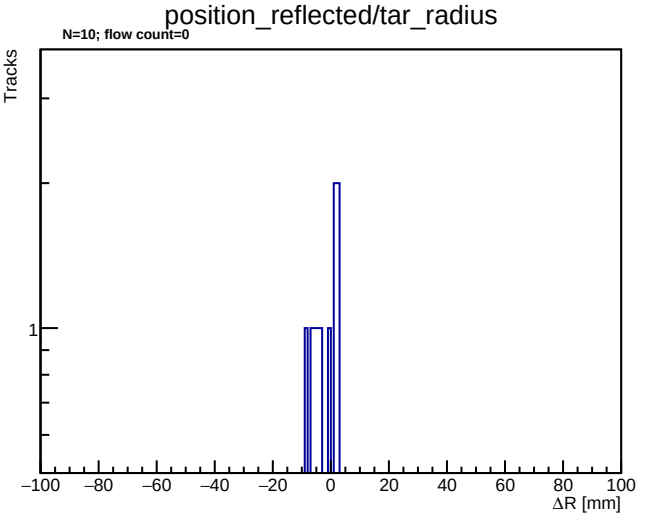

position_reflected/tar_dx
N=10; flow count=0
Tracks
1
10⁻¹
10⁻²
10⁻³
−600
−400
−200
0
200
400
600
Δx [mm]

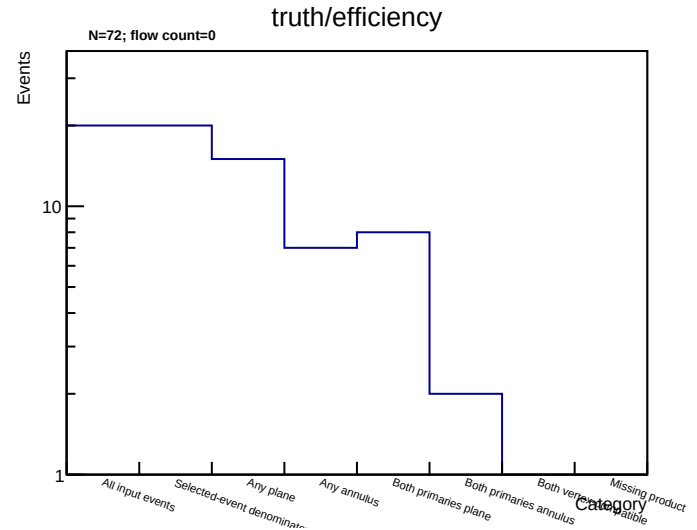

position_reflected/tar_dy
N=10; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δy [mm]

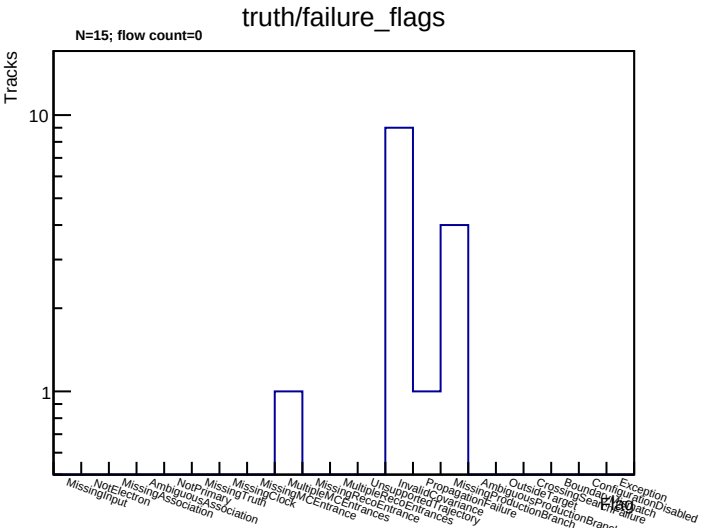

position_reflected/tar_phase
N=10; flow count=0
Tracks
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

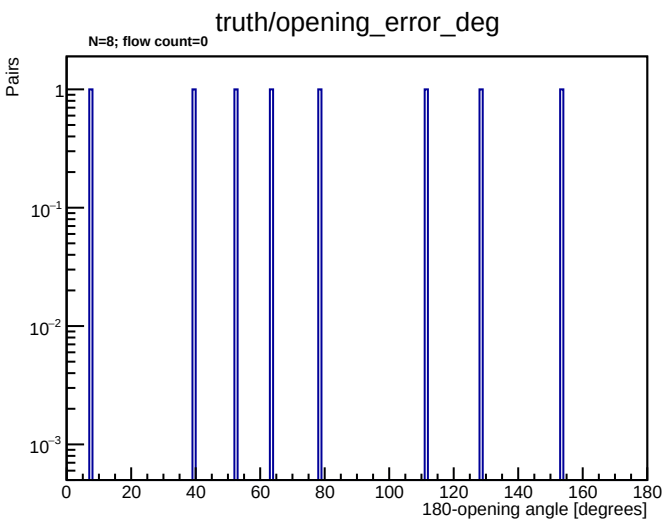

position_reflected/tar_phase_vs_tar_dr
N=10; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
$10^{-1}$
$10^{-2}$
$10^{-3}$

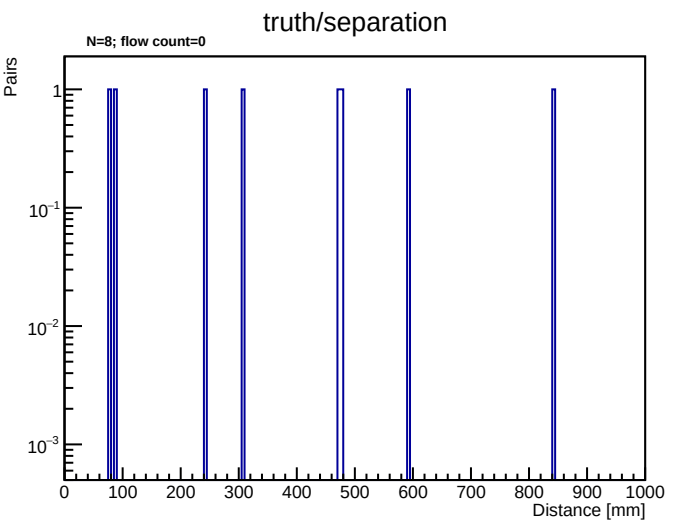

position_reflected/tar_pitch
N=10; flow count=0
Tracks
1
−500
−400
−300
−200
−100
0
100
200
300
400
500
Δθ [mrad]

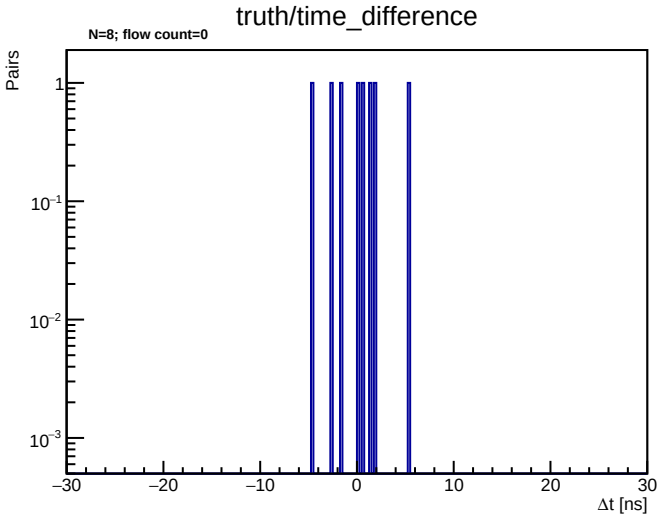

position_reflected/tar_radius
N=10; flow count=0
Tracks
1
−100
−80
−60
−40
−20
0
20
40
60
80
100
ΔR [mm]

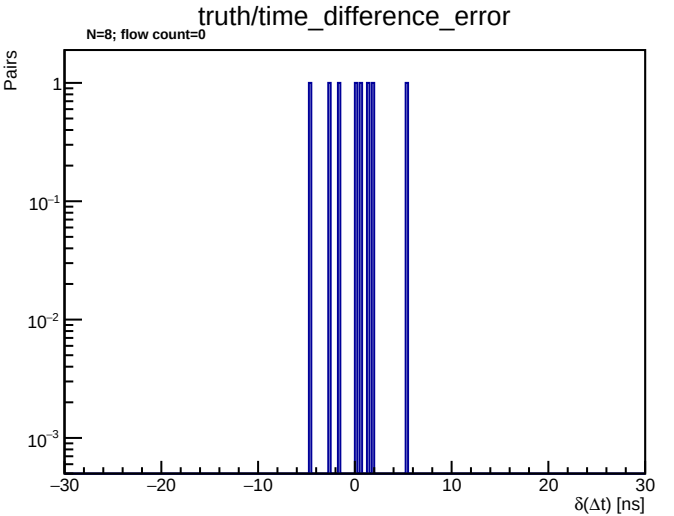

truth/efficiency
N=72; flow count=0
Events
10
1
All input events
Selected-event denomina
Any plane
Any annulus
Both primaries plane
Both primaries annulus
Both ver
Missing product
Category

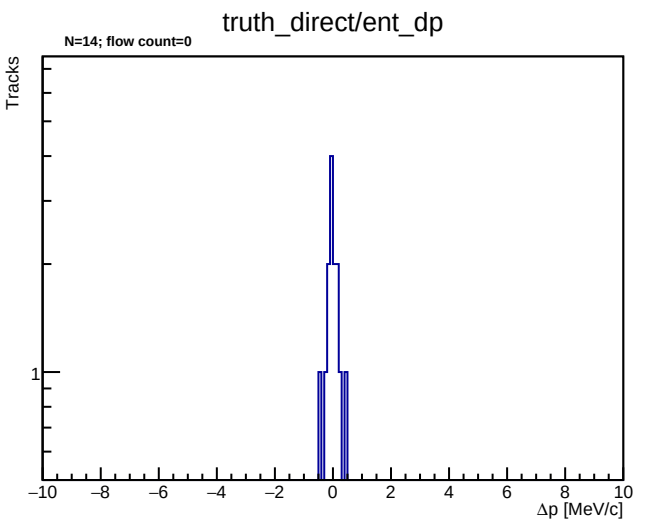

truth/failure_flags
N=15; flow count=0
Tracks
10
1
MissingInput
NotElectron
MissingAssociation
AmbiguousAssociation
NotPrimaryTruth
MissingTruth
MissingClock
MultipleMCEntrance
MissingMCEntrance
MultipleReceEntrances
MissingReceEntrances
UnsupportedTrajectory
InvalidCovariance
PropagationFailure
MissingProduction
AmbiguousProduction
OutsideTarget
CrossingSeam
BoundaryMismatch
ConfigurationDisabled
Exception
Failure

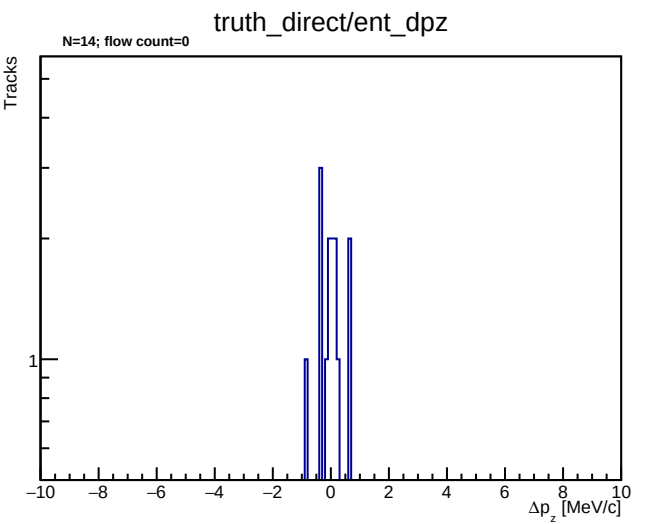

truth/opening_error_deg
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
20
40
60
80
100
120
140
160
180
180-opening angle [degrees]

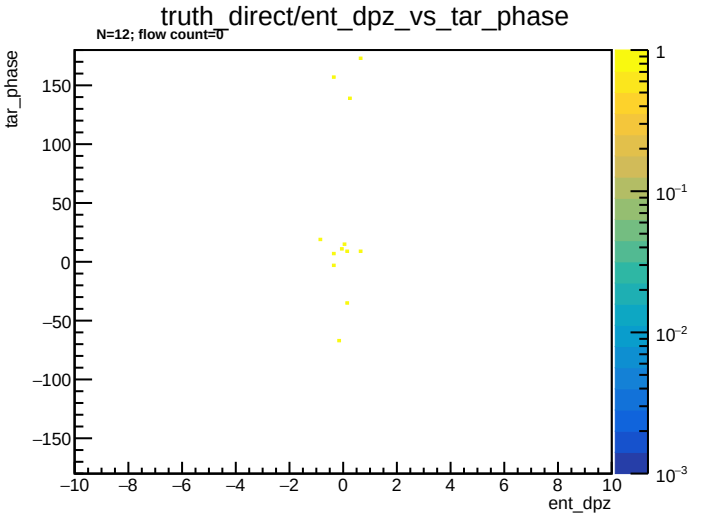

truth/separation
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
0
100
200
300
400
500
600
700
800
900
1000
Distance [mm]

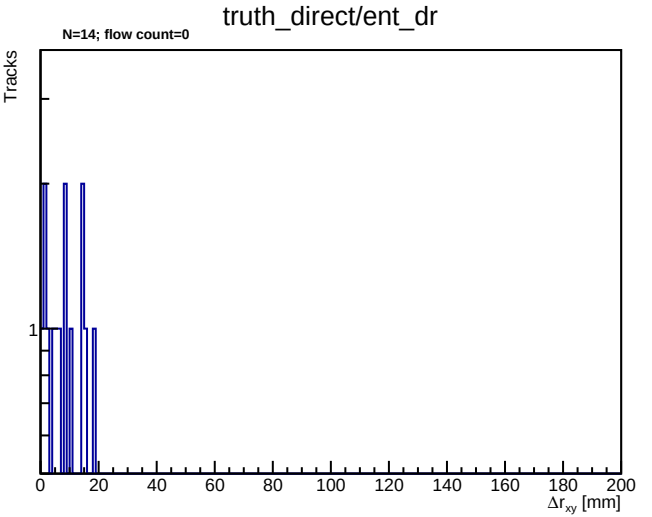

truth/time_difference
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−30 −20 −10 0 10 20 30
Δt [ns]

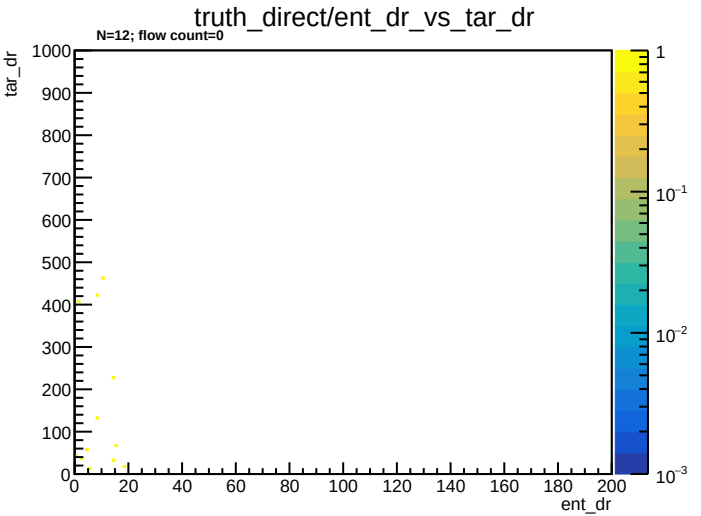

truth/time_difference_error
N=8; flow count=0
Pairs
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−30 −20 −10 0 10 20 30
δ(Δt) [ns]

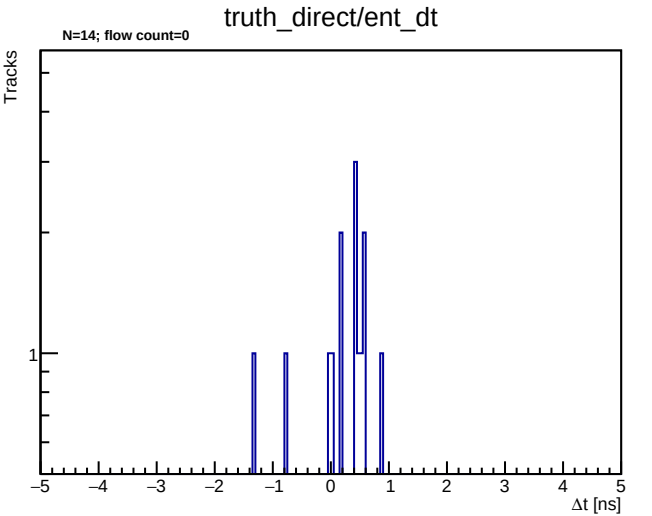

truth_direct/ent_dp
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

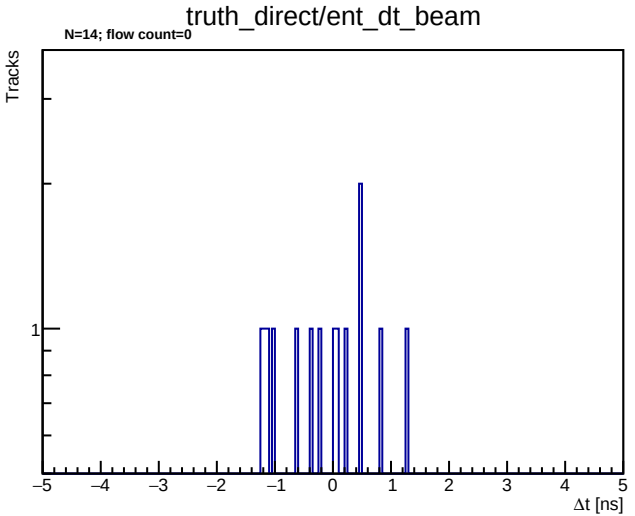

truth_direct/ent_dpz
N=14; flow count=0
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp$_z$ [MeV/c]

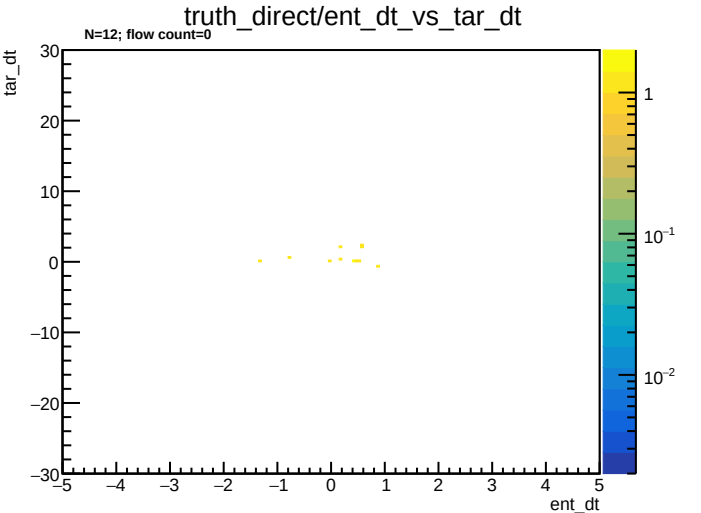

truth_direct/ent_dpz_vs_tar_phase
N=12; flow count=0
tar_phase
ent_dpz

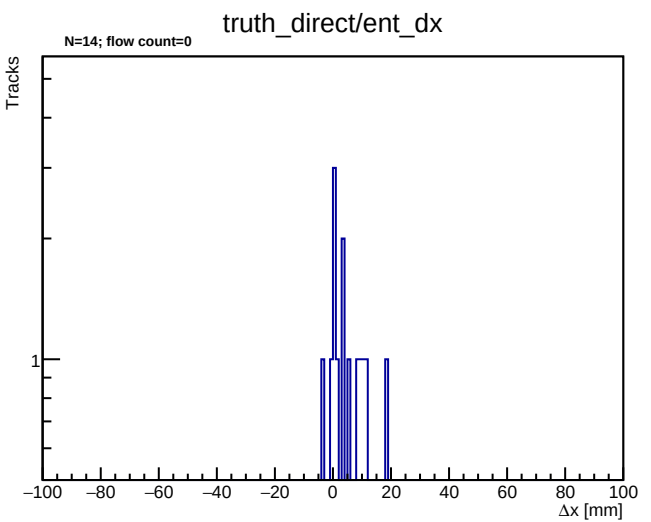

truth_direct/ent_dr
N=14; flow count=0
Tracks
Δr_xy [mm]

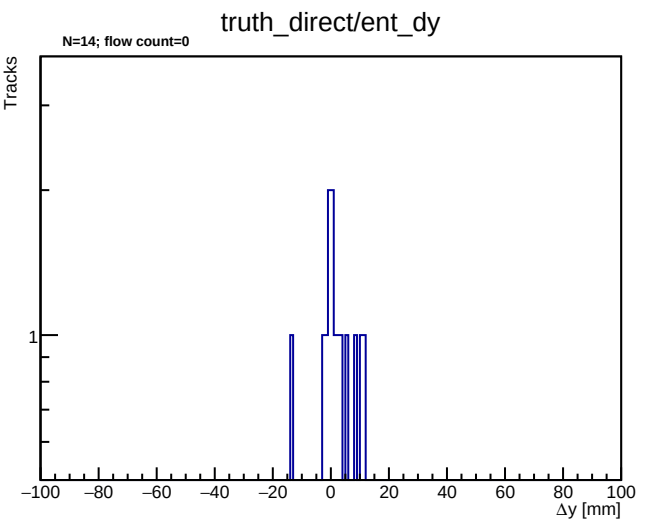

truth_direct/ent_dr_vs_tar_dr
N=12; flow count=0
tar_dr
ent_dr

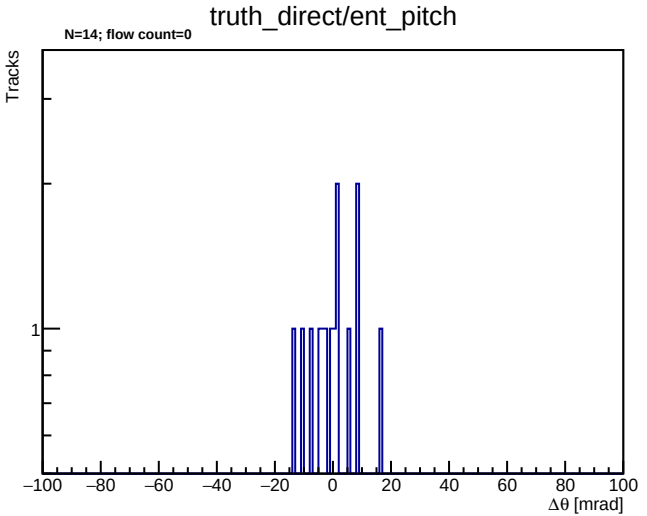

truth_direct/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

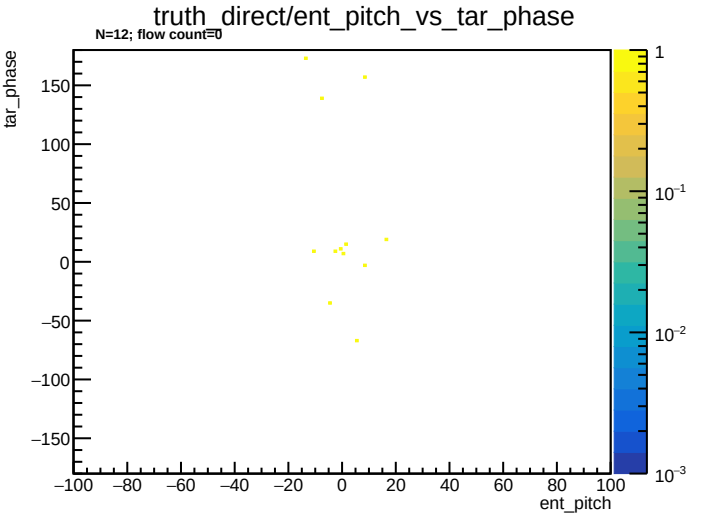

truth_direct/ent_dt_beam
N=14; flow count=0
Tracks
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

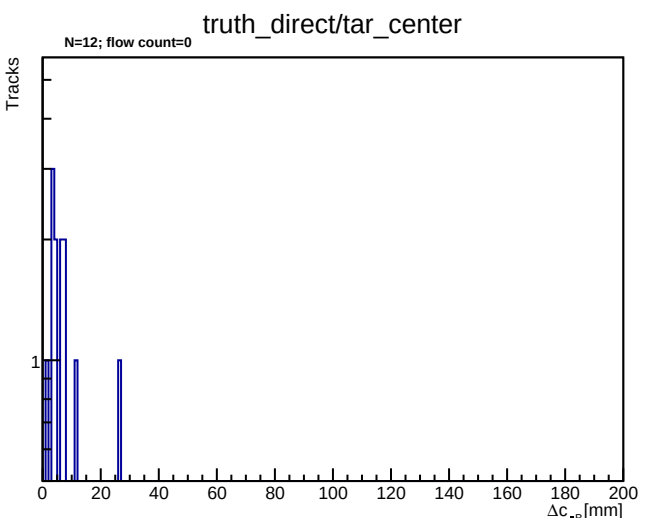

truth_direct/ent_dt_vs_tar_dt
N=12; flow count=0
tar_dt
30 20 10 0 −10 −20 −30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^−1
10^−2

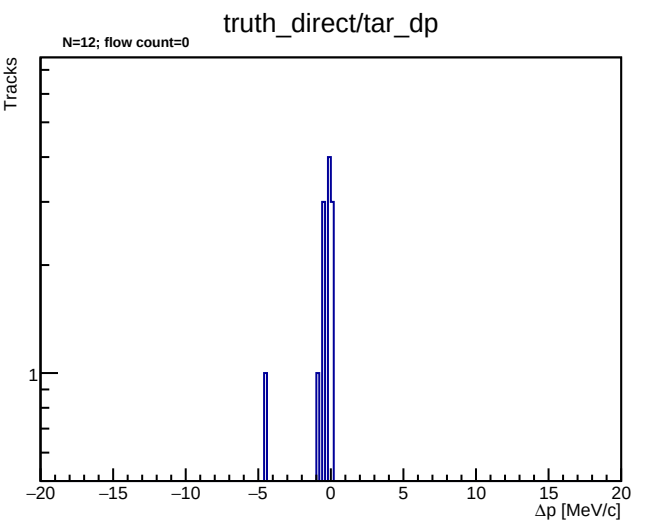

truth_direct/ent_dx
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

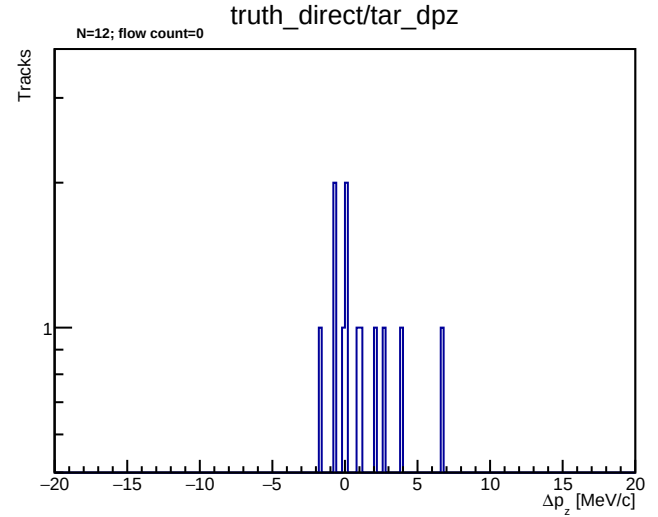

truth_direct/ent_dy
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

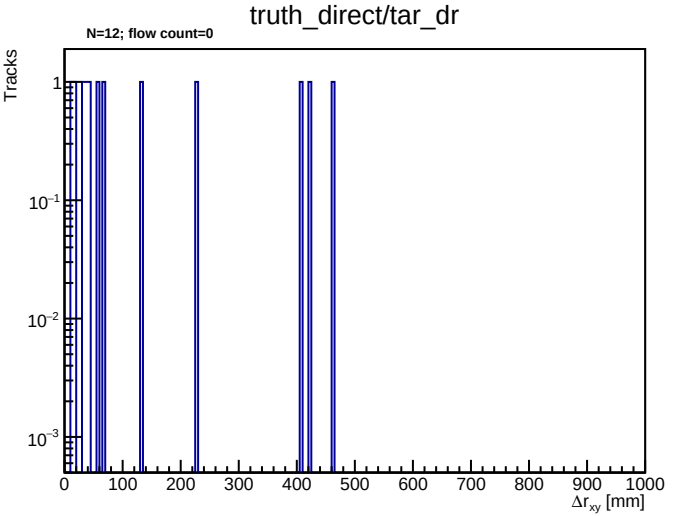

truth_direct/ent_pitch
N=14; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

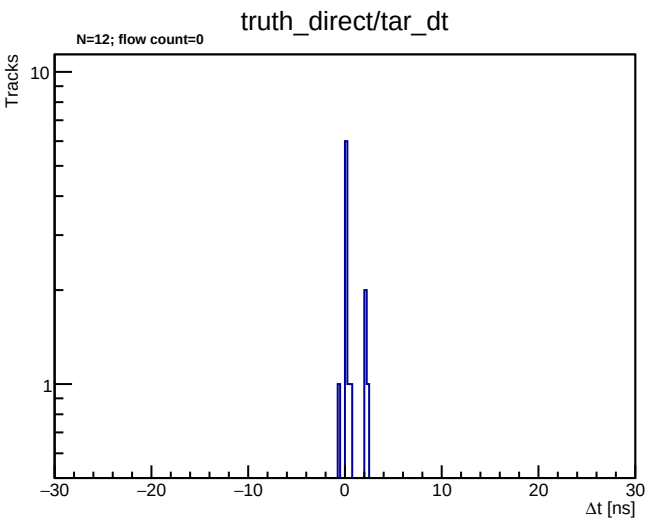

truth_direct/ent_pitch_vs_tar_phase
N=12; flow count=0
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
$10^{-1}$
$10^{-2}$
$10^{-3}$

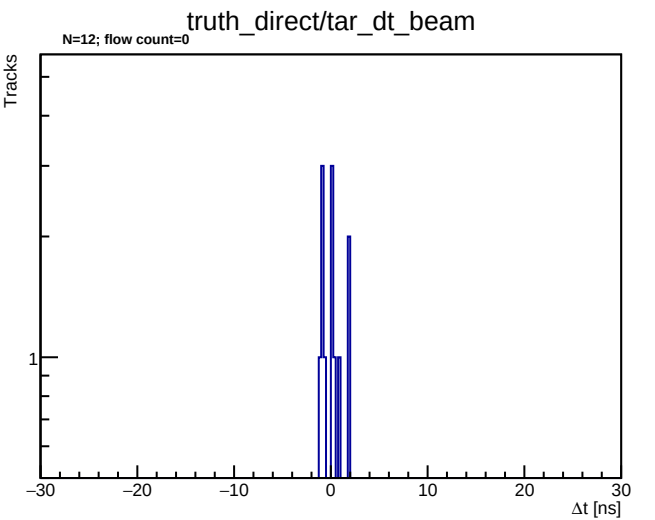

truth_direct/tar_center
N=12; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

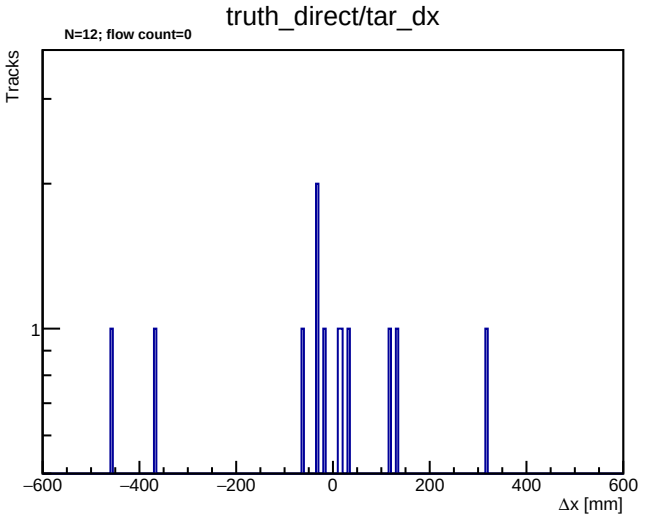

truth_direct/tar_dp
N=12; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

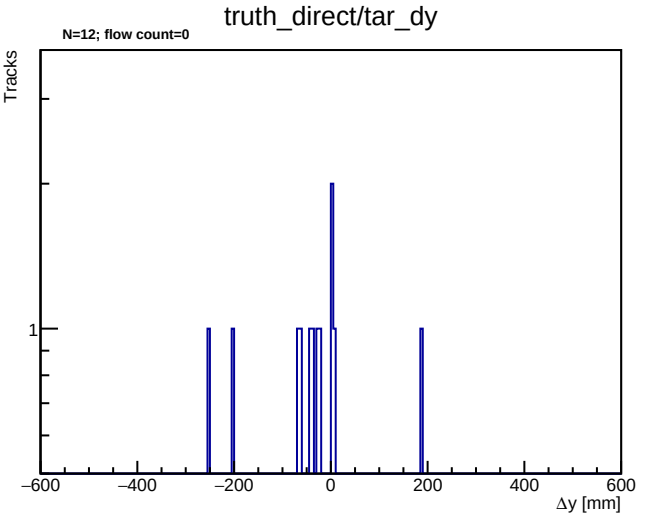

truth_direct/tar_dpz
N=12; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp_z [MeV/c]

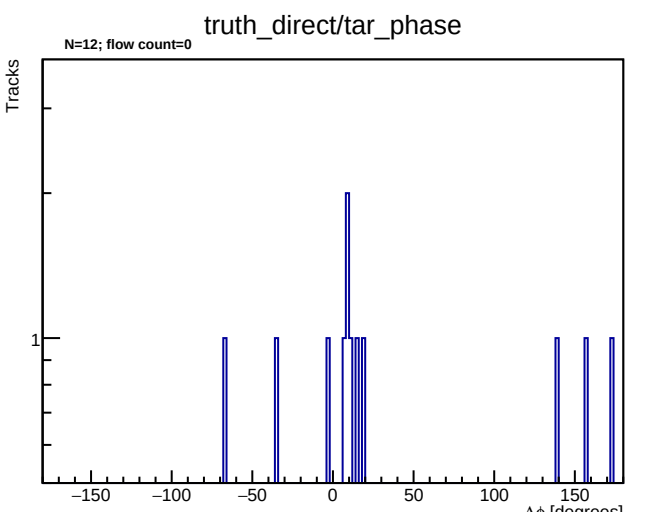

truth_direct/tar_dr
N=12; flow count=0
Tracks
1
10^-1
10^-2
10^-3
0 100 200 300 400 500 600 700 800 900 1000
Δr_xy [mm]

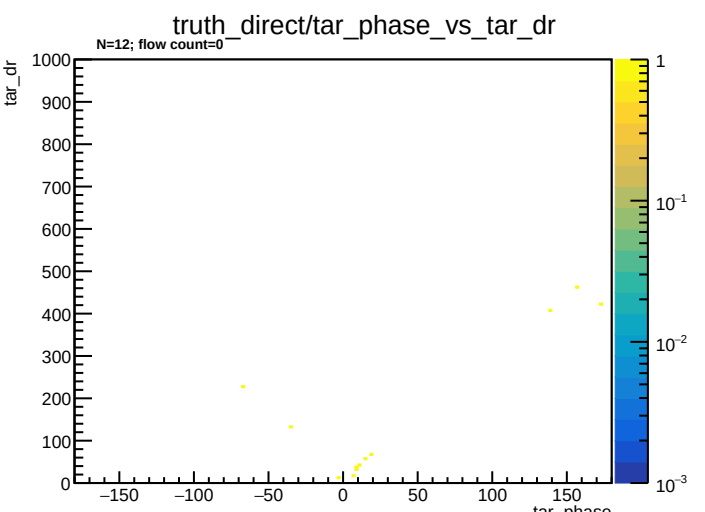

truth_direct/tar_dt
N=12; flow count=0
Tracks
10
1
−30 −20 −10 0 10 20 30
Δt [ns]

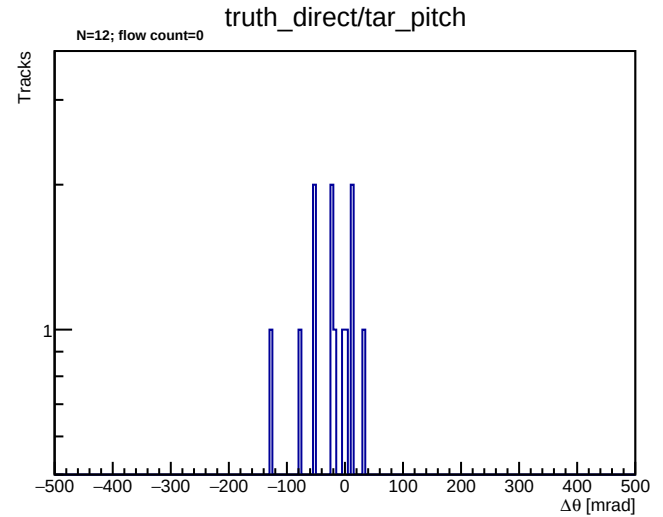

truth_direct/tar_dt_beam
N=12; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

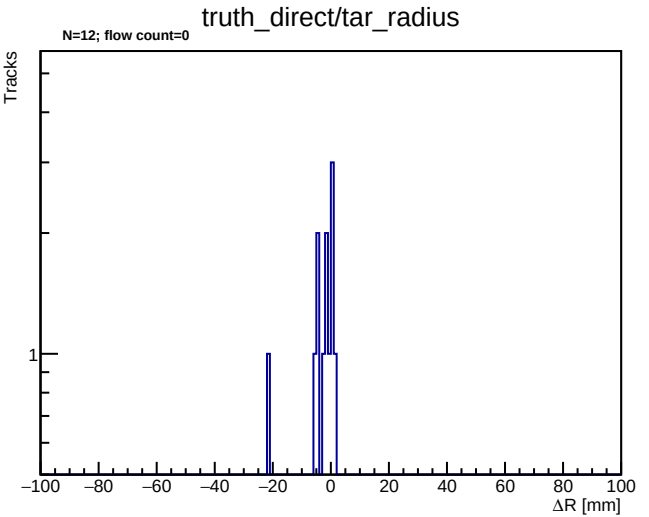

truth_direct/tar_dx
N=12; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δx [mm]

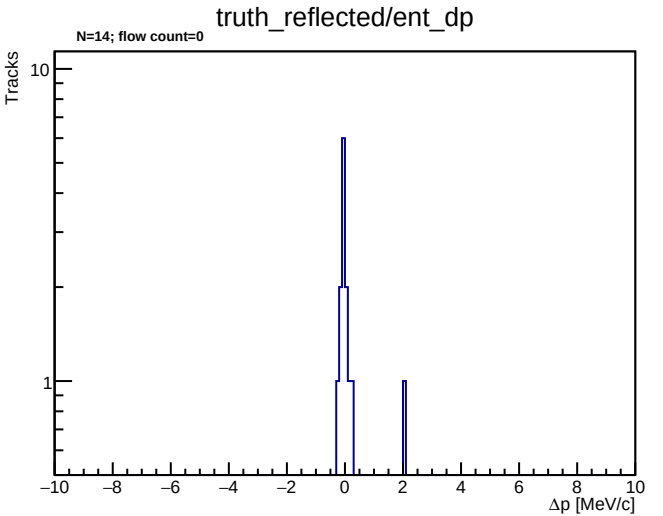

truth_direct/tar_dy
N=12; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δy [mm]

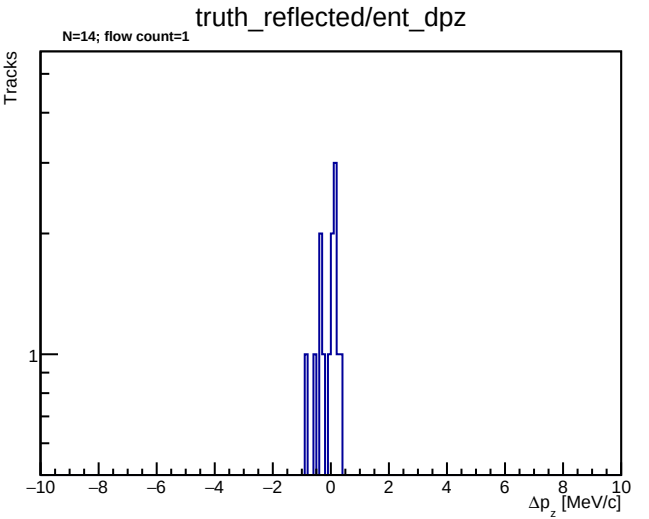

truth_direct/tar_phase
N=12; flow count=0
Tracks
1
−150
−100
−50
0
50
100
150
Δϕ [degrees]

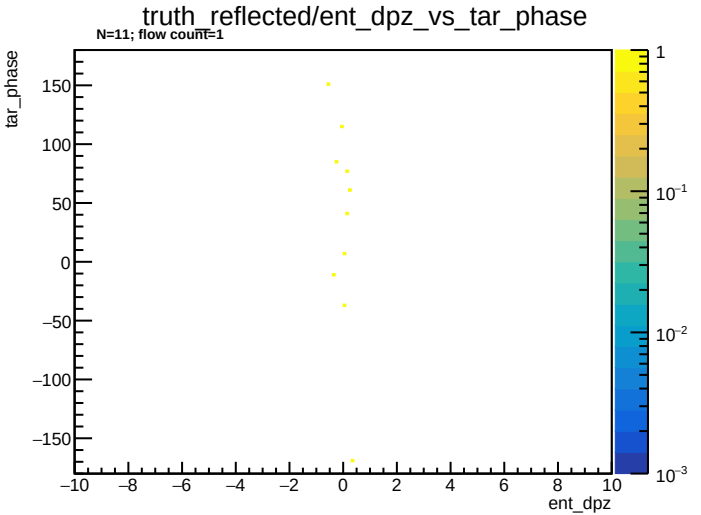

truth_direct/tar_phase_vs_tar_dr
N=12; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10⁻¹
10⁻²
10⁻³

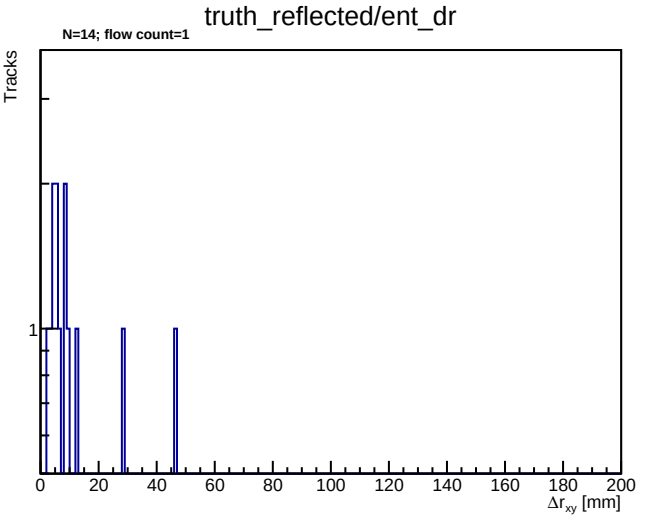

truth_direct/tar_pitch
N=12; flow count=0
Tracks
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]

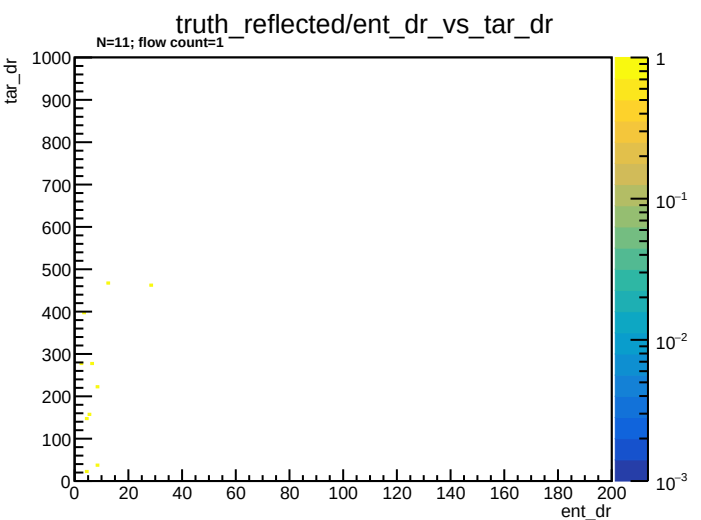

truth_direct/tar_radius
N=12; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]

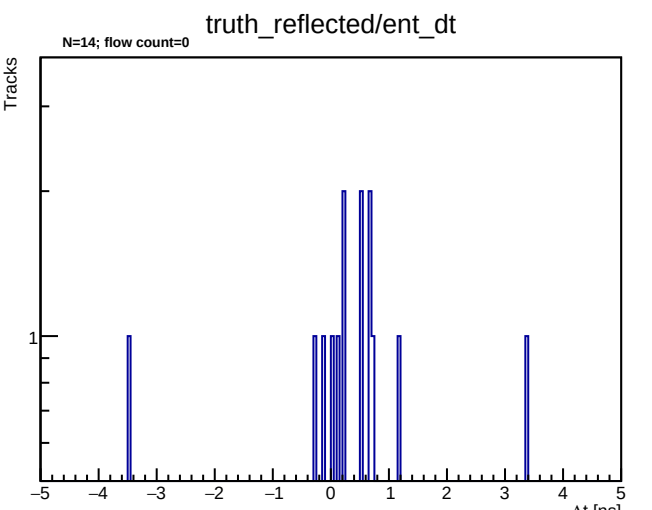

truth_reflected/ent_dp
N=14; flow count=0
Tracks
10
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp [MeV/c]

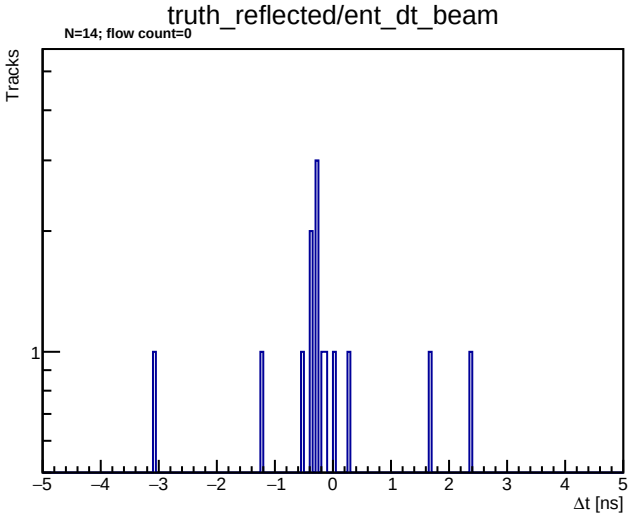

truth_reflected/ent_dpz
N=14; flow count=1
Tracks
1
−10 −8 −6 −4 −2 0 2 4 6 8 10
Δp_z [MeV/c]

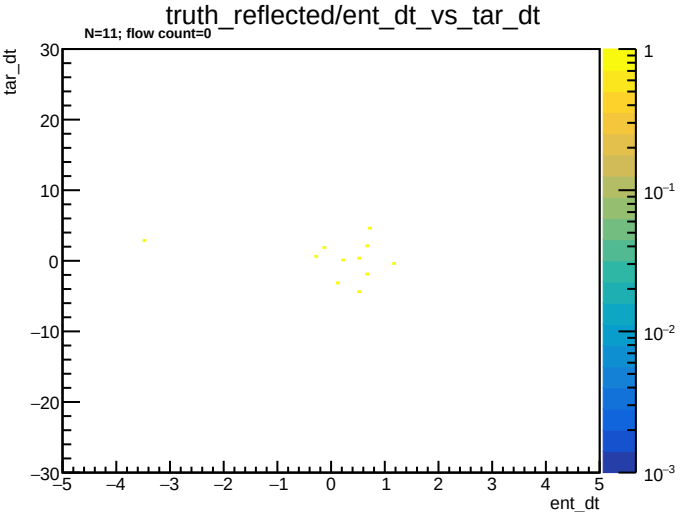

truth_reflected/ent_dpz_vs_tar_phase
N=11; flow count=1
tar_phase
ent_dpz

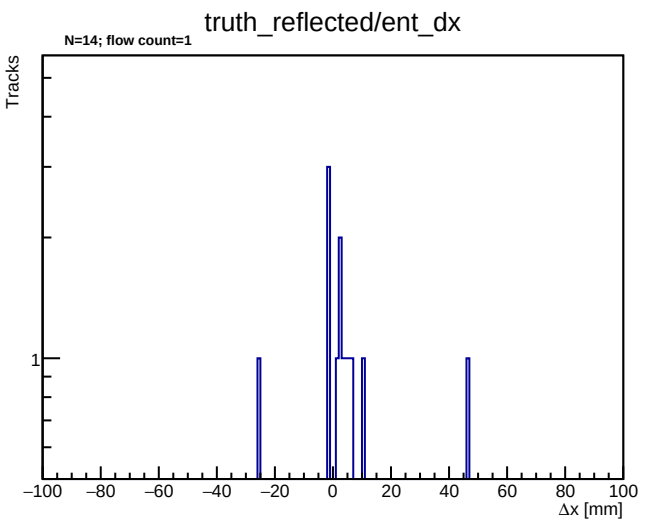

truth_reflected/ent_dr
N=14; flow count=1
Tracks
Δr_xy [mm]

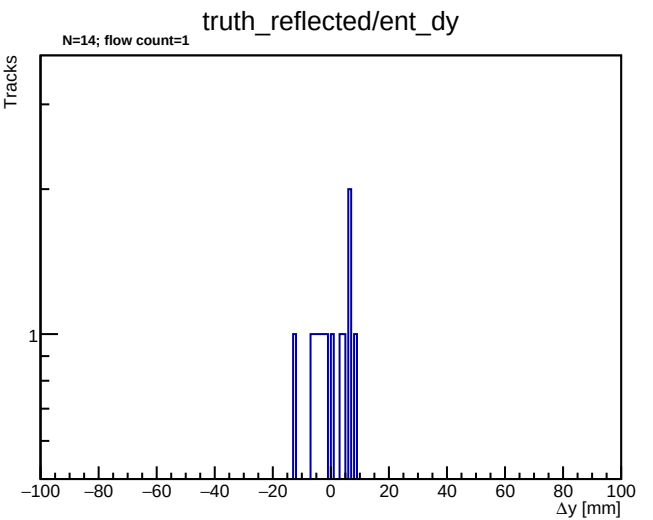

truth_reflected/ent_dr_vs_tar_dr
N=11; flow count=1
tar_dr
ent_dr

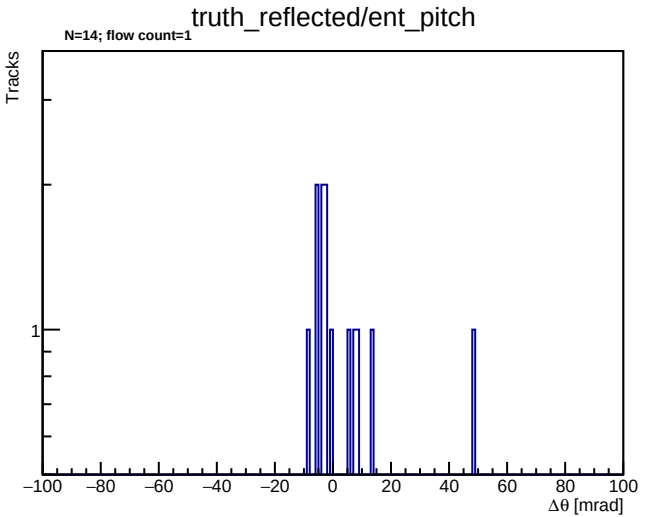

truth_reflected/ent_dt
N=14; flow count=0
Tracks
Δt [ns]

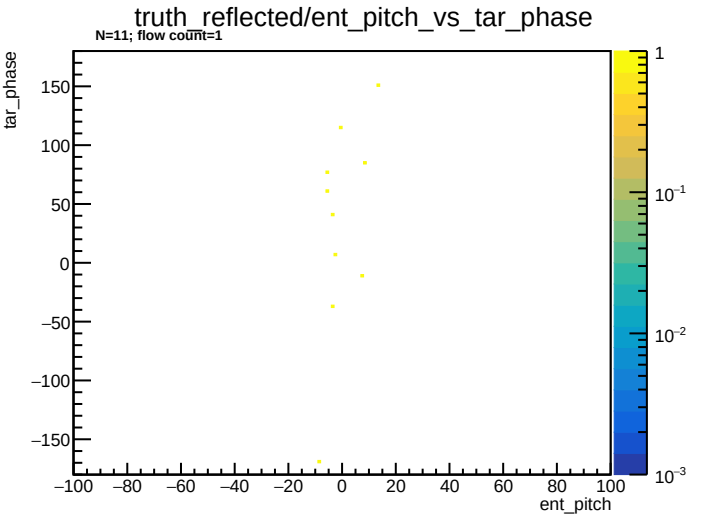

truth_reflected/ent_dt_beam
N=14; flow count=0
Tracks
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
Δt [ns]

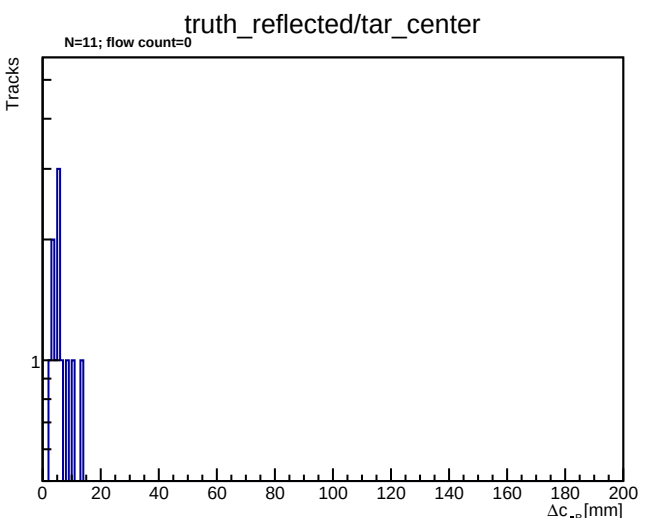

truth_reflected/ent_dt_vs_tar_dt
N=11; flow count=0
tar_dt
30 20 10 0 −10 −20 −30
−5 −4 −3 −2 −1 0 1 2 3 4 5
ent_dt
1
10^-1
10^-2
10^-3

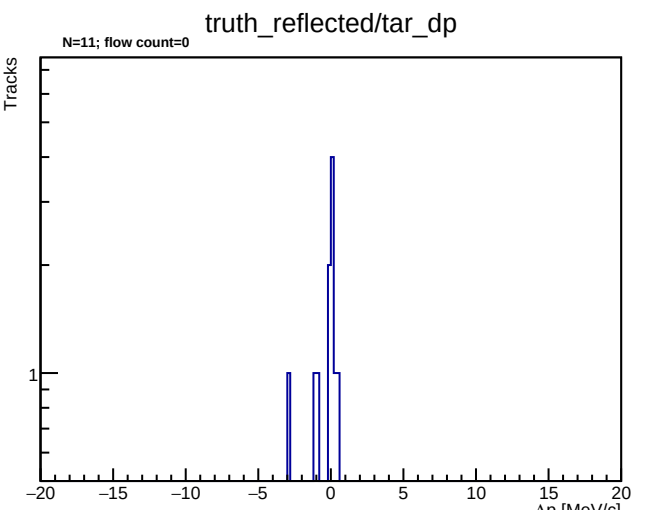

truth_reflected/ent_dx
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δx [mm]

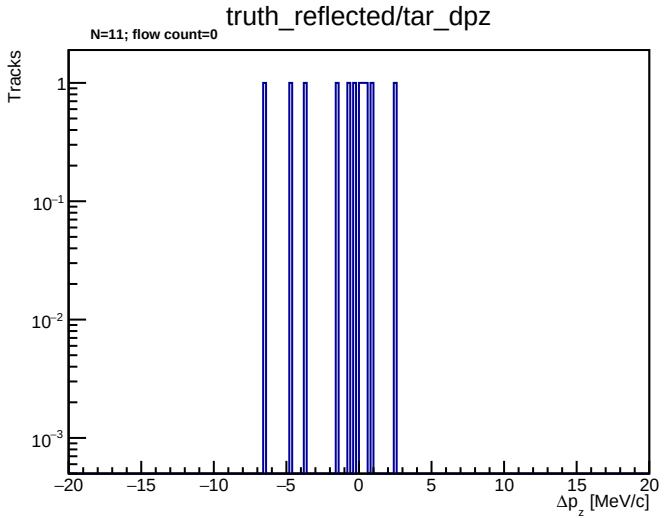

truth_reflected/ent_dy
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δy [mm]

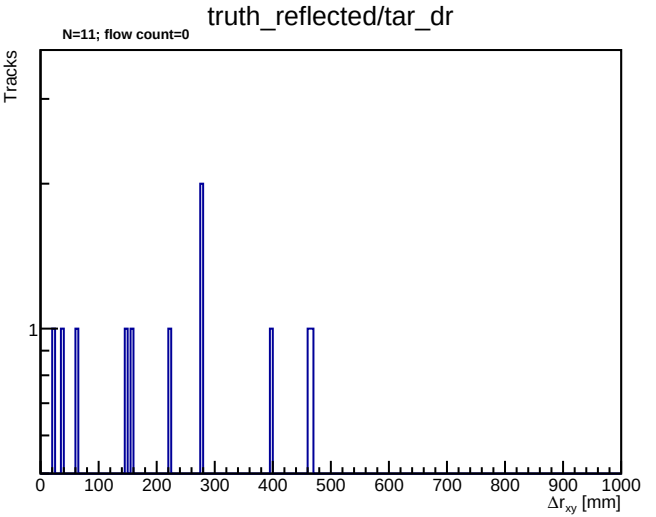

truth_reflected/ent_pitch
N=14; flow count=1
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
Δθ [mrad]

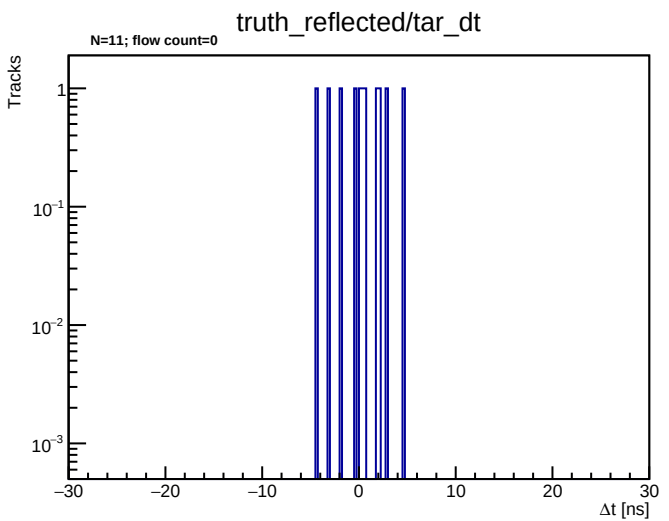

truth_reflected/ent_pitch_vs_tar_phase
N=11; flow count=1
tar_phase
150
100
50
0
−50
−100
−150
−100 −80 −60 −40 −20 0 20 40 60 80 100
ent_pitch
1
$10^{-1}$
$10^{-2}$
$10^{-3}$

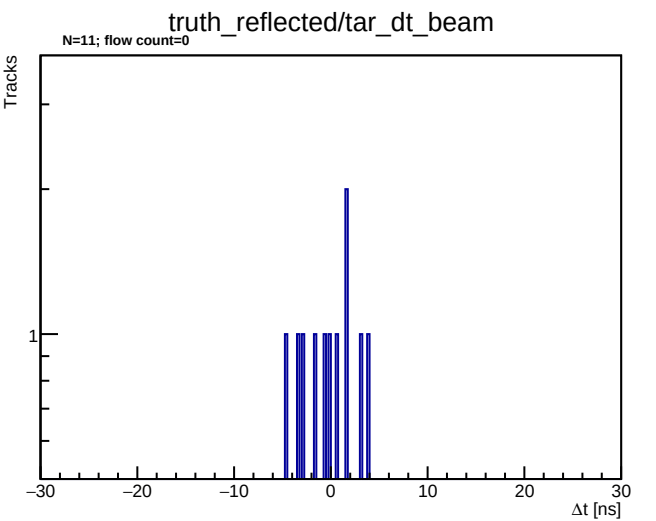

truth_reflected/tar_center
N=11; flow count=0
Tracks
1
0 20 40 60 80 100 120 140 160 180 200
Δc_B [mm]

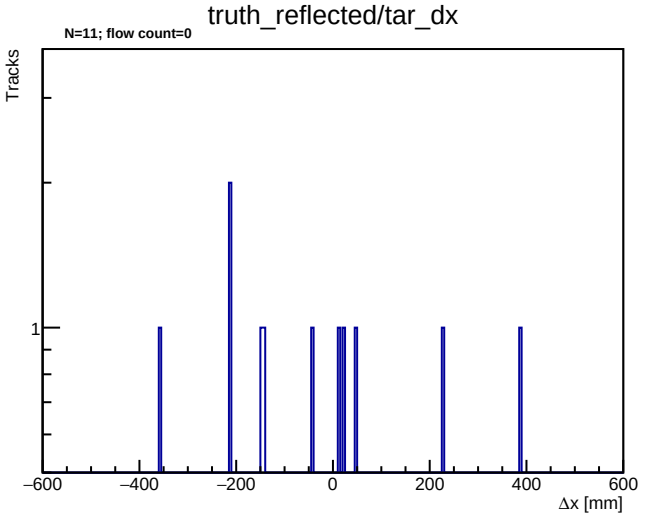

truth_reflected/tar_dp
N=11; flow count=0
Tracks
1
−20 −15 −10 −5 0 5 10 15 20
Δp [MeV/c]

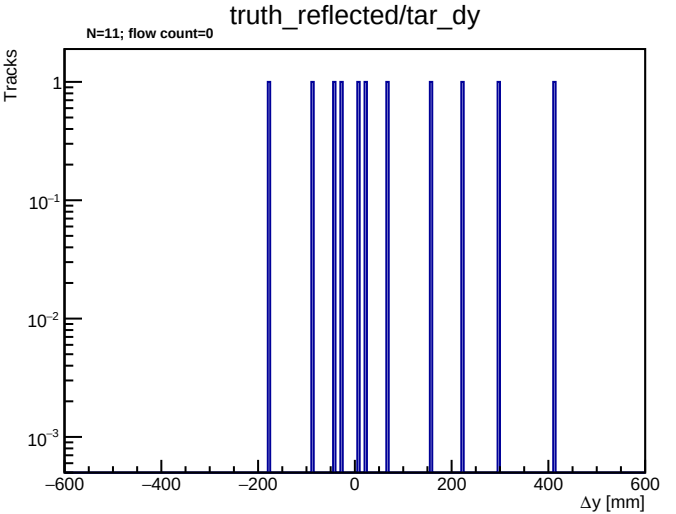

truth_reflected/tar_dpz
N=11; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−20 −15 −10 −5 0 5 10 15 20
$\Delta p_z$ [MeV/c]

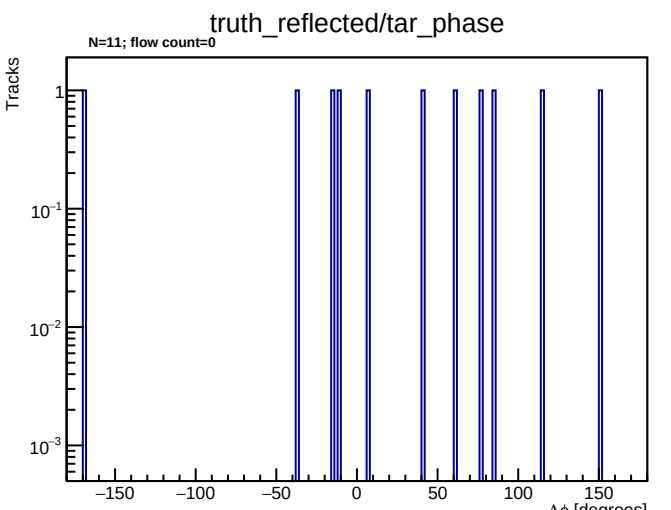

truth_reflected/tar_dr
N=11; flow count=0
Tracks
1
0 100 200 300 400 500 600 700 800 900 1000
$\Delta r_{xy}$ [mm]

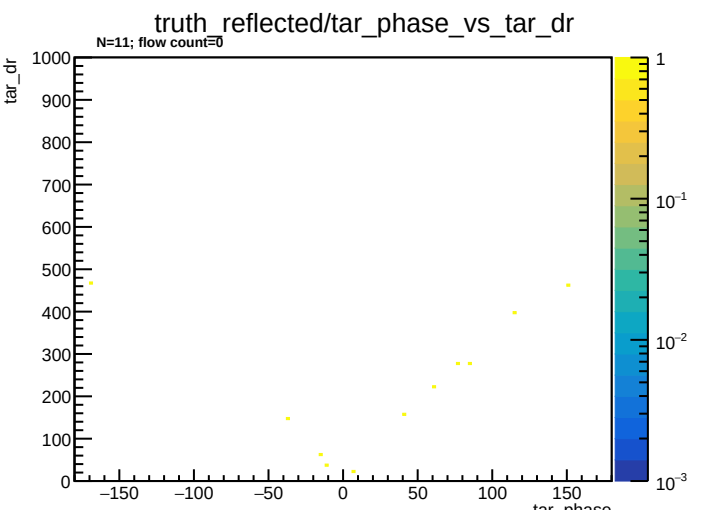

truth_reflected/tar_dt
N=11; flow count=0
Tracks
1
$10^{-1}$
$10^{-2}$
$10^{-3}$
−30 −20 −10 0 10 20 30
Δt [ns]

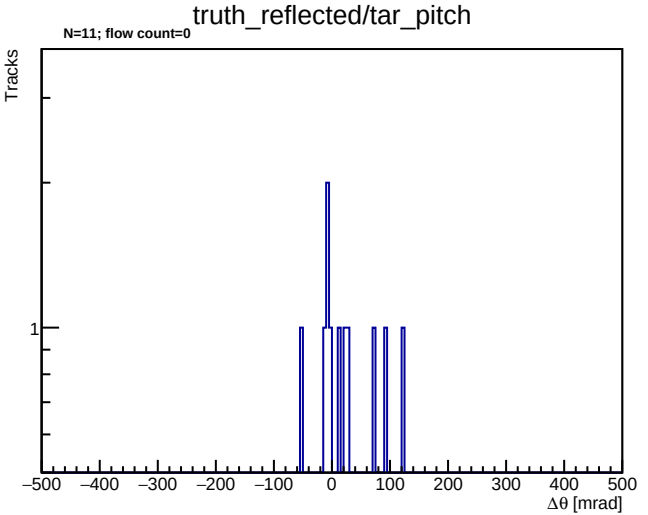

truth_reflected/tar_dt_beam
N=11; flow count=0
Tracks
1
−30 −20 −10 0 10 20 30
Δt [ns]

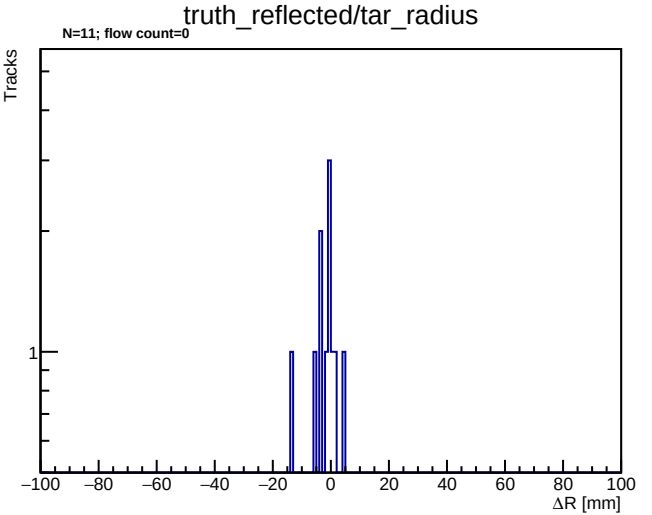

truth_reflected/tar_dx
N=11; flow count=0
Tracks
1
−600
−400
−200
0
200
400
600
Δx [mm]


truth_reflected/tar_dy
N=11; flow count=0
Tracks
1
10^-1
10^-2
10^-3
−600
−400
−200
0
200
400
600
Δy [mm]


truth_reflected/tar_phase
N=11; flow count=0
Tracks
1
10^-1
10^-2
10^-3
−150
−100
−50
0
50
100
150
Δϕ [degrees]


truth_reflected/tar_phase_vs_tar_dr
N=11; flow count=0
tar_dr
1000
900
800
700
600
500
400
300
200
100
0
−150
−100
−50
0
50
100
150
tar_phase
1
10^-1
10^-2
10^-3


truth_reflected/tar_pitch
N=11; flow count=0
Tracks
1
−500 −400 −300 −200 −100 0 100 200 300 400 500
Δθ [mrad]


truth_reflected/tar_radius
N=11; flow count=0
Tracks
1
−100 −80 −60 −40 −20 0 20 40 60 80 100
ΔR [mm]